Statistical Regularities in Cocktail Recipes:

An Exploratory Analysis of CocktailDB Adapted from the Framework of Universal Culinary Scaling Laws

Research-style exploratory data analysis report

Dataset: CocktailDB (ID, name, category, ingredients, nutrients, ratings, votes, preparation time, cuisine)

Methodological reference: Bagler, Tewari, Yadav, Singh, Bansal, Dargar, Goel & Sinha, *Universal statistical laws governing culinary design* (arXiv:2604.28021v1).

This report does not reproduce the paper's global-cuisine corpus. It tests which of the paper's statistical questions can be asked, and which patterns actually appear, in a cocktail-specific dataset.

16,895 raw rows · 14,916 unique names · 16,872 recipes in the frequency corpus · 14,466 normalized ingredients

How this study was conducted

This front section documents the computational workflow before any scientific results. It answers three practical questions: how the pipeline was built, what exploratory and preprocessing operations were applied to CocktailDB, and which libraries implemented each step. Later sections report findings; they do not re-describe the engineering.

Design principle

The professor's paper (Bagler et al., arXiv:2604.28021v1) supplied the *questions and estimators*, not a template of numbers to recover. Work therefore began by reading the paper's Introduction, Figures 1–4, Methods, and Extended Data, then mapping each paper quantity onto CocktailDB columns. Where a variable was absent (cooking techniques, utensils, instruction steps, reliable cuisine labels, NER attributes, per-serving nutrients), the analysis was marked **not testable** rather than approximated with a substitute. Where a variable existed (pipe-delimited ingredients, ratings, votes, prep time, nutrient strings), the paper's conceptual definition was implemented on this file and the empirical pattern was allowed to agree, differ, or be weak.

Project layout and reproducibility

All work lives under `eda_project/`. The original CSV is copied to `data/CocktailDB_raw.csv` and is never overwritten. Scripts write a cleaned table, long ingredient table, wide nutrient table, figures (PNG + vector PDF), CSV summaries, and JSON numerical payloads. The research PDF is assembled last by reading those JSON files, so exponents and sample sizes in the text are generated rather than typed by hand. The full run is `python scripts/run_all.py`, which executes scripts 01–11 in order.

Path	Role
data/	Raw CSV (untouched original copy), CocktailDB_clean.csv, ingredients_long.csv, nutrients_wide.csv
scripts/	Numbered pipeline: audit → clean → EDA → Zipf/Heaps/MA → nutrition → co-occurrence → multivariate → PDF
figures/	All numbered figures as PNG (300 dpi) and PDF (vector)
tables/ and tables/json/	CSV summaries and machine-readable results used by the report builder
report/	CocktailDB_EDA_Research_Report.pdf
README.md, requirements.txt	Reproduction notes and Python dependencies

Pipeline architecture (what each script does)

The pipeline is linear and idempotent: each stage reads the raw or cleaned table and writes artifacts for downstream stages. Shared helpers live in `paths.py`, `plot_style.py` (academic matplotlib defaults), `ingredient_norm.py`, `parsers.py` (votes, prep time, nutrients), and `stats_utils.py`.

Script	What it does
01_data_audit.py	Load raw CSV. Shape, dtypes, unique counts, missingness, exact duplicates, Name vs Name.1 identity, Name+Ingredients clones, same-name variants, rating–vote joint missingness, cuisine/category sparsity, paper-to-column mapping.
02_cleaning.py	Build the analytical table without mutating the raw file. Drop Name.1. Parse votes and prep minutes. Flag Rating=0 & Votes=0 as unrated. Split ingredients; apply four normalization levels. Extract nutrients. Define the frequency corpus.
03_basic_eda.py	Missingness bar and matrix. Recipe-size histogram, violin, ECDF. Top categories (non-missing only). Rating and vote distributions; rating vs log1p(votes). Prep-time distributions and Spearman associations.
04_ingredient_analysis.py	Recipe-incidence frequencies f_i (no double-counting within a recipe). Top-30 bars, frequency histogram, log–log rank–frequency plot.
05_zipf.py	f(r)~r^(−α) with log–log OLS over multiple rank windows; Clauset-style MLE on the frequency vector; likelihood-ratio checks vs log-normal and exponential.
06_heaps.py	100 random recipe orders. Cumulative unique ingredients V(R). Fit β on log–log V vs R (R≥20). Mean curve, 95% permutation band, β histogram.
07_menzerath.py	s_i=−log p(i), L=unique ingredients, y=mean information. Scatter, binned means, nonlinear least squares for y=a L^b exp(cL). Sensitivity across normalization and p(i) definitions.
08_nutrition.py	Parse two nutrient string formats; map Carbs↔Carbohydrates. Descriptives, completeness, calorie/macro plots, Spearman nutrient matrix, KS fits (log-normal, gamma, Weibull, Gaussian, exponential) on x>0.
09_cooccurrence.py	Top-40 co-occurrence heatmap; PMI with a minimum joint-count threshold; network of positive-PMI edges.
10_multivariate.py	Pre-specified Spearman pairs (complexity, popularity, nutrients) with bootstrap CIs. Optional PCA on complete-case standardized nutrients.

Script	What it does
12_extended_additions.py	Prep-time boxplots by top-10 categories; Bayesian vote-adjusted ratings; Zipf log-frequency boxplots by rank stratum.
11_generate_report.py	Compose this PDF (ReportLab): narrative, embedded figures, captions, comparison table. Numbers loaded from JSON.

First inspection (before any modeling)

The raw file has 16,895 rows and 9 columns. Header duplication produced pandas names `Name` and `Name.1`; these two series were compared programmatically and are identical (0 mismatches). Ingredients are 100% present as pipe-delimited free text (quantities still inside the string). Votes are text (comma grouping such as “1,216”). Prep time mixes “3 minutes”, “5 mins”, seconds, hours, days, duplicated phrases, ranges, and unitless numbers. Nutrients use two textual grammars: compact “184 Calories|3g Fat|8g Carbs” and colon-style “Calories: 239 kcal|Carbohydrates: 14 g|…”. The paper PDF was used to lock estimators: f_i = number of recipes containing ingredient i; Zipf via rank–frequency plus Clauset MLE; Heaps with shuffled incremental sampling; Menzerath–Altmann y(L) = a L^b exp(cL); nutrient log-space KS. Cuisine-level laws were abandoned at this stage because Cuisine is 97.7% missing.

Exploratory data analysis protocol

EDA was treated as a sequence of documented questions, each with an explicit observation count n, not as a gallery of charts. Univariate work covered recipe size, labelled categories, ratings, votes, and parsed preparation time. Bivariate work used Spearman rank correlation as the default association measure because votes, times, and nutrients are skewed; Pearson was reported only as a companion on log1p(votes) vs rating. Bootstrap 95% intervals (400 resamples) accompany key Spearman estimates. Heavy-tailed variables were shown on both linear and log (or log1p) axes. ECDFs were used for discrete recipe size. No 3-D plots and no pie charts. matplotlib only, white background, fixed color palette, 300 dpi PNG plus vector PDF. Every figure in the results sections carries: title, axis labels with units, sample size, and a five-part caption (what it shows, key insight, why it matters, statistical evidence, caveat).

EDA block	Operations
Missingness	Percent-missing bars; binary missingness matrix; combination patterns of Category/Nutrients/Ratings/Votes/Prep_Time/Cuisine; joint rating–vote table.
Duplicates	Exact row clones vs Name+Ingredients clones vs same name with different ingredients. Only the first two classes were removed from the frequency corpus.
Recipe structure	raw vs unique ingredient counts; repeats inside a recipe; histogram, violin/box, ECDF of L.
Categorical fields	Category restricted to non-missing rows; long-tail and cumulative share. Cuisine counted and then ruled out for stratified analysis.
Popularity	Rating histogram (bounded 0–5); votes linear + log1p; scatter rating vs log1p(votes); low-vote vs high-vote extremeness.
Prep time	Parsed minutes; raw + log10 histograms; Spearman vs ingredient count and vs votes. Extremes inspected, not auto-deleted.
Ingredients	Incidence ranking; hapax fraction; log–log rank–frequency; co-occurrence and PMI on a top-N set.
Scaling laws	Zipf range sensitivity; Heaps permutations; Menzerath binned NLS + residual/sensitivity checks.
Nutrition	Format harmonization; completeness; macros; Spearman matrix; candidate distribution KS; PCA if complete cases sufficed.

Preprocessing decisions (what we changed, and what we refused to change)

Kept untouched. The original CSV. Original Ratings, Votes, Prep_Time, Nutrients, and Ingredients strings remain on the cleaned table beside derived fields.

Dropped from models only. Redundant `Name.1`. 17 extra exact duplicate rows and 6 extra Name+Ingredients clones, solely so one scraped formula cannot inflate f_i. 1131 names with different ingredient lists were retained as variants.

Ratings / votes. Commas stripped; 0 malformed vote strings. All 116 rows with Rating=0 also had Votes=0; they were recoded as missing in `rating_analyzed` / `votes_analyzed`, not as genuine zero-star scores. No imputation of ratings, votes, categories, cuisine, or free text.

Preparation time. Tokenized into minutes (seconds/60, hours×60, days×1440; ranges averaged; duplicated phrases de-duplicated when the whole string repeated; unitless numbers treated as minutes). Unparsed strings: 0. Zeros and multi-hour/day values were flagged, not deleted.

Ingredients. Split on "|". Four documented normalization levels: *surface* (lowercase/whitespace only); *standard* (strip leading/trailing quantities and units; main analyses; Cointreau→orange liqueur and Kahlua→coffee liqueur only as obvious aliases); *moderate* (also garnish/prep descriptors); *core* (sensitivity only: ice variants and drop "fresh"). Lime, lime juice, and fresh lime juice were **not** collapsed. Frequency is distinct recipes, not mention counts.

Nutrients. Two parsers (colon labels and compact "value + unit + name"). Carbs and Carbohydrates merged only after confirming both are gram masses. Vitamin A remains in IU where reported. Serving size is not assumed. Log transforms and KS tests use strictly positive values; zeros are counted separately and excluded from log-space.

Libraries and statistical tools

The environment is CPython 3.13. Analyses are ordinary scientific Python; no black-box AutoML. ReportLab builds the paginated PDF. Pillow measures figure aspect ratios so embedded plots are not stretched.

Library	Use in this project
pandas / numpy	Tables, missingness, grouping, frequency corpus, I/O of CSV/JSON.
matplotlib	All research figures (histograms, ECDF, log–log plots, heatmaps, PCA scree/projection).
scipy	Spearman/Pearson, skewness, linregress (log–log OLS), curve_fit (Menzerath–Altmann), kstest, candidate distribution MLE.
powerlaw (Alstott)	Clauset-style discrete power-law MLE on ingredient frequencies; likelihood-ratio tests vs log-normal and exponential.
scikit-learn	StandardScaler + PCA on complete-case nutrient rows only (descriptive, not prediction).
networkx	Positive-PMI co-occurrence graph layout.
statsmodels	Available in the environment; core association tests used scipy to keep estimators transparent.
reportlab / Pillow	Academic PDF layout, tables, captions, page numbers; non-distorted figure embedding.
re / unicodedata	Ingredient quantity/unit stripping; vote commas; prep-time tokens; nutrient format detection.

Quality checks after the run: re-execution from the raw CSV; figure files present; JSON values match captions; NaNs excluded from correlations and log transforms; duplicate formulas not allowed to inflate frequencies; log(x) only for x>0; outliers investigated rather than bulk-deleted; no causal language in captions.

Executive Summary

Dataset. CocktailDB contains **16,895 recipe rows** and nine columns. The two name fields are identical. There are **14,916 unique recipe names**. After removing 17 exact duplicate rows and 6 additional Name+Ingredients duplicates, **16,872 recipes** enter frequency analyses. Same-name recipes with different ingredient lists (1131 names) were retained as variants. Major missingness: Category 59.0%, Nutrients 66.3%, Ratings 52.5%, Votes 52.6%, Prep_Time 58.6%, Cuisine 97.7%. Ingredients are complete.

1. Ingredient usage is concentrated: the most common normalized ingredient is “vodka” (2,191 recipes; 13.0% of the corpus). 73.5% of the vocabulary occurs in only one recipe.

2. Rank–frequency curves are heavy-tailed. A log–log OLS Zipf exponent of $\alpha \approx 1.00$ was obtained on ranks 5–500 (descriptive $R^2 = 0.991$). A Clauset-style power-law fit to the frequency distribution is reported separately and is not treated as automatic confirmation of Zipf’s law.

3. Heaps’ law is clearly sublinear: mean $\beta = 0.738$ (100 permutations; 95% interval [0.725, 0.751]). This is close to the paper’s global $\beta \approx 0.56$ but is estimated on a smaller, cocktail-only corpus and is not a target to match.

4. Menzerath–Altmann: Spearman correlation between recipe size L and mean ingredient information y(L) is $\rho = 0.101$. The paper’s decreasing-then-rising global pattern is not assumed; the fitted MA parameters and R^2 are reported in §6.3.

5. Recipe size is small relative to the paper’s food corpus: mean unique ingredients = 4.67 (median 4.0; n = 16,872).

6. Ratings, where observed, are high and left-censored toward 5 stars (mean 4.32, median 4.45, n = 7,890). All 116 Rating=0 rows have Votes=0 and were treated as unrated.

7. Votes are heavy-tailed. Spearman association of votes with rating is $\rho = -0.058$ (n = 7878). This is an association, not evidence that votes cause ratings.

8. Cuisine cannot support cuisine-level laws: only 393 non-missing values (2.33% of the frequency corpus), with labels that mix drink types and nationalities.

9. Category is available for 6,925 recipes (41.0% of the frequency corpus); category comparisons are conditional on that subset.

10. Nutrient strings exist for 5,694 recipes (66.3% missing). Log-space KS comparisons are reported per nutrient; log-normality is not claimed globally. Best KS candidates include: calories: lognorm; carbohydrates: gamma; protein: weibull; fat: lognorm; sugar: gamma; sodium: lognorm.

11. Strongest complete-case Spearman associations among the pre-specified multivariate pairs: ingredient count vs calories (ρ=0.25, n=5683), ingredient count vs votes (ρ=-0.19, n=7878), ingredient count vs protein (ρ=0.14, n=4988).

12. Cooking techniques, utensils, instruction steps, and 26-cuisine stratification from the reference paper cannot be reproduced because those variables are absent.

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Abstract

Bagler and colleagues argue that traditional recipes behave as a compositional symbolic system: ingredient frequencies are Zipf-like, culinary vocabulary grows sublinearly (Heaps' law), recipe size trades off with average ingredient information (Menzerath–Altmann), and macronutrient amounts are approximately log-normal. This study asks whether those regularities appear in CocktailDB, a corpus of 16,895 cocktail-related recipe rows with pipe-delimited ingredients and partially observed ratings, votes, preparation times, categories, cuisines, and nutrient strings. Methods were adapted, not copied: frequency is the number of distinct recipes containing a normalized ingredient; Heaps curves use 100 random recipe orders; Menzerath–Altmann uses y(L)=mean[−log p(i)]; nutrient log-normality is tested only on positive extracted values. Technique- and cuisine-stratified analyses from the paper are not testable here. Cocktail recipes are short (median 4 unique ingredients). Ingredient frequencies are heavy-tailed (OLS Zipf $\alpha \approx 1.00$ on a restricted rank range). Vocabulary growth is sublinear ($\beta \approx 0.74$). The Menzerath–Altmann relationship is estimated rather than assumed to match the global food corpus. Nutrient coverage is incomplete and two text formats had to be harmonized. The contribution is an honest mapping from a global culinary-law framework onto a cocktail dataset, including negative and non-testable results.

1. Introduction

The reference paper treats recipes as symbolic compositions analogous to sentences: discrete culinary units are combined under cultural and practical constraints. In a corpus of 118,083 traditional recipes from 26 cuisines, the authors report Zipf-like ingredient usage (global rank–frequency exponent $\alpha \approx 1.53$; cuisine-mean ≈ 1.39), Heaps' law with $\beta \approx 0.56$ (cuisine-mean ≈ 0.54), a Menzerath–Altmann relation y(L)=a L^b exp(cL) with a global minimum near $L \approx 13$, and log-normal macronutrient concentrations. Those claims concern a NER-structured, cuisine-labelled food corpus, not cocktails.

Cocktails are a narrower compositional domain: typically few ingredients, repeated base spirits, juices, liqueurs, and garnishes, and recipe sources that often mix blogs, databases, and user reviews. If Zipf/Heaps structure is truly generic to culinary design, a cocktail corpus should show heavy-tailed reuse and sublinear vocabulary growth even without global cuisine coverage. If instead those laws depend on savory cooking, long ingredient lists, or NER-normalized taxonomies, cocktail data may differ—and that difference is scientifically informative.

What can and cannot be reproduced. Reproducible in spirit: ingredient frequency, Zipf rank–frequency analysis, Heaps growth, Menzerath–Altmann on recipe size versus mean ingredient information, nutrient distributional tests where values exist, and recipe-size histograms. Not reproducible: named-entity decomposition of quantity/unit/state; cooking-technique and utensil vocabularies; instruction-step complexity; geographic maps; 26-cuisine exponent distributions; per-serving RecipeDB nutrient standardization; and the paper's generative simulations as a reproduction. Cuisine in CocktailDB is 97.7% missing and is not a substitute for the paper's cuisine axis.

2. Research Objectives

The objectives are: (i) audit CocktailDB and map paper variables onto observed columns; (ii) quantify missingness without imputing ratings, votes, or text fields; (iii) standardize ingredients with a documented hierarchy and without collapsing lime / lime juice / fresh lime juice; (iv) describe recipe size, category, rating, vote, and preparation-time distributions on observed subsets; (v) test Zipf, Heaps, and Menzerath–Altmann structure; (vi) parse nutrients, assess completeness, and compare candidate distributions; (vii) add co-occurrence and multivariate association analyses that the paper does not emphasize; (viii) state where the cocktail corpus agrees, differs, or cannot be compared.

3. Dataset Description

The file contains 16,895 rows and 9 columns: Name, Name.1, Category, Ingredients, Nutrients, Ratings, Votes, Prep_Time, Cuisine. Name equals Name.1 on every row (mismatches = 0). Name.1 is redundant and is dropped from models. Ingredients are pipe-delimited free-text phrases with quantities and units still inside the string (unlike the paper's NER attributes). There is no identifier column beyond the recipe name. Votes are stored as text (including comma grouping). Ratings are numeric on a 0–5 scale.

Paper quantity	CocktailDB field	Status
Recipe document	Row; Name	Present
Ingredient identity (NER)	Derived from Ingredients	Approximate; not NER
Frequency f_i	Distinct recipes containing i	Implemented
Recipe size L	unique_ingredient_count	Implemented
Cuisine (26 classes)	Cuisine	Present but 97.7% missing
Cooking techniques	—	Absent

Paper quantity	CocktailDB field	Status
Utensils / steps	—	Absent
Macronutrients per serving	Nutrients text	Partial; serving not documented
Cook time	Prep_Time	Partial; heterogeneous units

4. Data Quality and Preprocessing

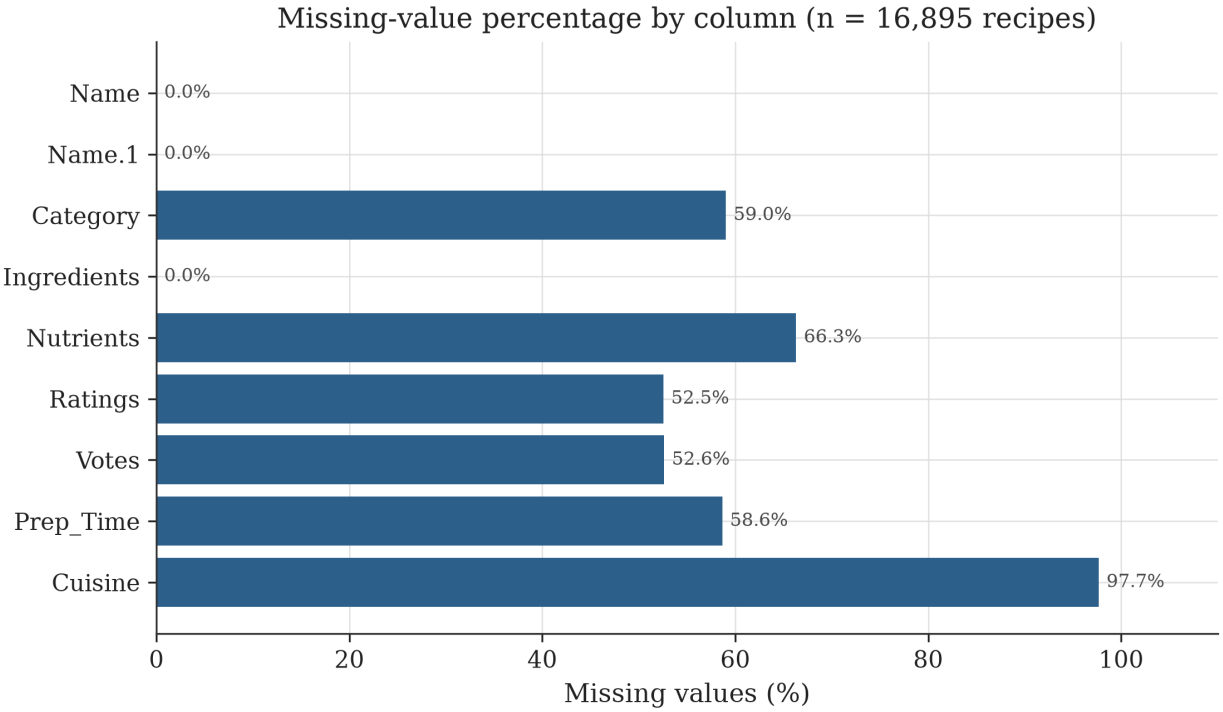
4.1 Audit of duplicates and redundancy

Exact duplicate rows: 17 extras (33 rows involved). These are scrape/storage repeats and were dropped from the frequency corpus. Name+Ingredients duplicates beyond the first: 23. These are the same formula under the same title and were also dropped for frequency counts so that one recipe cannot inflate f_i. Duplicate names overall: 1979 extra rows; 1144 names appear more than once. Of those, 1131 names have different ingredient strings (legitimate variants, retained) and 13 names repeat the same ingredients. Name and Name.1 are identical; Name.1 is excluded from analysis.

4.2 Missingness

Ratings and votes are almost a joint field: both present in 8,005 rows, both missing in 8,878, ratings without votes in 12, votes without ratings in 0. This is systematic missingness of a popularity block, not independent gaps. Cuisine is too sparse for stratified laws (393 non-missing). Category missingness (59.0%) means category plots are not globally representative.

Figure 1. Missing-value percentage by column.



What this figure shows. *Each bar is the percentage of the 16,895 rows with a null in that column. Ingredients and both name fields are complete; Cuisine is almost empty.*

Key insight. *Popularity (ratings/votes), category, prep time, and nutrients are majority-missing. Cuisine missingness is 97.7%.*

Why it matters. *Any comparison that ignores missingness would pretend CocktailDB is a complete design matrix comparable to the paper's curated corpus.*

Statistical evidence. *n = 16,895 rows. Highest missingness: Cuisine 97.7%, Nutrients 66.3%, Category 59.0%, Prep_Time 58.6%, Ratings 52.5%.*

Caveat. *Null encoding is taken as missing; empty strings were not separately treated as a second missing class for these fields.*

Figure 2. Missingness pattern (downsampled rows).



What this figure shows. *Black cells are missing. Rows are shown at a regular stride to make blocks visible. Ingredients stay complete while Cuisine is almost continuously missing.*

Key insight. *Missingness is structured: large contiguous stretches lack category/nutrients/ratings together, consistent with heterogeneous source tables concatenated into one file.*

Why it matters. *Systematic missingness implies that nutrient and rating analyses describe source subsets, not a random sample of all cocktail recipes.*

Statistical evidence. *Displayed stride chosen so that ~900 row-samples represent 16,895 recipes. Joint rating–vote presence n = 8,005.*

Caveat. *Downsampling can hide single-row anomalies; combination counts in the appendix use the full matrix.*

4.3 Ratings, votes, and preparation time

All 116 rows with Rating = 0 also have Votes = 0. They are coded as unrated (rating_analyzed and votes_analyzed missing), not as genuine zero-star quality. The original columns are preserved. Votes such as “1,216” were parsed by removing commas; malformed votes = 0. Prep times were parsed into minutes (seconds/60, hours×60, days×1440; ranges averaged; unitless numbers treated as minutes). On the frequency corpus, prep-time minutes have min 0.00, median 3.00, mean 7.95, IQR 0.00, P90 5.00, P95 10, max 20,170 (n = 6976). Extremes include multi-hour and multi-day values and one 336-hour string; they were retained and flagged rather than deleted. Unparsed prep strings: 0. Zeros: 44.

4.4 Ingredient normalization

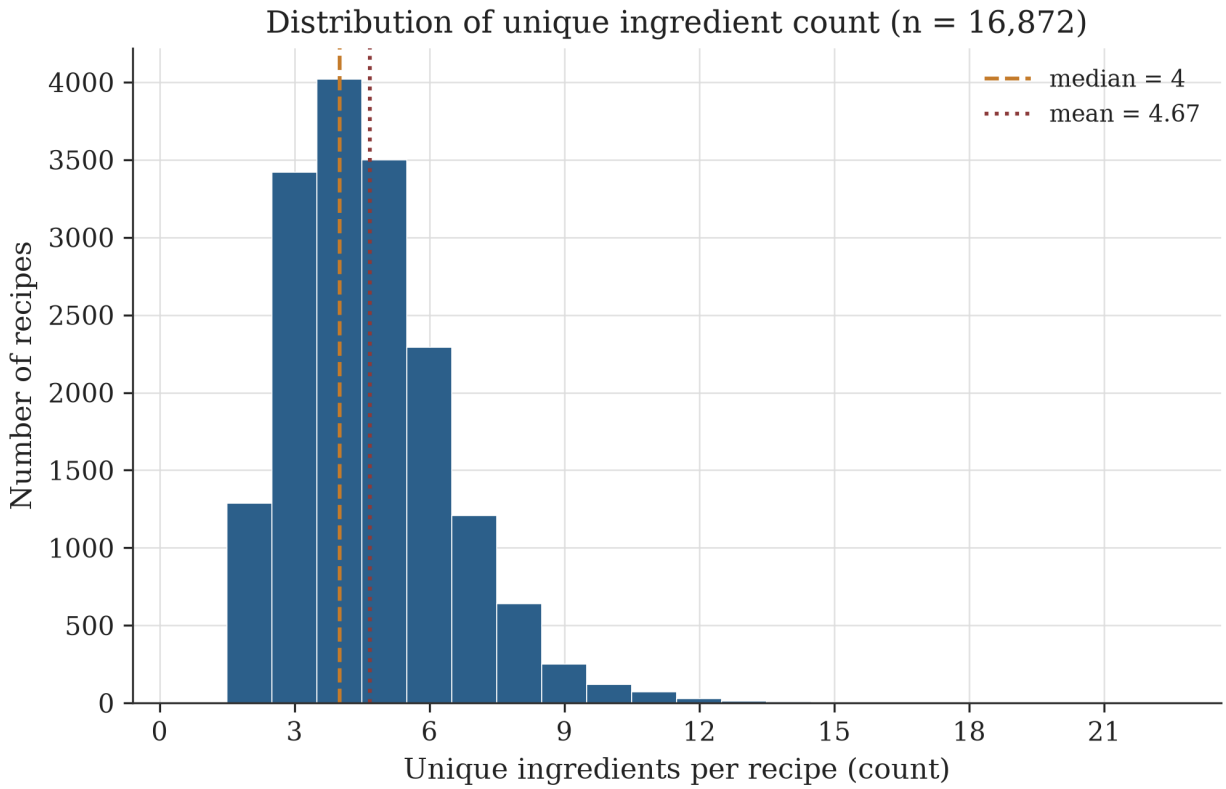
Four levels were computed. Main analyses use **standard** (lowercase, whitespace, strip leading/trailing quantities and units; Cointreau→orange liqueur and Kahlua→coffee liqueur only as obvious aliases). Moderate additionally strips garnish/prep descriptors. Core is sensitivity-only (ice variants; drop “fresh”). Vocabulary at standard level: 14,466 unique strings from 79,411 mentions. Surface unique strings: 31,180. Core unique: 12,480. Lime versus lime juice is not collapsed.

5. Exploratory Data Analysis

5.1 Recipe structure

Observation: cocktail formulas are short. Mean unique ingredients = 4.67, median = 4.00, IQR = 3.00, range 1–22 (n = 16,872). The paper’s global food corpus peaked near 8–12 ingredients with mean about 10; cocktails sit well below that characteristic scale. Interpretation: drink recipes are composed around a small number of components. Limitation: unique_ingredient_count depends on normalization; residual quantity fragments would bias counts upward.

Figure 3. Distribution of unique ingredients per recipe.



What this figure shows. *Histogram of L on the frequency corpus (n = 16,872). A vertical median and mean are overlaid.*

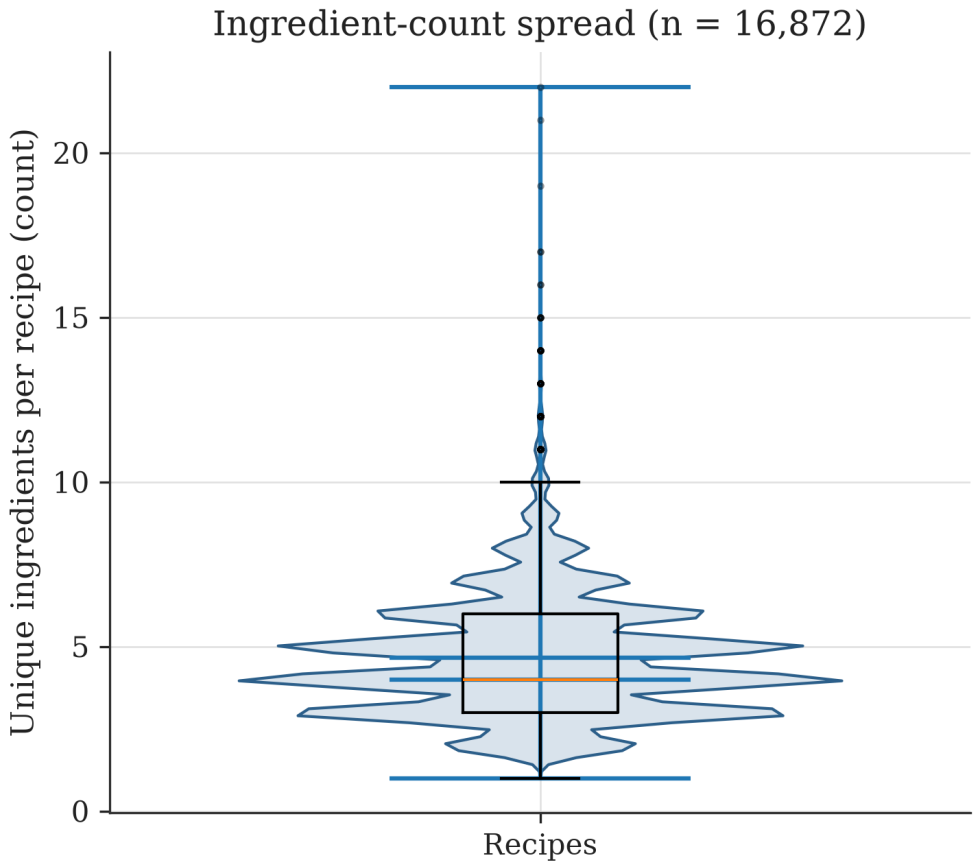
Key insight. *Most recipes use about 4 unique ingredients; the right tail reaches 22.*

Why it matters. *Recipe complexity in this corpus is primarily low-dimensional, which will later constrain Menzerath–Altmann (few large-L bins).*

Statistical evidence. *n = 16,872; mean = 4.67; median = 4.00; SD = 1.80.*

Caveat. *L is unique normalized ingredients, not raw pipe tokens.*

Figure 4. Spread of recipe size.



What this figure shows. *Violin plus boxplot of unique ingredient counts.*

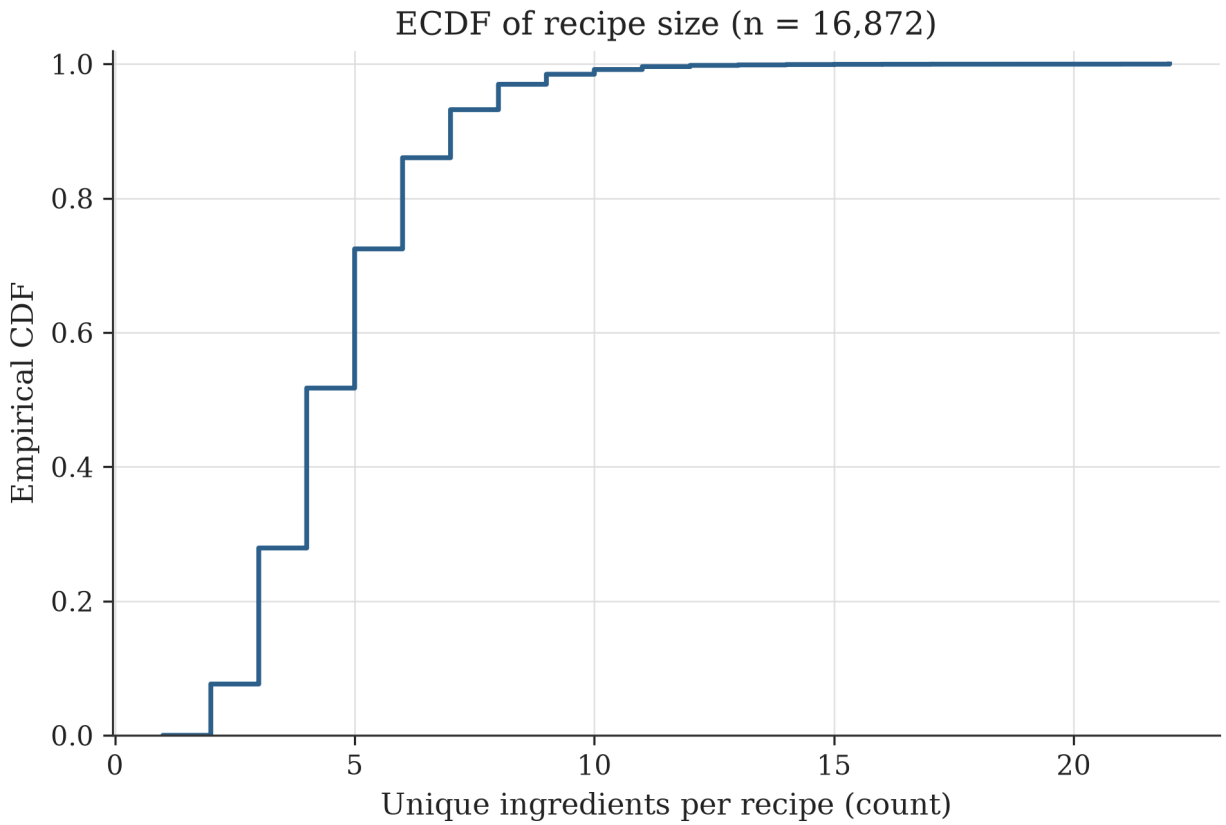
Key insight. *Dispersion is modest (IQR 3.00); outliers exist but do not dominate the mass.*

Why it matters. *Outlier long recipes exist (punches, pitchers, garnished showpieces) but are uncommon.*

Statistical evidence. *n = 16,872; min = 1; max = 22.*

Caveat. *Boxplot whiskers follow the default matplotlib convention; outliers are plotted, not removed.*

Figure 5. ECDF of recipe size.



What this figure shows. *The empirical CDF of L shows how quickly probability mass accumulates at small sizes.*

Key insight. *A large fraction of recipes have L at or below the median of 4.*

Why it matters. *ECDFs avoid binning artifacts in a discrete, skewed count variable.*

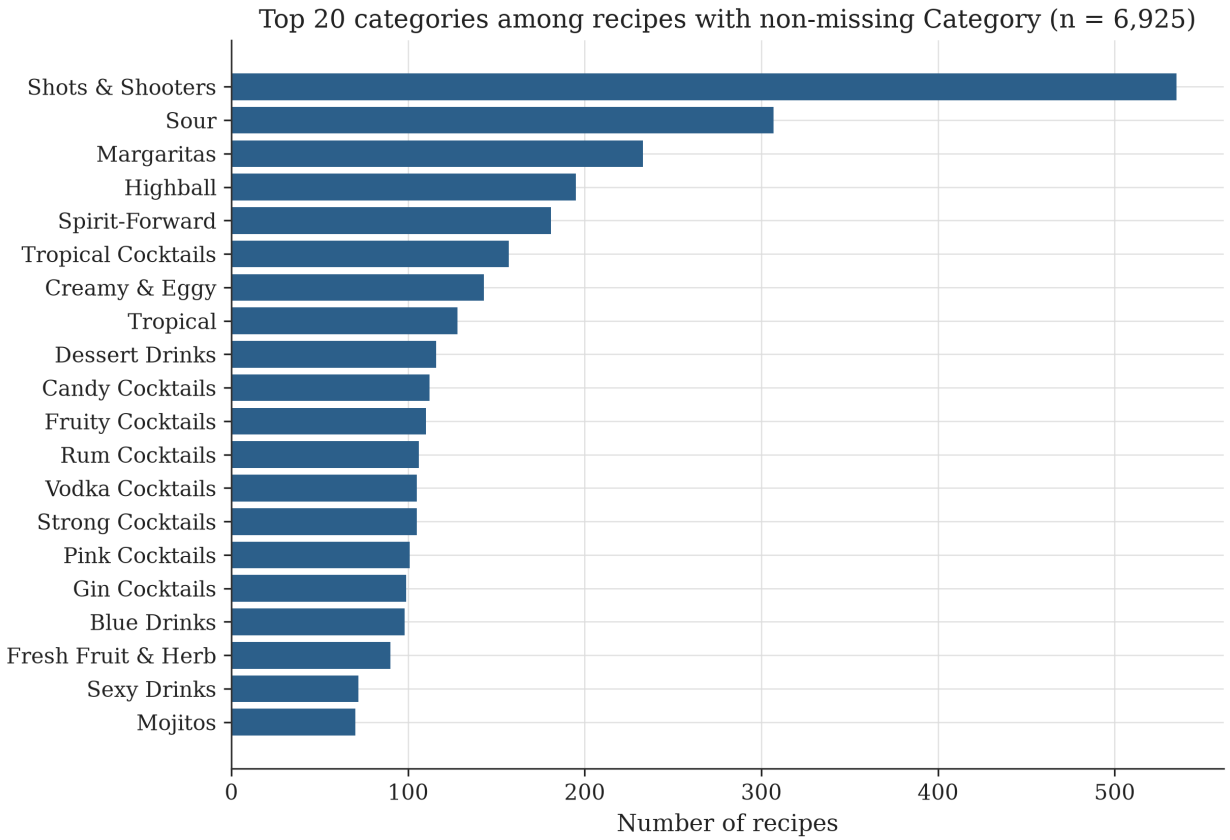
Statistical evidence. *n = 16,872.*

Caveat. *Ties at integer L produce vertical jumps.*

5.2 Categories

Category is observed for 6,925 recipes (41.0% of the frequency corpus), with 429 distinct labels. The top five categories account for 21.0% of labelled recipes; the top 20 account for 44.2%. Labels mix glassware, flavor families, and spirit families (shots, sours, margaritas, highballs). Missing Category is not treated as a category. Conclusions are conditional on the labelled subset.

Figure 6. Top 20 categories by recipe count.



What this figure shows. *Counts among recipes with non-missing Category (n = 6,925).*

Key insight. *The leading labelled class is dominated by a long-tailed remainder: 429 labels overall.*

Why it matters. *Category is a noisy ontology, useful for describing the labelled slice, not for claiming a global taxonomy of cocktails.*

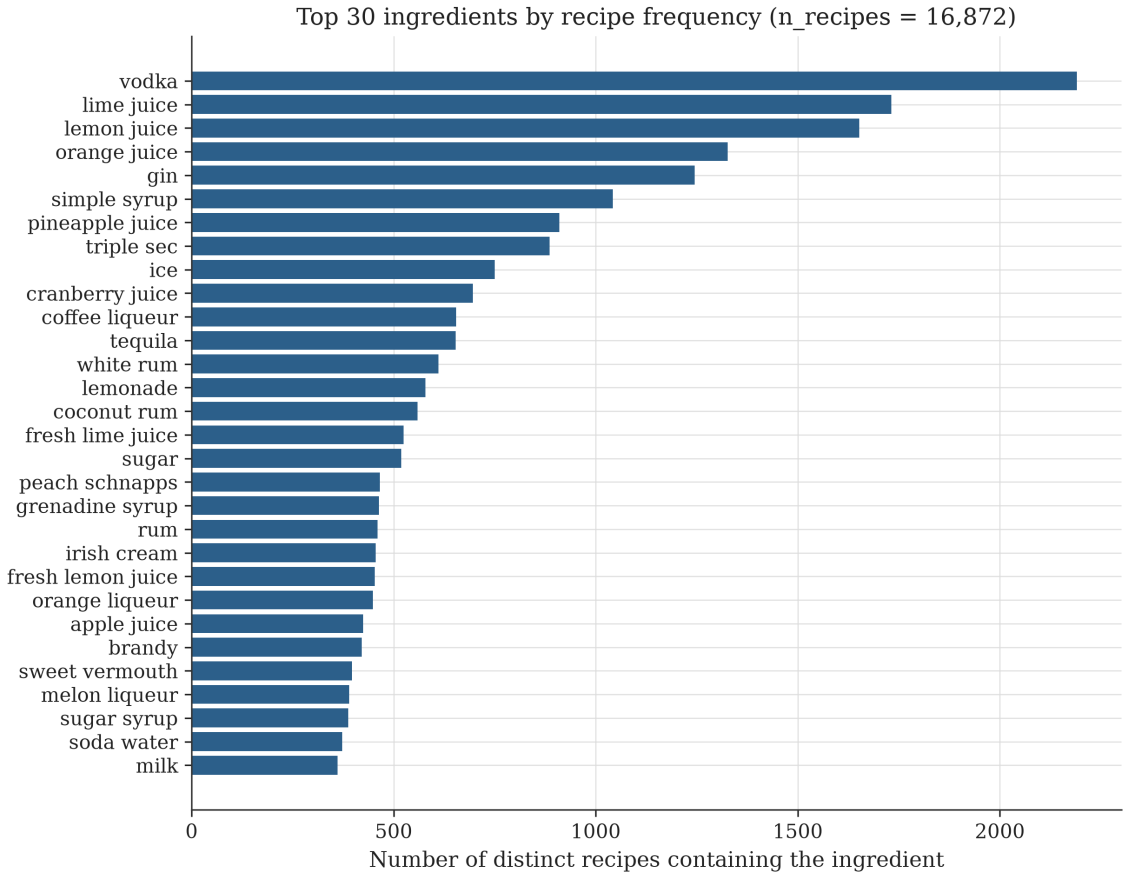
Statistical evidence. *n_labelled = 6,925; coverage = 41.0% of 16,872 frequency-corpus recipes; top-20 share = 44.2%.*

Caveat. *Do not compare categories as if every recipe were labelled.*

5.3 Ingredients

Using standard normalization, 14,466 distinct ingredients appear in 16,872 recipes. “vodka” occurs in 2,191 recipes (13.0%). Hapax ingredients (frequency 1) are 10,626 (73.5% of vocabulary). This is the same conceptual frequency as the paper: f_i = number of distinct recipes containing i, with within-recipe repeats not double-counted.

Figure 13. Top 30 ingredients by recipe frequency.



What this figure shows. *Bars are f_i for the 30 most common standard-normalized ingredients (n_recipes = 16,872).*

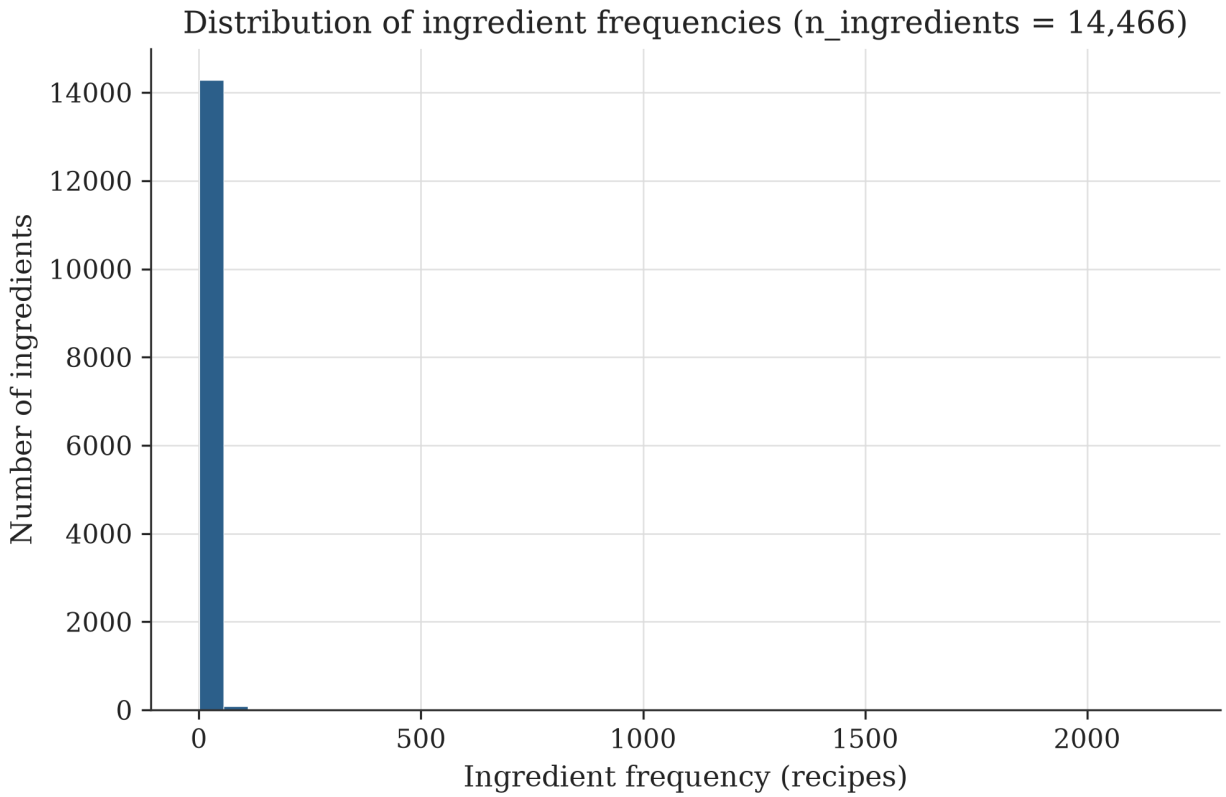
Key insight. *A small set of bases, juices, sweeteners, bitters, and ice-like items dominate. The leader is vodka with f = 2,191.*

Why it matters. *Preferential reuse of a core bar vocabulary is the cocktail analogue of the paper's salt/onion/butter/flour/oil dominance in food.*

Statistical evidence. *n_recipes = 16,872; vocabulary = 14,466; top-10 share of all frequency mass = 15.8%.*

Caveat. *Normalization aliases (e.g., Cointreau as orange liqueur) slightly inflate some totals; lime juice is kept distinct from lime.*

Figure 14. Distribution of ingredient frequencies.



What this figure shows. *Most ingredients are rare; a thin tail of high-frequency items sits far to the right.*

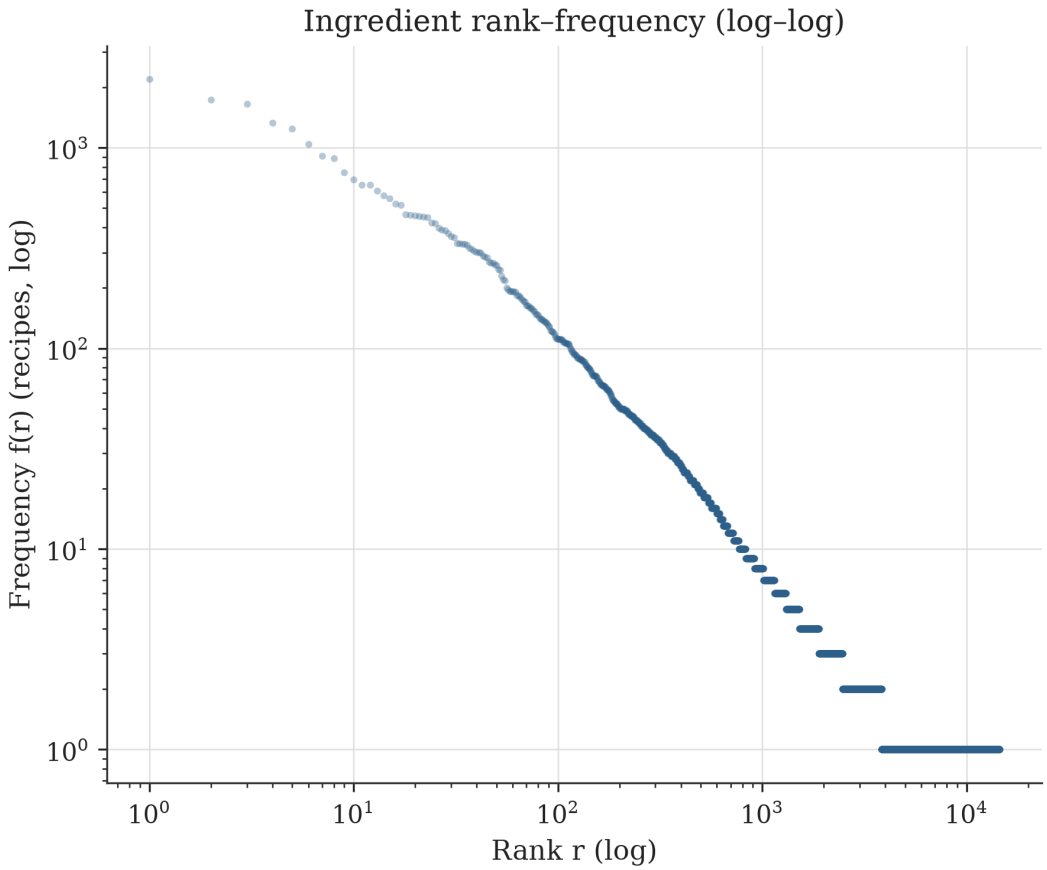
Key insight. *73.5% of ingredients appear once. Mean frequency = 5.45; median = 1.00.*

Why it matters. *This is the size distribution that Zipf and Heaps later summarize: reuse of few items plus a long tail of novelty.*

Statistical evidence. *n_ingredients = 14,466.*

Caveat. *The linear-axis histogram compresses the tail; the rank–frequency plot is the appropriate scaling view.*

Figure 15. Rank–frequency plot on log–log axes.



What this figure shows. *Each point is an ingredient: rank r versus recipe frequency f(r).*

Key insight. *The cloud declines over several decades and is curved rather than a single perfect line, which already warns against an unrestricted power-law claim.*

Why it matters. *Heavy tails mean a bartender's core shelf covers many recipes while rare modifiers create diversity.*

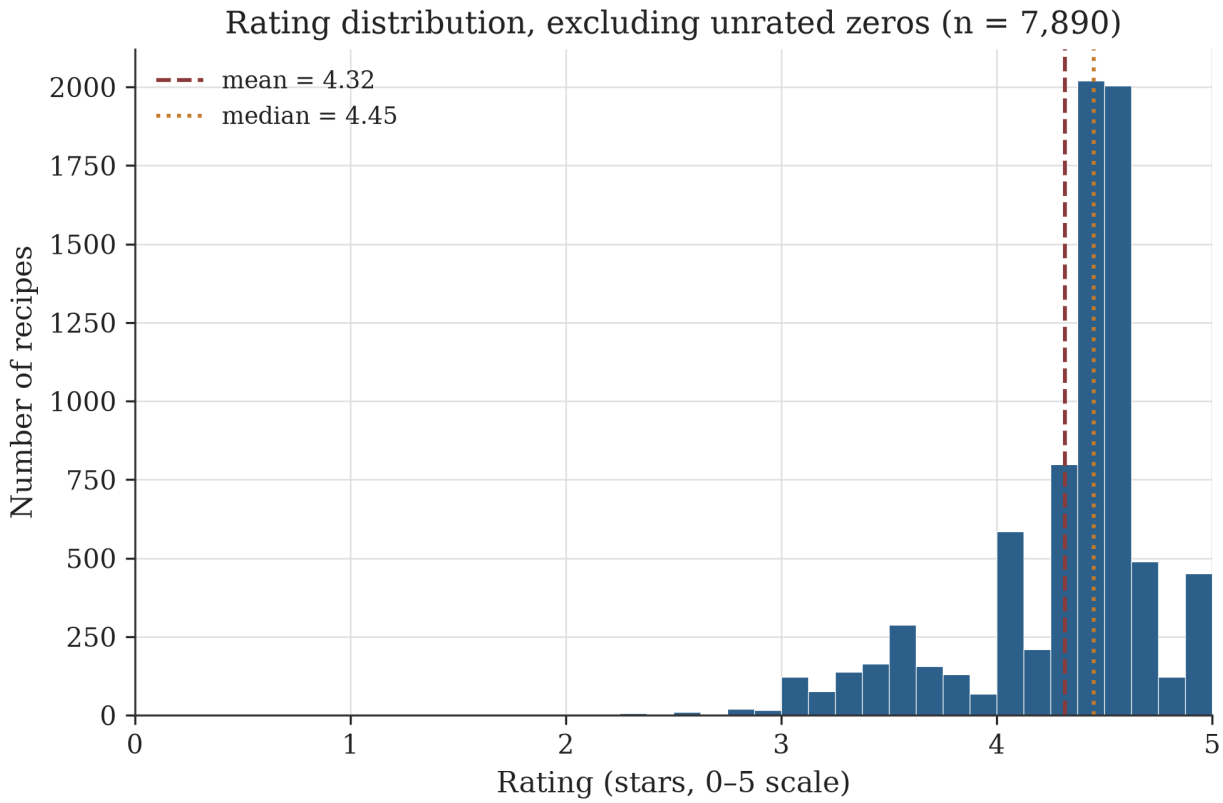
Statistical evidence. *n = 14,466 ingredients; n_recipes = 16,872.*

Caveat. *Visual linearity is not a statistical test; see Figure 16.*

5.4 Ratings and popularity

Ratings (excluding unrated zeros): n = 7,890, mean = 4.317, median = 4.450, SD = 0.442, IQR = 0.320, skewness = -1.441. The support is bounded by 5, so unbounded continuous distribution fitting is not used as a primary model. Votes: n = 7,878, median = 30, mean = 38.4, max = 4,348, skewness = 25.10. Pearson(rating, log1p votes) r = -0.060, p = < 0.0001, n = 7878. Spearman(rating, votes) ρ = -0.058, p = < 0.0001, n = 7878; bootstrap 95% CI [-0.08157368286926583, -0.030059730233818565]. Spearman is primary because votes are heavy-tailed. Low-vote recipes (≤3 votes, n = 540) have more extreme ratings (share of ratings ≤2 or ≥4.9 = 60.4%) than high-vote recipes (≥50, n = 1325, share 0.5%).

Figure 7. Rating distribution.



What this figure shows. *Histogram of analyzed ratings on 0–5 (n = 7,890), excluding Rating=Votes=0.*

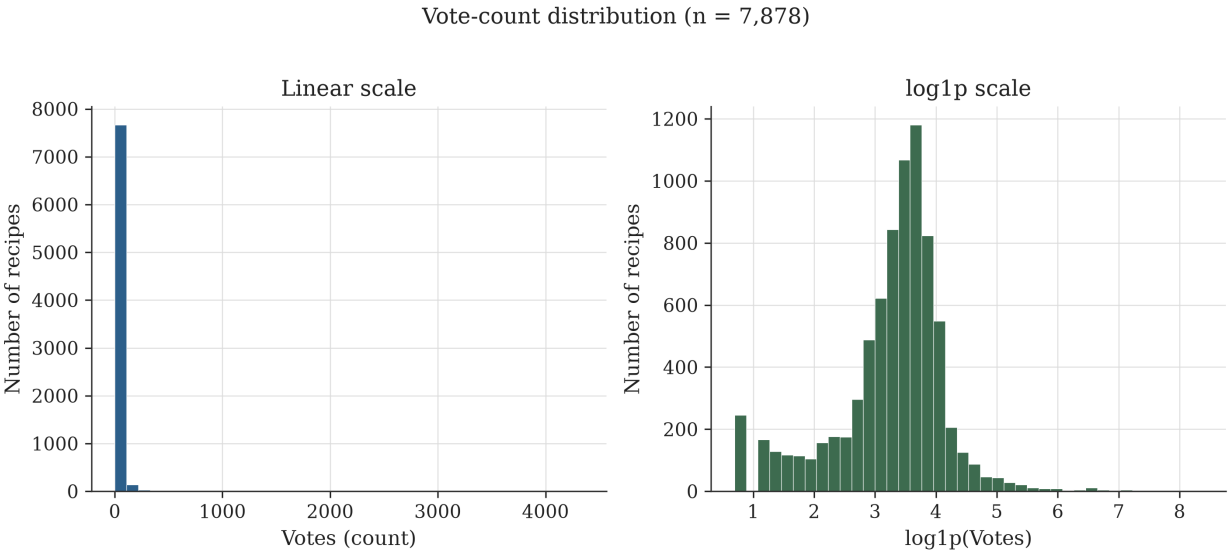
Key insight. *Mass piles near 4–5 stars (median 4.45). The scale is compressed at the top.*

Why it matters. *High means do not imply uniformly excellent recipes; they can reflect selection, interface defaults, and small-n 5-star ratings.*

Statistical evidence. *mean = 4.317; SD = 0.442; IQR = 0.320; skewness = -1.441.*

Caveat. *Bounded discrete/ordinal ratings; unrated zeros excluded by design.*

Figure 8. Votes on linear and log1p scales.



What this figure shows. *Left: raw vote counts. Right: log1p(Votes). n = 7,878.*

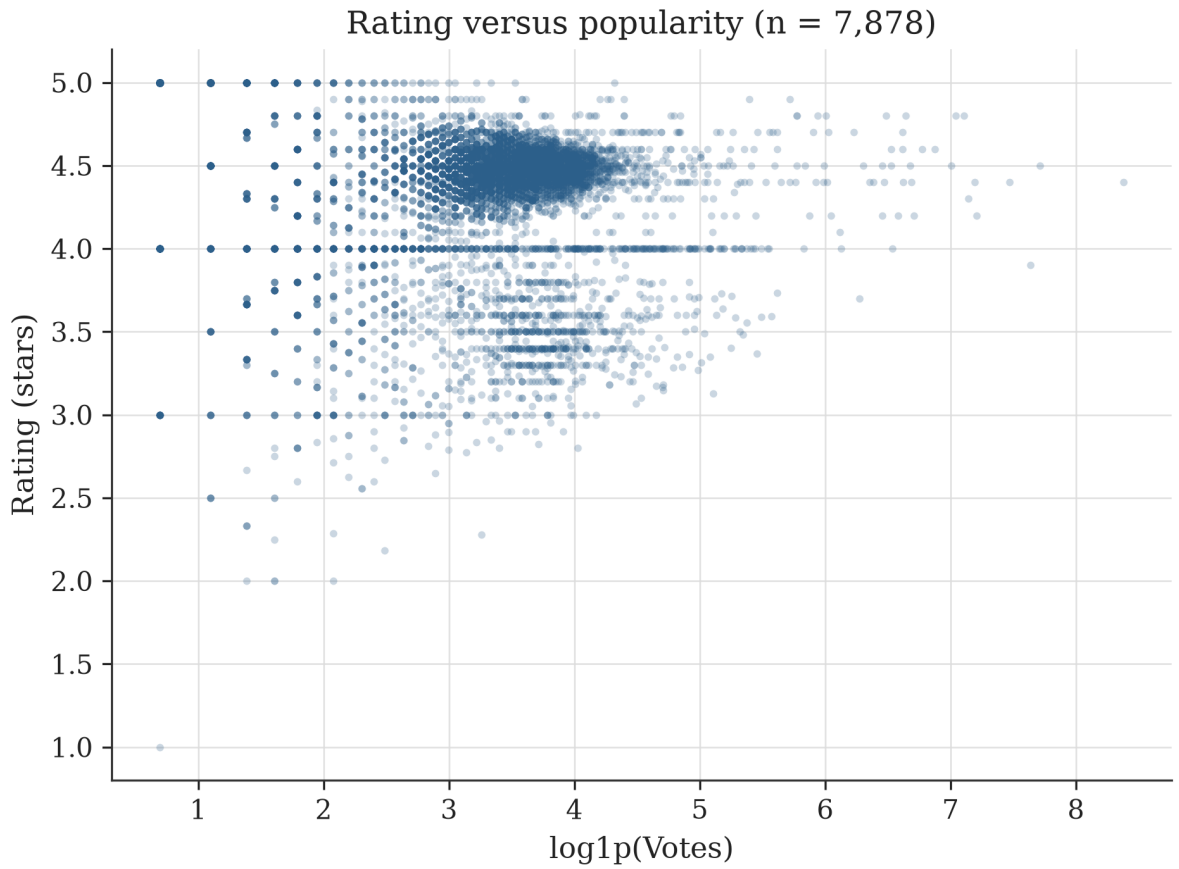
Key insight. *Popularity is long-tailed (mean 38.4 ■ median 30; max 4,348).*

Why it matters. *A few recipes attract most attention; vote counts are a poor proxy for objective quality.*

Statistical evidence. *skewness = 25.10; P90 = 58.*

Caveat. *Votes exist only for the popularity-observed subset.*

Figure 9. Rating versus log1p(Votes).



What this figure shows. *Each point is a recipe with both an analyzed rating and vote count.*

Key insight. *Association is positive but modest (Spearman ρ = -0.058). Low-n ratings fan out to 1 and 5.*

Why it matters. *Highly voted recipes are not guaranteed higher ratings; they do tend to occupy a narrower high-rating band, consistent with averaging.*

Statistical evidence. *Spearman ρ = -0.058, p = < 0.0001, n = 7878; Pearson(r, log1p votes) = -0.060.*

Caveat. *No causal claim: better-known recipes may be better, or only more rated.*

Vote-adjusted ratings

Following the extended recipes EDA notebook, a Bayesian-style shrinkage estimator is adj = (V·R + m·C) / (V + m) with m = 10 and C = global mean analyzed rating = 4.316 (n = 7,878 recipes with both rating and votes after unrated-zero exclusion). This does not create new information; it down-weights 1-vote 5-star recipes relative to well-voted ones.

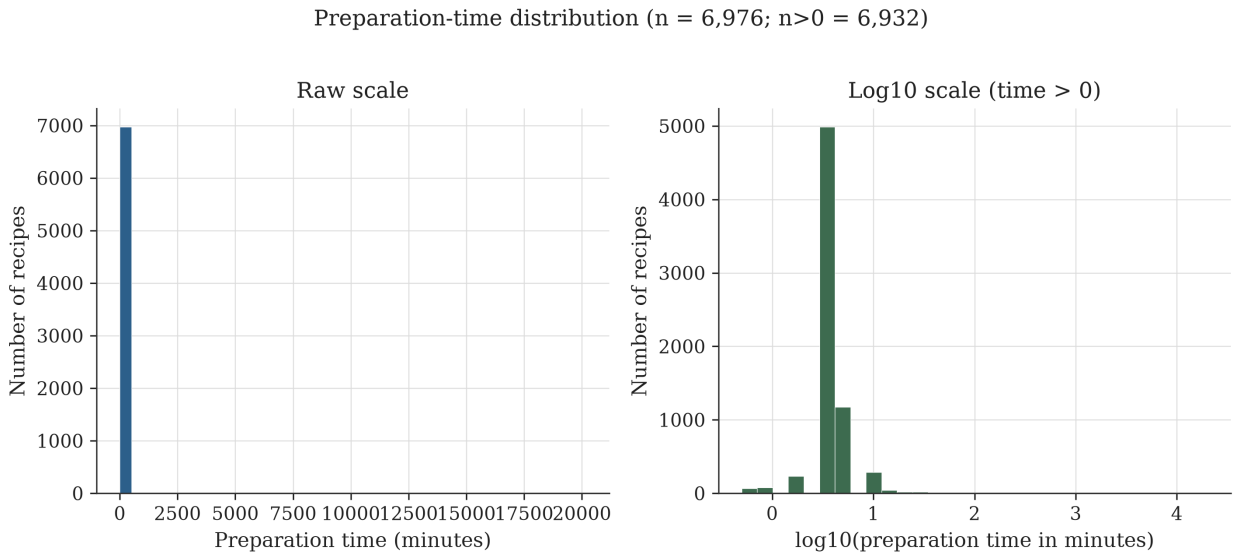
Name	Rating	Votes	Adj. rating
El Diablo Cocktail	5.00	74	4.919
Simple Syrup	4.90	304	4.881
Grandma's Apple Pie 'Ala Mode' Moonshine	4.90	219	4.875
Skeleton Key Cocktail	5.00	33	4.841
Whiskey Slush	4.90	80	4.835
Easy Frozen Margaritas	4.90	72	4.829
Sex on the Beach	4.90	70	4.827
Irish Old Fashioned	4.90	66	4.823
Vruca Rakija Recipe	5.00	27	4.815
Easy Mojitos	5.00	24	4.799

Table. Top 10 recipes by vote-adjusted rating (m = 10).

5.5 Preparation time

Parsed prep time (minutes) on the frequency corpus where available: n = 6,976, median = 3.00, mean = 7.95, max = 20,170. Spearman(prep time, ingredient count) ρ = 0.352, p = < 0.0001, n = 6976. Spearman(prep time, votes) ρ = -0.290, p = < 0.0001, n = 6082. Neither association is interpreted as a production function or a popularity mechanism.

Figure 10. Preparation-time distribution.



What this figure shows. *Raw minutes (left) and log10 of positive times (right). n = 6,976.*

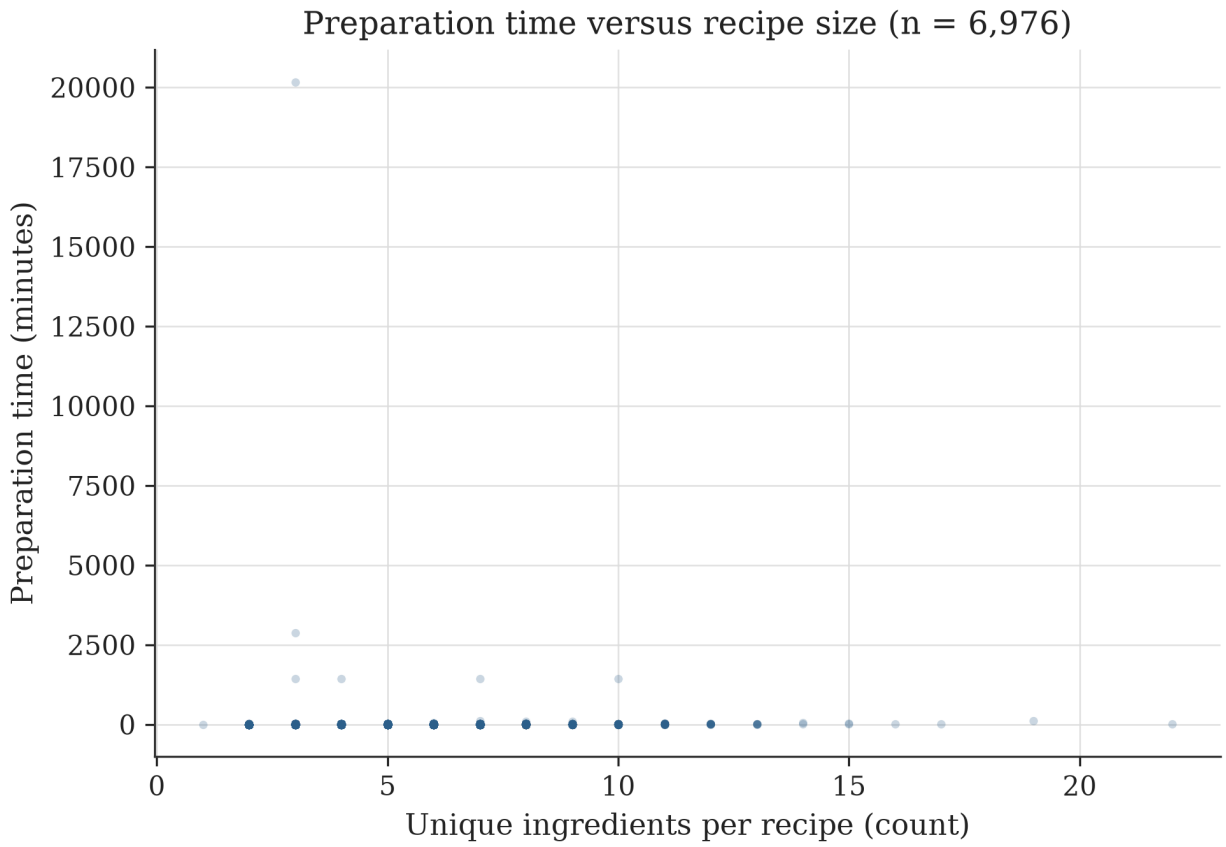
Key insight. *Typical times are a few minutes (median 3.0), with a long right tail including hours/days.*

Why it matters. *Most cocktails are quick to assemble; extreme values may mix infusion/rest times with active prep or parsing artifacts (e.g., duplicated phrases, 336 hours).*

Statistical evidence. *mean = 7.95; P95 = 10; max = 20,170.*

Caveat. *Unitless numbers were assumed to be minutes; that assumption is documented, not verified by the source.*

Figure 11. Preparation time versus ingredient count.



What this figure shows. *Scatter of minutes against unique ingredient count.*

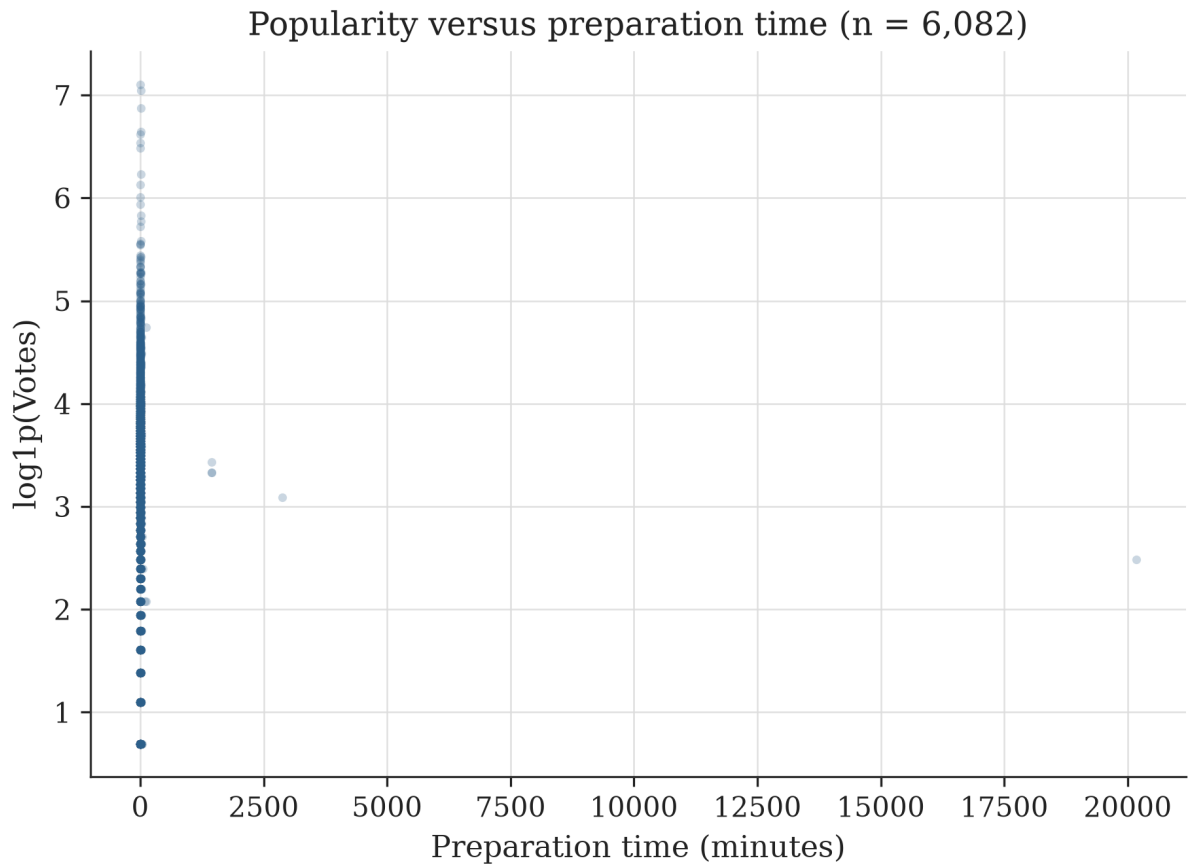
Key insight. *The rank correlation is 0.352 (n = 6976).*

Why it matters. *A weak relationship would mean longer drinks are not simply those with more ingredients (batching, chilling, or infusion can dominate).*

Statistical evidence. *Spearman ρ = 0.352, p = < 0.0001; 95% bootstrap CI [0.3324295891922727, 0.37284437718635].*

Caveat. *Outliers in time can dominate Pearson correlation; Spearman is used.*

Figure 12. Popularity versus preparation time.



What this figure shows. *log1p(Votes) against parsed minutes.*

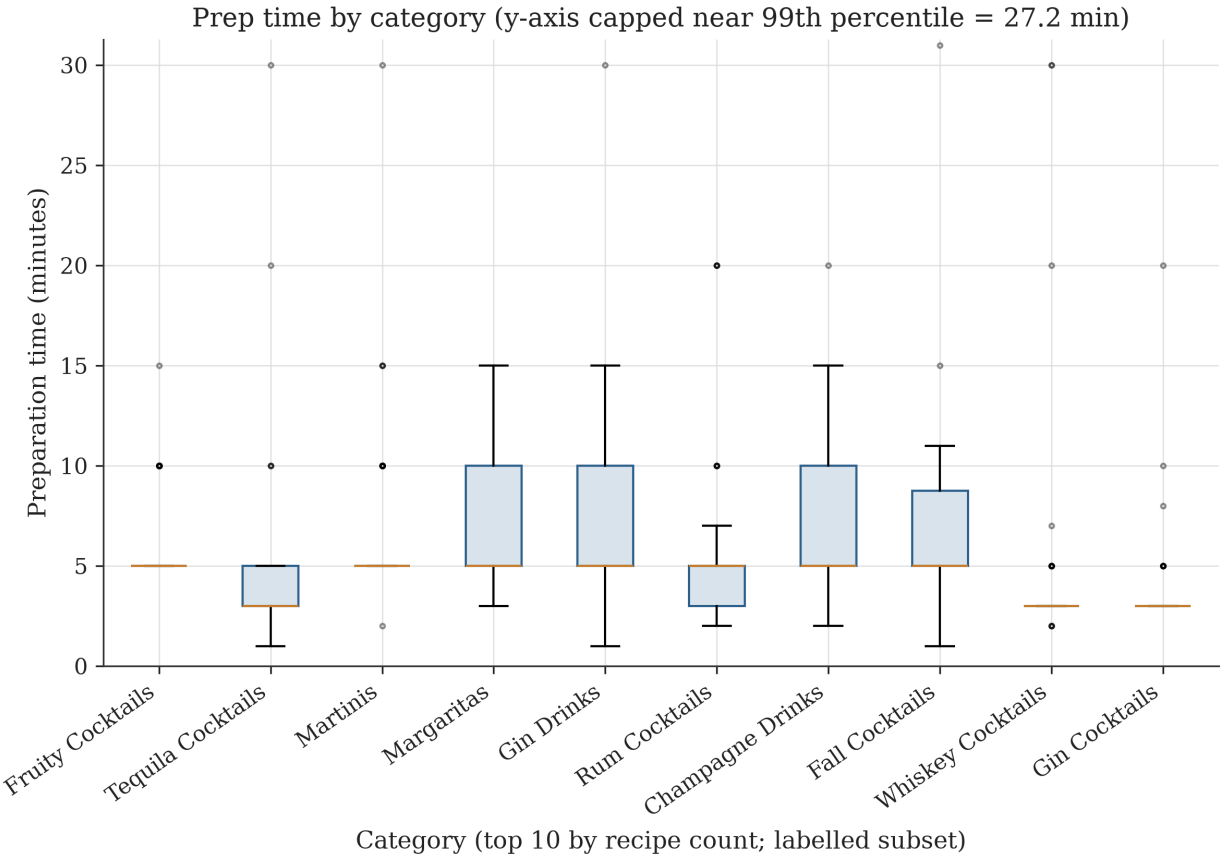
Key insight. *Spearman ρ = -0.290 (n = 6082).*

Why it matters. *Even a statistically nonzero ρ would not show that faster drinks become popular; source confounding is likely.*

Statistical evidence. *ρ = -0.290, p = < 0.0001; CI [-0.3173572925706958, -0.26082386797614776].*

Caveat. *Complete cases only; prep and votes are both missing for many rows.*

Figure 31. Preparation time by top 10 categories (boxplot).



What this figure shows. *Boxplots of parsed minutes in the ten most frequent non-missing categories (n = 629 recipes with both category and prep time). The vertical axis is clipped near the 99th percentile (27.2 min) so a few extreme infusions do not compress the boxes.*

Key insight. *Medians remain short in every leading category (typical medians Fruity Cocktails: 5.0 min, Tequila Cocktails: 3.0 min, Martinis: 5.0 min, Margaritas: 5.0 min). Outliers exist in several classes.*

Why it matters. *Category-level prep time is the comparison the extended EDA notebook emphasized and that this report previously omitted. Categories are the ten most frequent among recipes that have both Category and parsed prep time (not the global top-10 labels, because category and prep time rarely co-occur).*

Statistical evidence. *Top categories = Fruity Cocktails, Tequila Cocktails, Martinis, Margaritas, Gin Drinks, Rum Cocktails, Champagne Drinks, Fall Cocktails, Whiskey Cocktails, Gin Cocktails. Per-category n = {'Fruity Cocktails': 108, 'Tequila Cocktails': 67, 'Martinis': 65, 'Margaritas': 61, 'Gin Drinks': 57, 'Rum Cocktails': 56, 'Champagne Drinks': 55, 'Fall Cocktails': 54, 'Whiskey Cocktails': 53, 'Gin Cocktails': 53}.*

Caveat. *Do not treat these ten bars as a global ranking of cocktail types; Category coverage is incomplete and the y-cap hides the most extreme points.*

5.6 Nutritional characteristics

Nutrient strings are present for 5,694 / 16,895 recipes (66.3% missing). Two formats occur: compact “184 Calories|3g Fat|8g Carbs” and colon-style “Calories: 239 kcal|Carbohydrates: 14 g|…”. Carbs and Carbohydrates were mapped to one carbohydrates variable after checking that both are gram masses in the observed text. The file does not state per-serving standardization. Completeness and moments for each extracted nutrient are in §6.4 and the appendix tables. PCA, if performed, uses complete cases only.

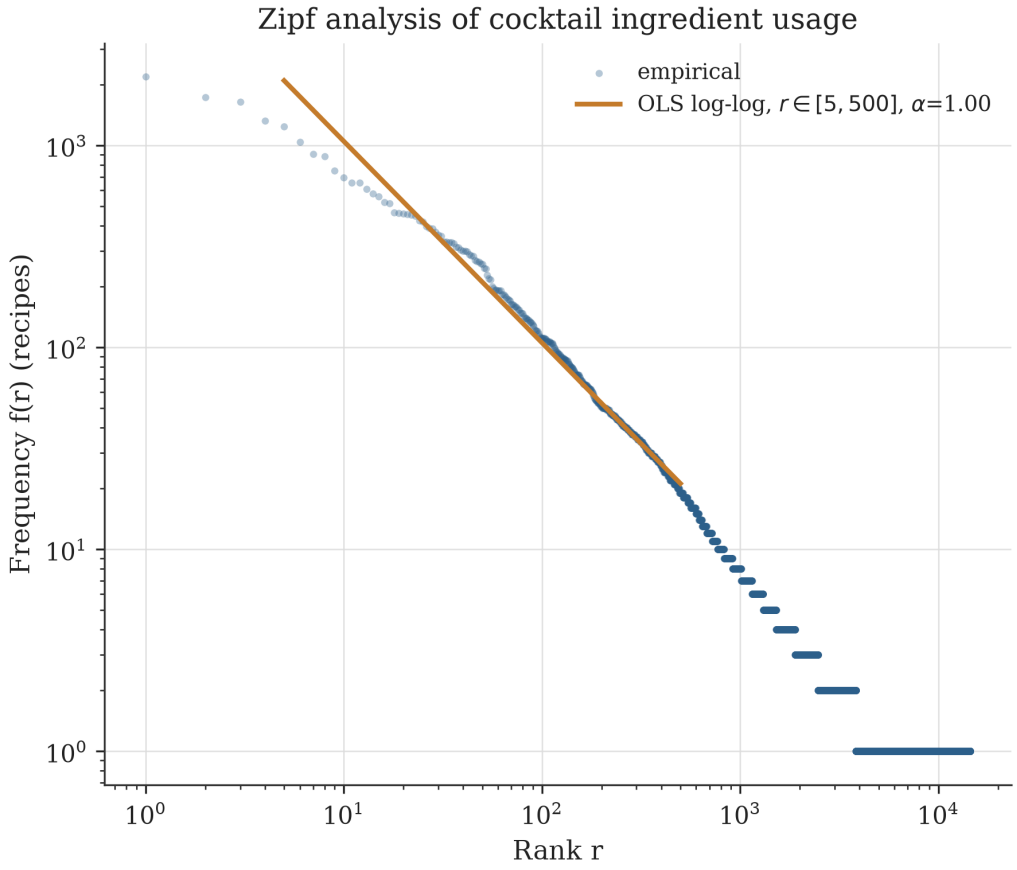
6. Statistical Laws in Cocktail Recipes

6.1 Zipf analysis

Model: f(r) ~ r^{-α} on ingredient ranks. Primary descriptive fit is log–log OLS on r ∈ [5, 500], giving α = 0.998, R^2 = 0.991, n_points = 496. The unrestricted full-range OLS α = 0.878 with R^2 = 0.884, which is a reminder that curvature at the head and far tail moves the exponent. Following Clauset et al. as in the paper’s Methods, a discrete power law was also fitted to the vector of frequencies (P(f) ~ f^{-α_freq}), which is not identical to the rank–frequency exponent. MLE α_freq = 1.848, x_min = 2.00, KS = 0.356, n_tail = 3840. Likelihood-ratio comparisons versus alternatives: lognormal: R=-3.99, p=6.61e-05; exponential: R=14.66, p=1.22e-48. Positive R favors the power law in the Alstott implementation; p-values near 1 do not automatically validate a power law over the whole range. A heavy tail in cocktails means a small core bar (spirits, citrus,

sugar, bitters, ice) is reused constantly, while a long tail of liqueurs, garnishes, and branded products supplies diversity. The fit is reported as Zipf-like, not as a proven unique generative law.

Figure 16. Empirical rank–frequency distribution and fitted Zipf relation.



What this figure shows. *Points: all ingredients. Line: OLS log–log fit on the primary rank window.*

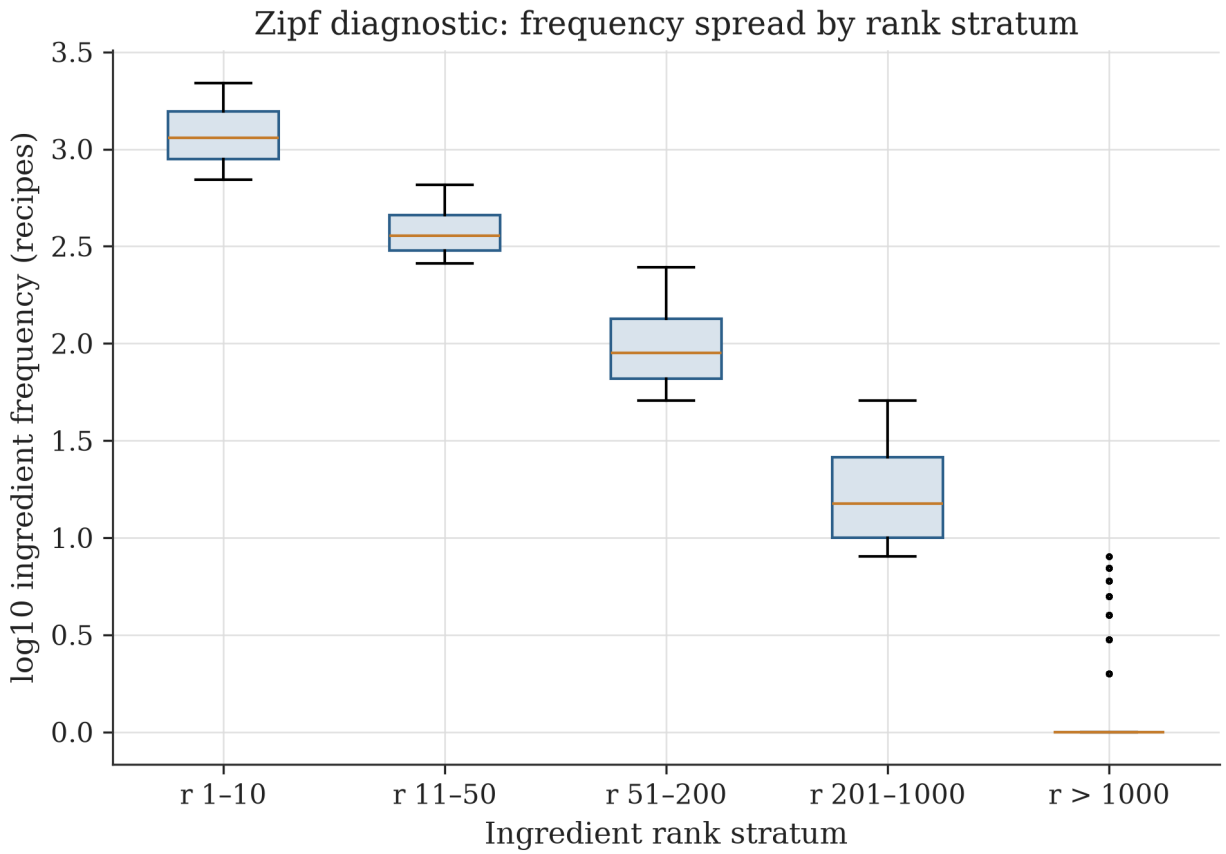
Key insight. *$\alpha \approx 1.00$ on $r \in [5, 500]$. The empirical curve is heavy-tailed but not a perfect straight line.*

Why it matters. *If a few ingredients dominate, cocktail "vocabulary" is reusable and predictable; the tail still allows innovation.*

Statistical evidence. *OLS $\alpha = 0.998$, $R^2 = 0.991$, $n = 496$. Clauset $\alpha_{freq} = 1.848$, xmin = 2.00.*

Caveat. *R^2 is descriptive. Range sensitivity is tabulated in zipf_range_sensitivity.csv. Do not equate α with the paper's 1.53 without noting corpus differences.*

Figure 32. Zipf diagnostic: log-frequency boxplots by rank stratum.



What this figure shows. *Each box is the distribution of log10 recipe-frequency for ingredients in a rank window (1–10, 11–50, 51–200, 201–1000, >1000).*

Key insight. *Median log-frequency falls across strata (r 1–10: 3.06, r 11–50: 2.55, r 51–200: 1.95, r 201–1000: 1.18, r > 1000: 0.00), which is the boxplot view of the same heavy tail shown on the log–log rank plot.*

Why it matters. *This is a compact way to see that a handful of head ingredients sit far above a long body of rare items, without claiming a unique power-law exponent.*

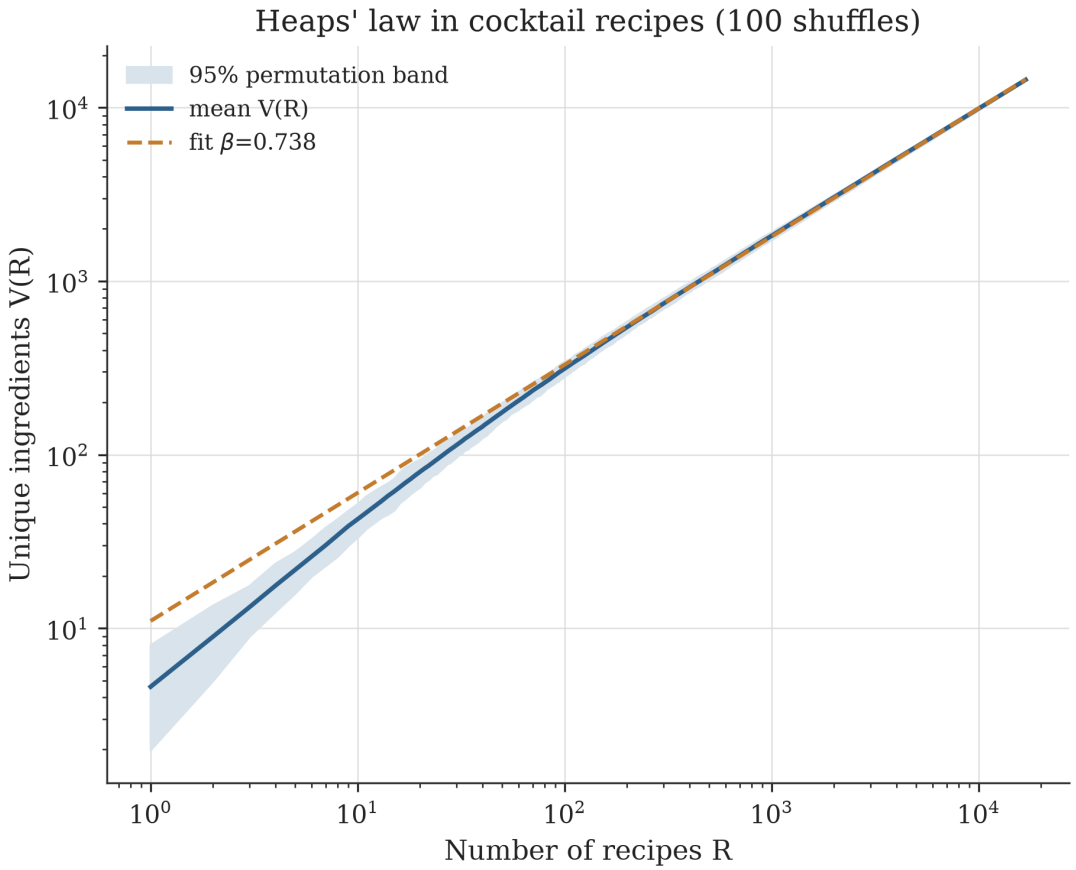
Statistical evidence. *Stratum sizes n = {'r 1–10': 10, 'r 11–50': 40, 'r 51–200': 150, 'r 201–1000': 800, 'r > 1000': 13466}.*

Caveat. *Rank bins are analyst choices. They illustrate Zipf-like decay; they are not a substitute for the MLE/OLS fits in Figure 16.*

6.2 Heaps analysis

V(R) is the number of distinct standard-normalized ingredients after R recipes. 100 random permutations were used. Mean β = 0.7385 (SD 0.0066; 95% permutation interval [0.7254, 0.7515]); mean log–log R^2 = 0.9996. $\beta < 1$ is sublinear growth: new recipes keep introducing ingredients, but the marginal discovery rate declines. The paper's global $\beta \approx 0.56$ is a reference point, not a target. Cocktail $\beta \approx 0.74$ can differ because of domain (drinks vs meals), corpus size (16,872 vs 118,083), normalization, and source mix. Sensitivity: moderate β = 0.720; core β = 0.719.

Figure 17. Heaps curve with permutation band.



What this figure shows. *Mean V(R) and 95% band across 100 shuffles, log–log axes, with a power-law overlay.*

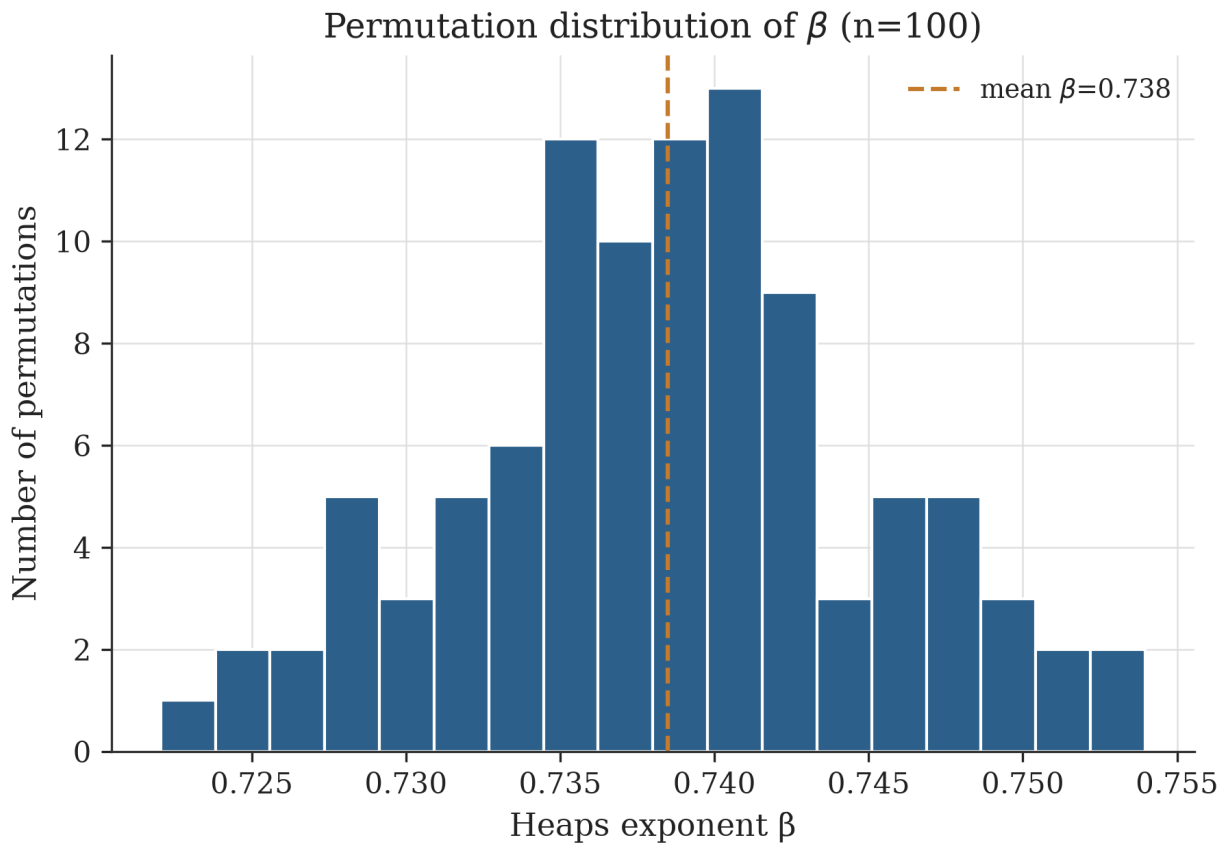
Key insight. *Growth is sublinear (β ≈ 0.738); the band is tight, so ordering noise is small relative to the trend.*

Why it matters. *Cocktail innovation continues (the curve still rises at the corpus end) but with diminishing new-ingredient returns.*

Statistical evidence. *β_mean = 0.7385; 95% interval [0.7254, 0.7515]; final mean V = 14,466; n_recipes = 16,872.*

Caveat. *Fits use $R \geq 20$ to reduce the discrete start-up; K is a nuisance scale parameter.*

Figure 18. Distribution of β across permutations.



What this figure shows. *Histogram of log–log OLS exponents, one per shuffle.*

Key insight. *β estimates concentrate tightly around 0.738 (SD 0.0066).*

Why it matters. *The Heaps exponent is a stable corpus property, not an artifact of one recipe order.*

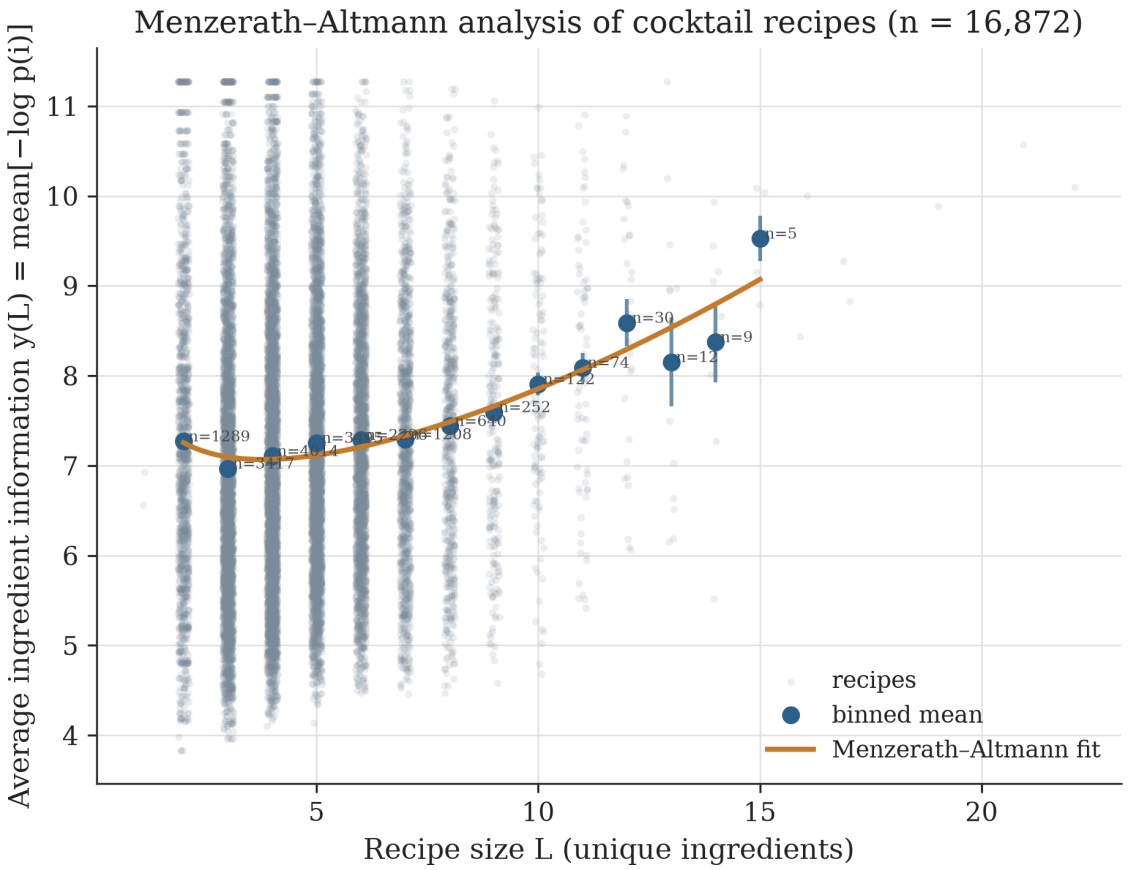
Statistical evidence. *n_shuffle = 100; mean R^2 = 0.9996.*

Caveat. *Least-squares on log V vs log R is the paper's estimator; it is not maximum likelihood.*

6.3 Menzerath–Altmann analysis

Following the paper, p(i) = f_i / Σ f_j, s_i = –log p(i), L = unique ingredients, y = mean s_i in the recipe. On standard normalization, n = 16,872, Spearman(L, y) = 0.101. Binned MA fit y(L)=a L^b exp(cL): a = 7.446, b = -0.157, c = 0.042, R^2 = 0.898, n_bins = 14. The paper's global fit (a=13.16, b=–0.26, c=–0.019, R^2=0.99, minimum near L≈13) lives on a length range cocktails barely populate. If y(L) is flat, increasing, or poorly fitted, that is a result: short drink recipes may not exhibit the same complexity–length trade-off as 10-ingredient meals. Sensitivity (Spearman): moderate 0.066; core 0.051; p(i)=f_i/N_recipes 0.101.

Figure 19. Average ingredient information versus recipe size.



What this figure shows. *Grey points: recipes. Blue: binned means with SE bars and bin counts. Orange: MA functional fit on bins with n≥5.*

Key insight. *Spearman ρ(L, y) = 0.101. The MA R^2 on bins is 0.898.*

Why it matters. *A decreasing y(L) would mean longer drinks pad with common ingredients; an increasing or flat curve would differ from the paper's efficiency–expressivity story.*

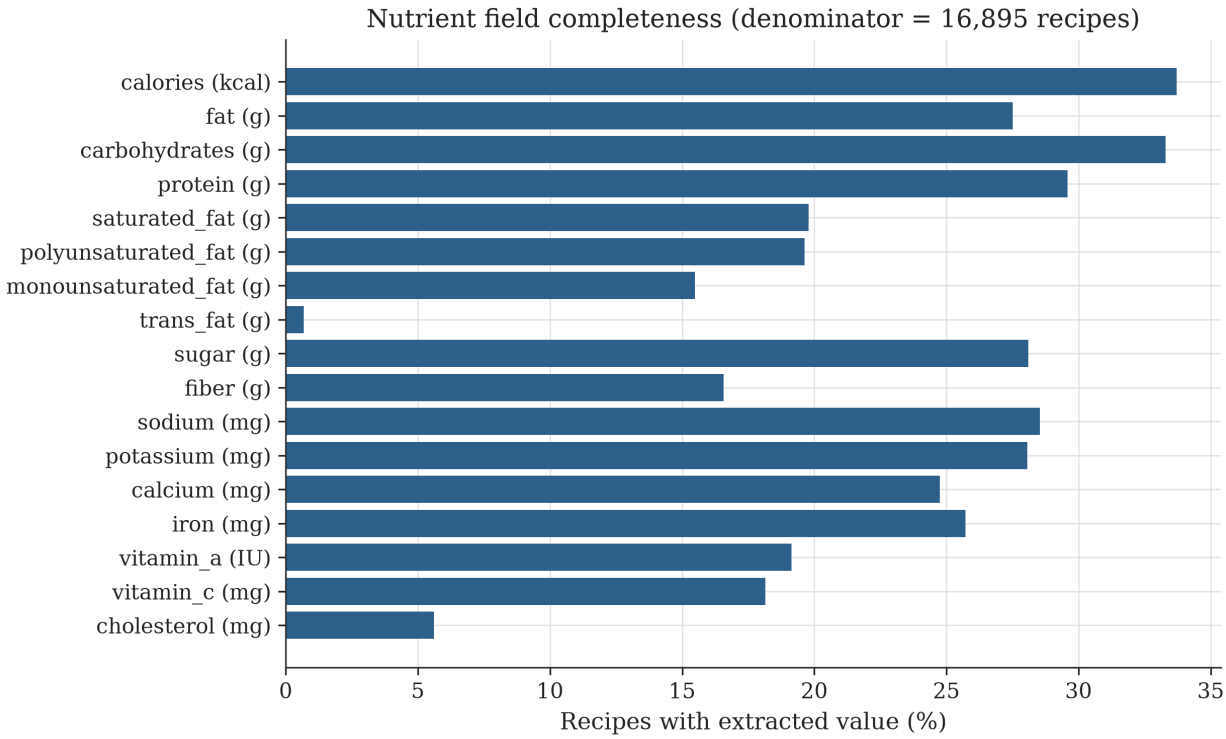
Statistical evidence. *n_recipes = 16,872; L range 1–22 (median 4.0). a,b,c = 7.45, -0.16, 0.042; R^2 = 0.898.*

Caveat. *Few recipes have L near 13, so a turning point at the paper's scale cannot be identified even if the functional form is forced.*

6.4 Nutrient distribution analysis

Positive-valued extracted nutrients were compared to log-normal, gamma, Weibull, Gaussian, and exponential laws via KS distance (smaller is better). Calories: n+ = 5692, log-skewness = -1.084, best = lognorm. Carbohydrates: n+ = 5614, log-skewness = -1.551, best = gamma. The paper reported mean KS 0.0727 for log-normal on meal macronutrients. Cocktail nutrients are sparser, possibly not per serving, and include zeros (especially protein/fat) that are excluded from log tests. Log-normality is supported only where KS and log-skewness jointly agree—not because a histogram looks bell-shaped.

Figure 22. Nutrient completeness.



What this figure shows. *Percent of all 16,895 recipes with a parsed value for each nutrient.*

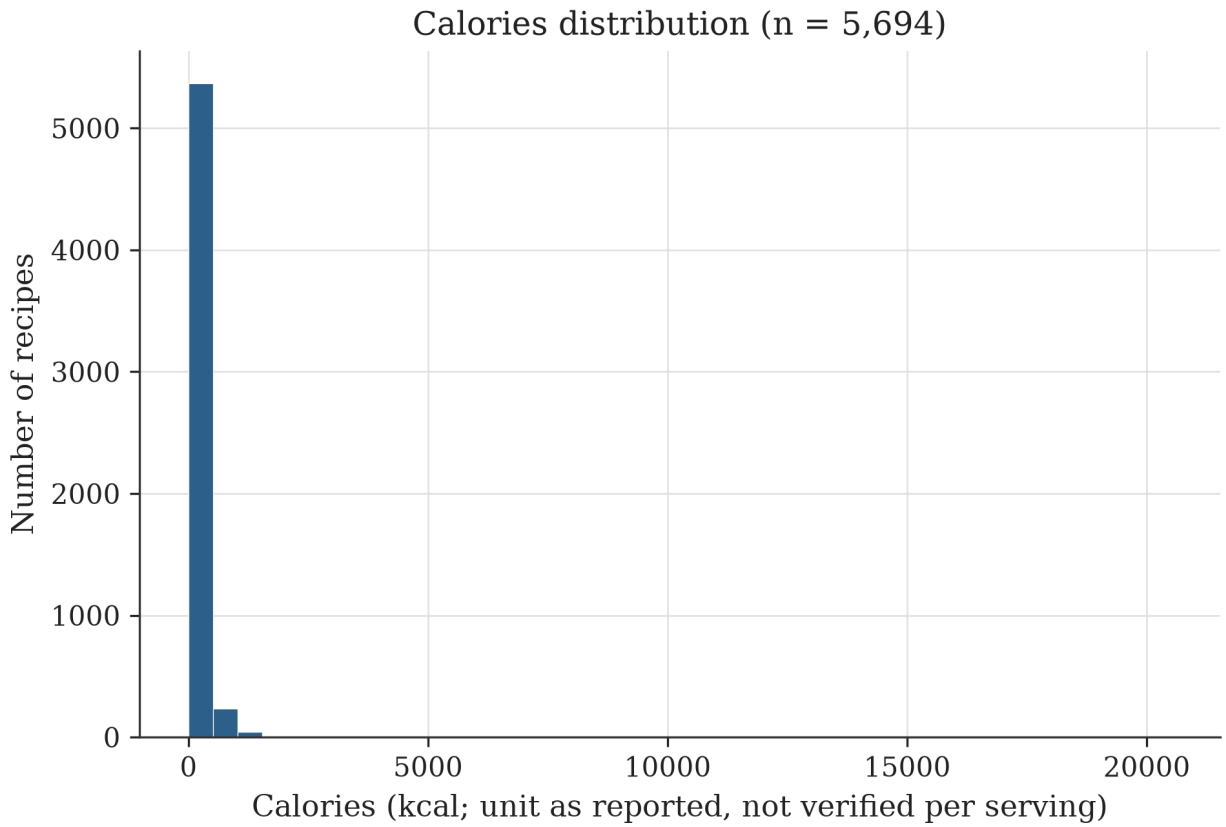
Key insight. *Even calories, the most common field, cover only a minority of the full table; micronutrients are rarer.*

Why it matters. *Nutrient EDA is a study of the nutrient-reporting subset, which may be biased toward recipe sites that compute nutrition panels.*

Statistical evidence. *Nutrient strings present: 5,694 (33.7% of rows have a non-null Nutrients field).*

Caveat. *A non-null string can still omit individual nutrients.*

Figure 23. Calories distribution.



What this figure shows. *Histogram of extracted calorie values as reported.*

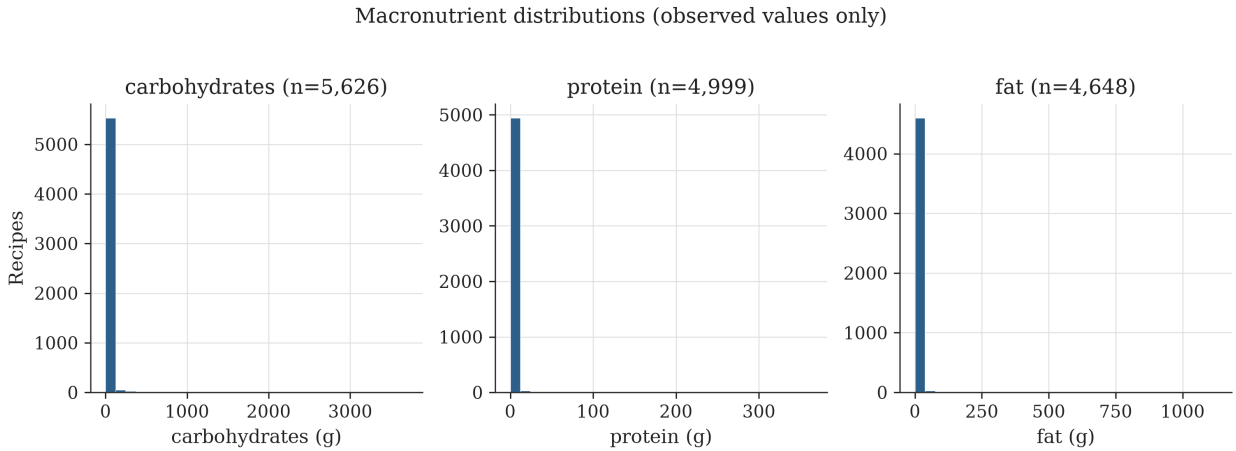
Key insight. *The distribution is right-skewed with extreme high values that may reflect pitchers, multiple servings, or parsing/source errors—not typical single-glass cocktails.*

Why it matters. *Calories cannot be treated as a standardized per-drink nutrient without serving metadata.*

Statistical evidence. *See nutrient_descriptives.csv for n, mean, median, SD, IQR, min, max.*

Caveat. *Zeros and missing are excluded from this histogram if non-finite.*

Figure 24. Macronutrient distributions.



What this figure shows. *Side-by-side histograms of carbohydrates, protein, and fat in grams where extracted.*

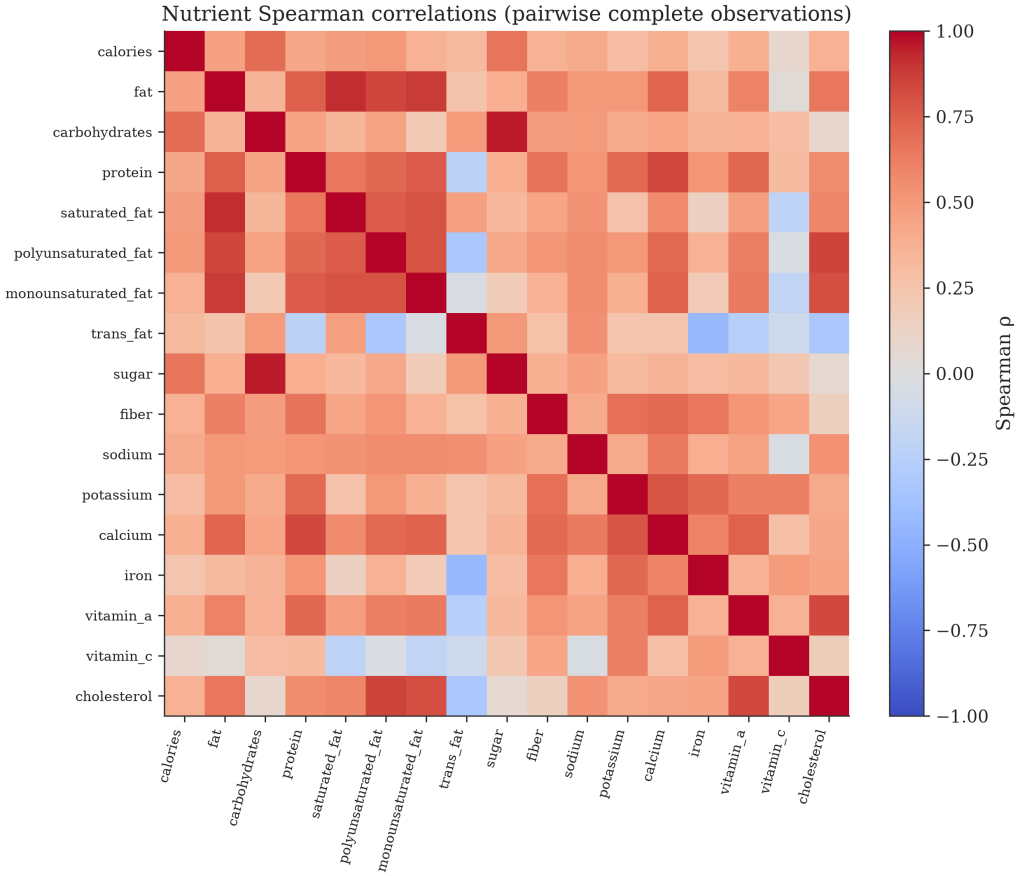
Key insight. *Protein is often near zero in drinks; carbohydrates (sugars/juices) and occasional fats (cream, egg, dairy liqueurs) differ sharply.*

Why it matters. *Cocktail macronutrient structure is not interchangeable with cooked-meal macronutrients in the paper.*

Statistical evidence. *Sample sizes differ by nutrient; see Figure 22 and descriptives table.*

Caveat. *Units are grams as parsed; cream-based drinks drive fat tails.*

Figure 25. Nutrient Spearman correlation matrix.



What this figure shows. *Pairwise complete-case Spearman correlations among nutrients with ≥80 observations.*

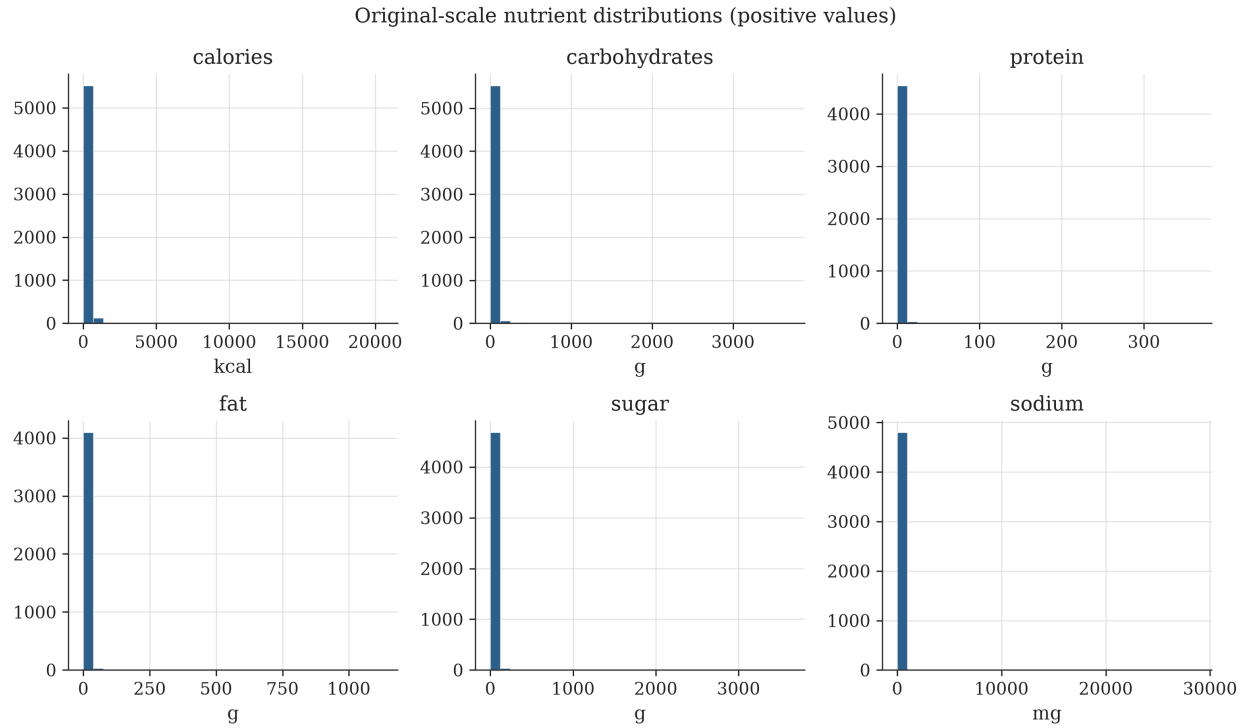
Key insight. *Positive associations among energy-related fields (calories, carbohydrates, sugar, fat) are expected arithmetically if they co-occur on the same panel, not as a biological law.*

Why it matters. *Correlations are descriptive of reported panels.*

Statistical evidence. *Spearman chosen because of skew and outliers.*

Caveat. *Pairwise deletion means each cell has its own n.*

Figure 26. Original-scale nutrient distributions.



What this figure shows. *Positive observations for calories, carbohydrates, protein, fat, sugar, and sodium when n+ is sufficient.*

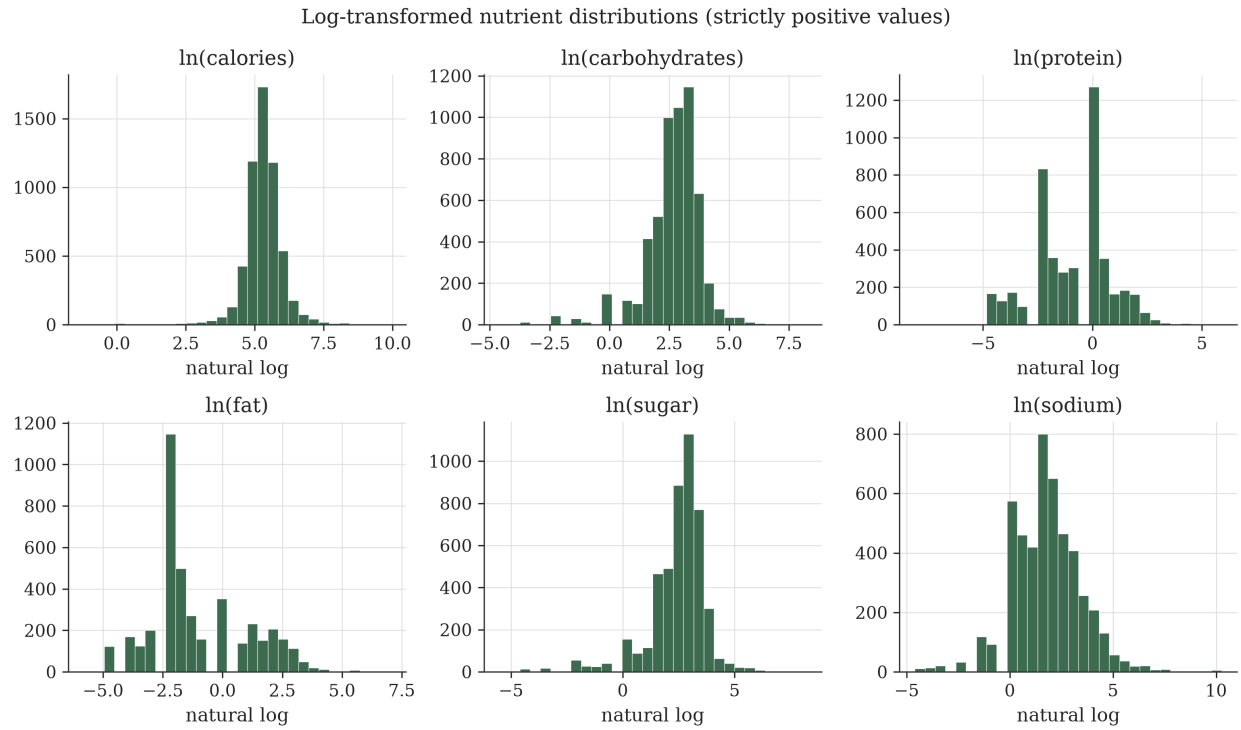
Key insight. *All are right-skewed on the original scale.*

Why it matters. *Skew is why the paper examines log space.*

Statistical evidence. *n+ listed in 08_nutrition.json per nutrient.*

Caveat. *Zeros excluded.*

Figure 27. Log-transformed nutrient distributions.



What this figure shows. *Natural logs of strictly positive values.*

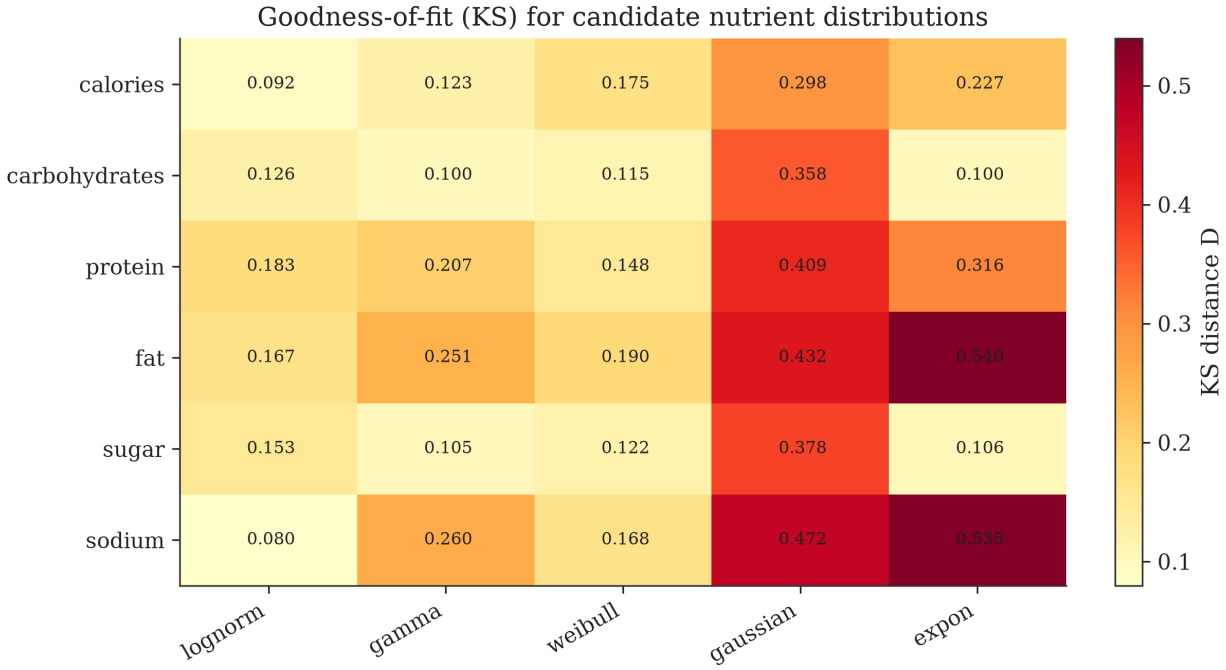
Key insight. *Log-skewness near 0 would support log-normal symmetry; observed values include calories -1.084 and carbohydrates -1.551.*

Why it matters. *Translational invariance across macronutrients (paper Fig. 3b) is only weakly testable with three incomplete cocktail macros.*

Statistical evidence. *Log transform applied only for x>0.*

Caveat. *Visual unimodality is not a KS test.*

Figure 28. Distribution-fit comparison (KS distances).



What this figure shows. *Rows: nutrients. Columns: candidate laws. Cell text is KS D.*

Key insight. *Best fits by nutrient: calories: lognorm; carbohydrates: gamma; protein: weibull; fat: lognorm; sugar: gamma; sodium: lognorm. Log-normal is not automatically best for every cocktail nutrient.*

Why it matters. *This is the paper's GOF logic applied honestly to drinks data.*

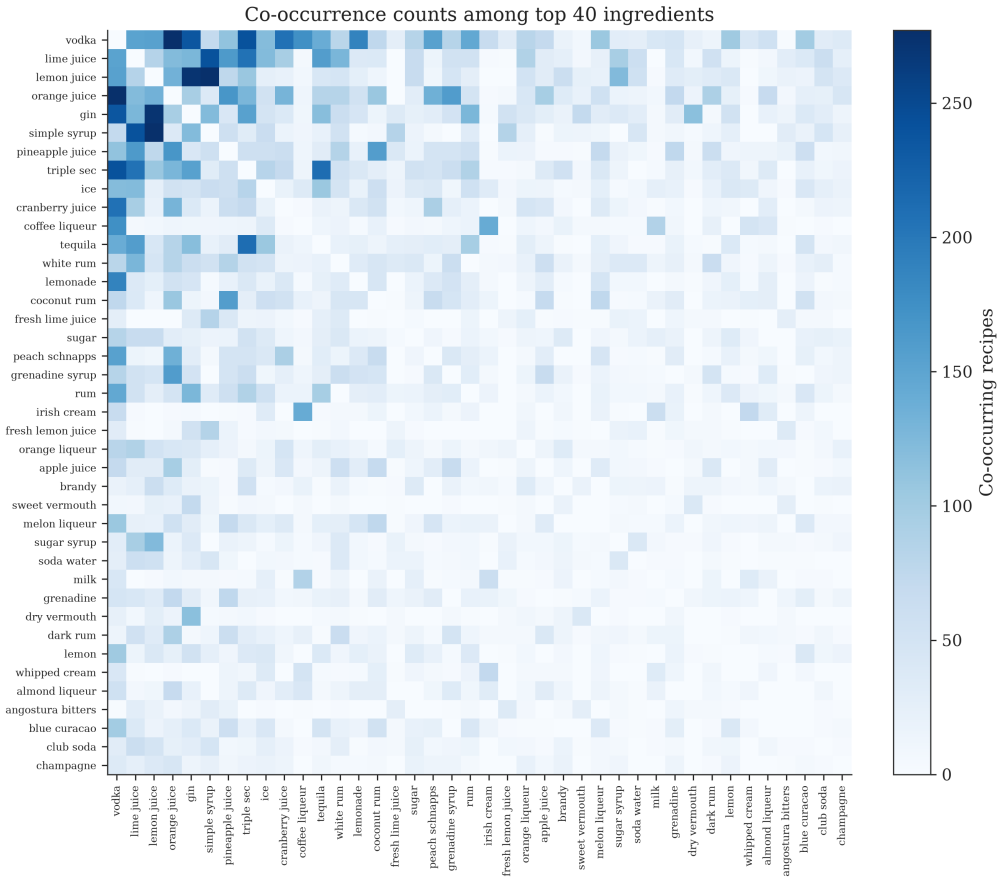
Statistical evidence. *D is the KS statistic on positive observations with floc=0 for skewed candidates where used.*

Caveat. *KS on large n flags tiny deviations; ranking of D is more relevant than a binary p-value.*

7. Ingredient Co-occurrence

Beyond the paper, co-occurrence was counted among the top 40 ingredients. PMI uses p(i,j)/(p(i)p(j)) with a minimum co-occurrence of 25 recipes to limit rare-event inflation. The network retains positive PMI edges (n_edges = 166). Clusters reflect recipe conventions (e.g., spirit+citrus+sweetener, bitters+sugar, cream+chocolate liqueurs) rather than measured flavor-compound pairing. Co-occurrence is not causality and is not food-pairing chemistry.

Figure 20. Co-occurrence heatmap (top ingredients).



What this figure shows. *Cell (i,j) is the number of recipes containing both ingredients among the top 40.*

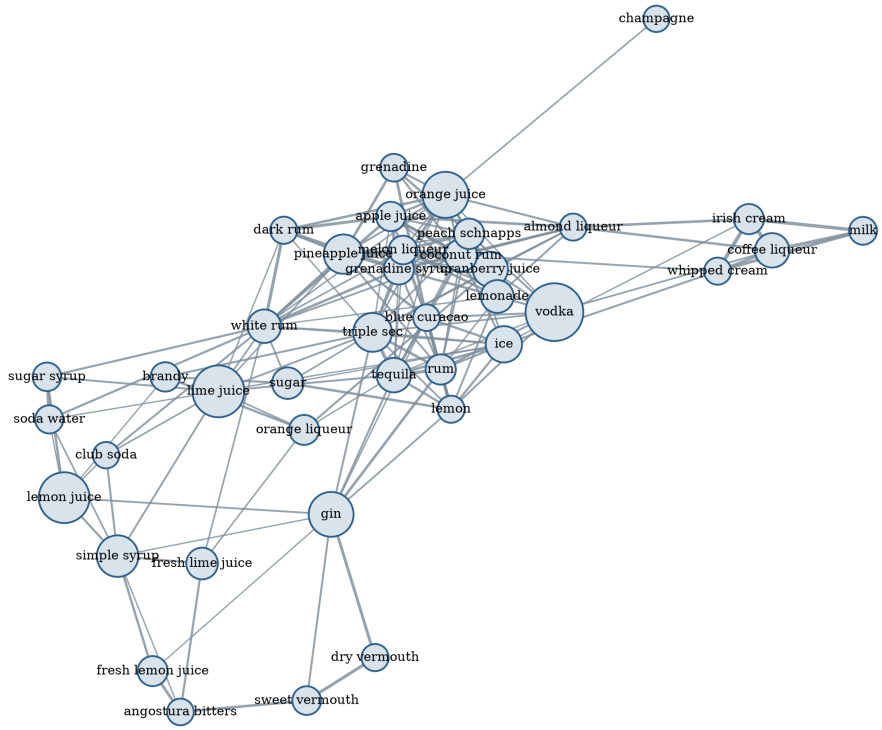
Key insight. *Darker cells mark bar-standard pairs (citrus with spirits; sugar with citrus; ice with many bases).*

Why it matters. *These pairs describe recipe incidence, the combinatorial skeleton of the cocktail corpus.*

Statistical evidence. *n_recipes in long table = 16,872.*

Caveat. *Top-N selection conditions on frequency, so rare but culturally important pairs are absent.*

Figure 21. Positive-PMI association network.



What this figure shows. *Nodes: top 40 ingredients (size ~ frequency). Edges: co-occurrence ≥ 25 and PMI > 0.4.*

Key insight. *Highest-PMI edges include: irish cream–whipped cream (PMI=2.09), coffee liqueur–irish cream (PMI=2.08), lemon–blue curacao (PMI=1.87), irish cream–milk (PMI=1.83), triple sec–tequila (PMI=1.82), coffee liqueur–milk (PMI=1.82).*

Why it matters. *Network clusters are usage cliques, useful for generating hypotheses, not for asserting compatibility.*

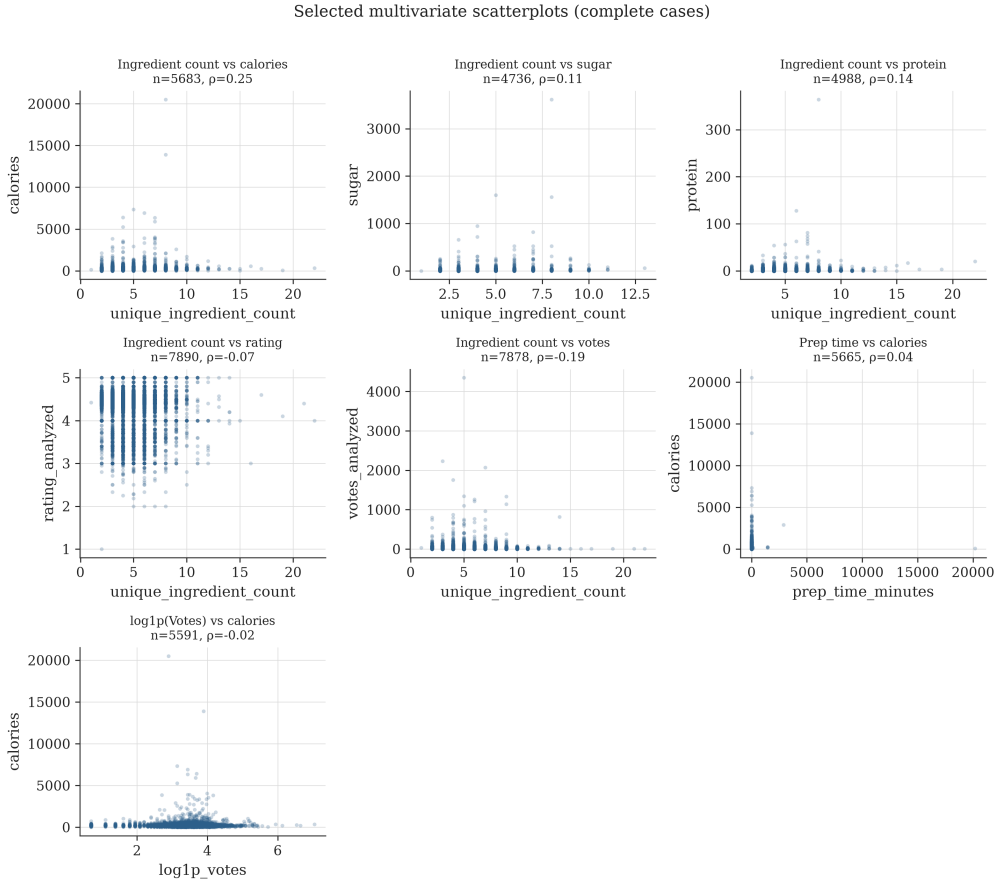
Statistical evidence. *n_edges = 166.*

Caveat. *Thresholds are analyst choices; they hide weak ties.*

8. Multivariate Relationships

Pre-specified pairs linked complexity (ingredient count, prep time), popularity (votes, ratings), and nutrients (calories, sugar, protein). Spearman correlations with bootstrap CIs are stored in multivariate_spearman.csv. Largest |ρ| among those pairs: ingredient count vs calories (ρ=0.25, n=5683), ingredient count vs votes (ρ=-0.19, n=7878), ingredient count vs protein (ρ=0.14, n=4988). Effect sizes are interpreted over p-values because n can be large in some complete-case slices and tiny in nutrient slices. No pair is treated as a causal graph.

Figure. Selected multivariate scatterplots.



What this figure shows. *Complete-case scatters for ingredient count, calories, sugar, protein, rating, votes, and prep time.*

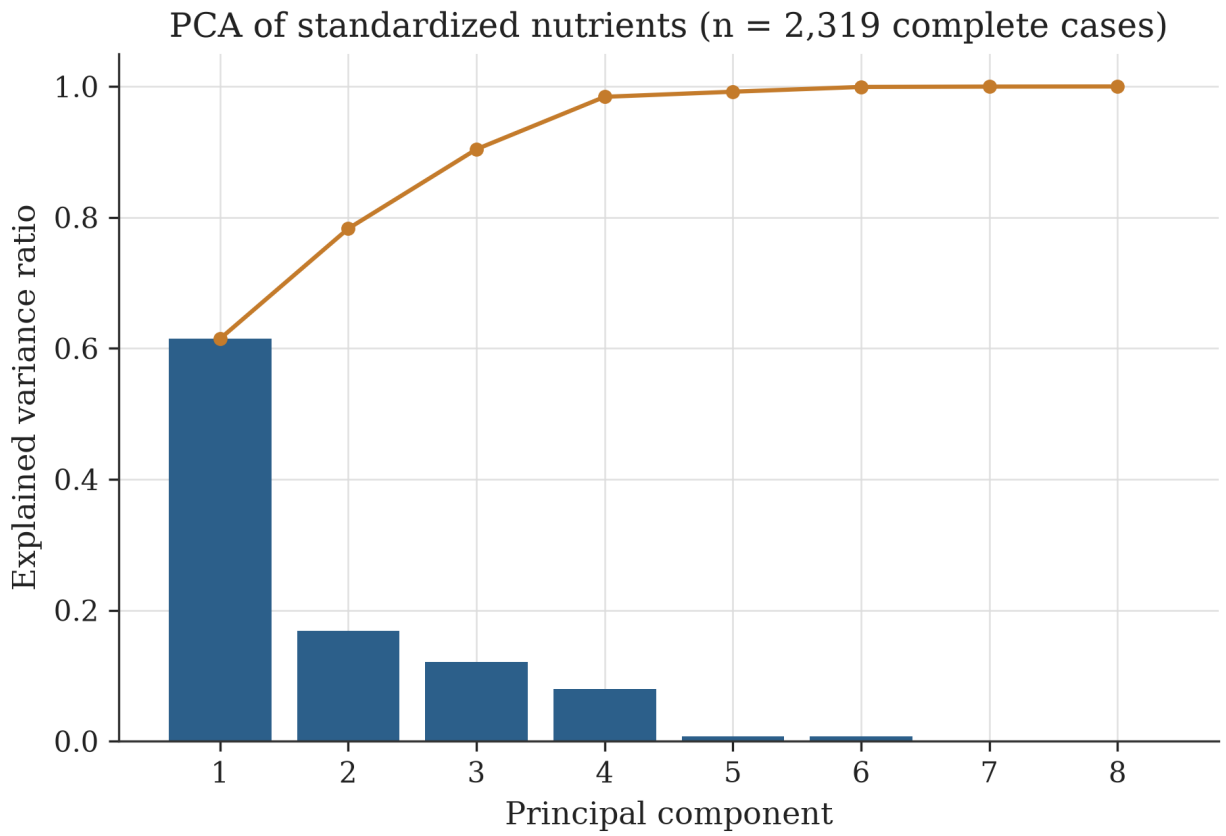
Key insight. *Visual structure is weak to moderate; nutrient axes are dominated by skew and a smaller n.*

Why it matters. *Cocktail 'complexity' is not a strong linear predictor of reported nutrition or popularity in this file.*

Statistical evidence. *See multivariate_spearman.csv; strongest listed above.*

Caveat. *Each panel has its own complete-case n.*

Figure 29. PCA explained variance.



What this figure shows. *Scree of standardized nutrients on n = 2,319 complete cases, variables = ['calories', 'fat', 'carbohydrates', 'protein', 'sugar', 'fiber', 'sodium', 'saturated_fat'].*

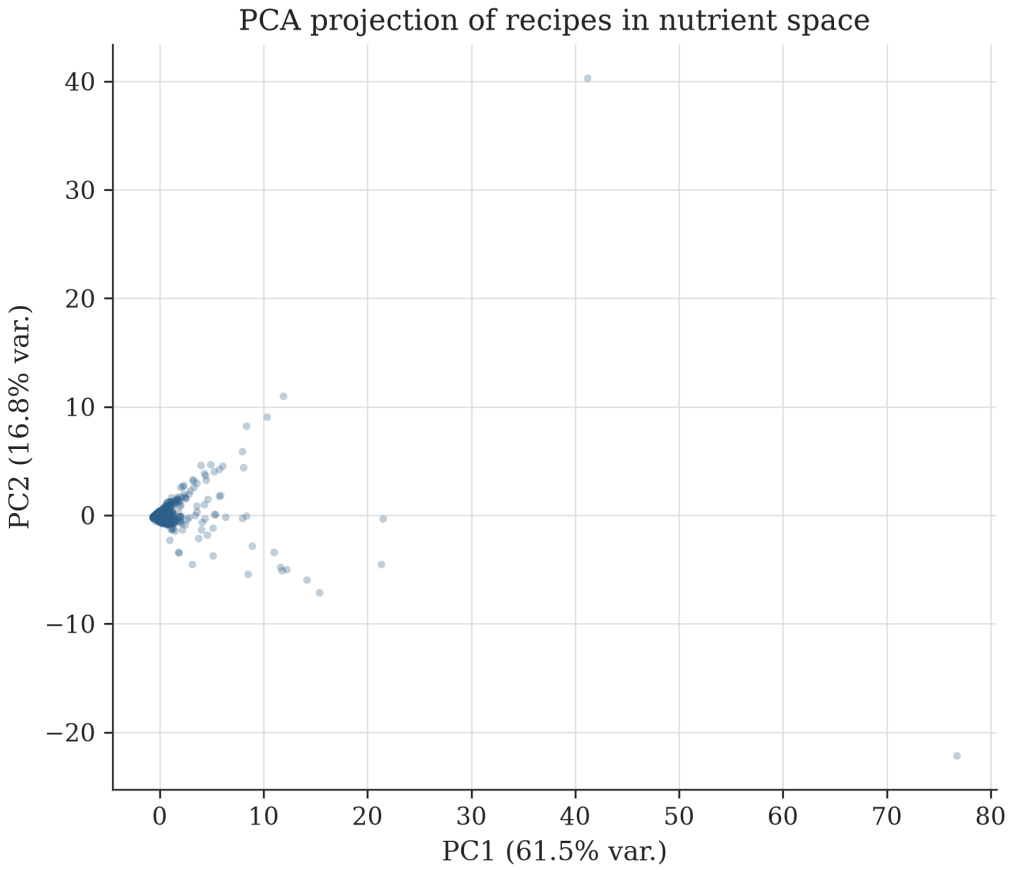
Key insight. *PC1 explains 61.5% and PC2 16.8%.*

Why it matters. *PCA is a descriptive compression of co-reported nutrient panels, not a latent 'health' index.*

Statistical evidence. *Cumulative variance in first two PCs = 78.3%.*

Caveat. *Complete-case selection biases toward richer nutrition labels.*

Figure 30. PCA projection.



What this figure shows. *Each point is a recipe in the complete nutrient subset.*

Key insight. *Spread along PC1 typically aligns with overall magnitude (calories/macros) if loadings share sign.*

Why it matters. *PC1 loadings: {'calories': 0.4308058376461811, 'fat': 0.3910828444355299, 'carbohydrates': 0.3655308884951331, 'protein': 0.40766508205135343, 'sugar': 0.3545004777225405, 'fiber': 0.28136545080174763, 'sodium': 0.08351134622978185, 'saturated_fat': 0.3870990040166974}.*

Statistical evidence. *n = 2,319.*

Caveat. *Do not cluster-interpret without a pre-registered clustering analysis (not performed).*

9. Comparison with the Reference Paper

Comparison with Universal Statistical Laws Governing Culinary Design. The paper is a methodological reference. CocktailDB is not required to match its exponents.

Phenomenon	Paper	This dataset	Method	Evidence	Interpretation
Zipf	f(r)~r^{-α}, α_global≈1.53; cuisine mean≈1.39; Clauset MLE on frequencies; LR vs log-normal/exponential	OLS α≈1.00 on r=5–500; Clauset α_freq≈1.85, xmin=2.0	Rank–frequency OLS + Clauset MLE	Heavy tail yes; perfect power law not assumed	Partially consistent: Zipf-like reuse in a smaller drink vocabulary
Heaps	V(R)~R^β, β≈0.56 (100 shuffles; methods also mention 10 for error bars)	β≈0.74 (100 shuffles; 95% [0.73, 0.75])	Permutation + log–log OLS	mean R²=1.000	Consistent in sign (sublinear); similar magnitude is coincidental unless replicated on other drink corpora
Menzerath–Altmann	y(L)=a L^b e^{cL}; a=13.16,b=−0.26,c=−0.019; R²=0.99; min L≈13	ρ(L,y)=0.10; binned R²=0.90; L median=4.0	Information s_i=−log p(i); NLS	Short L range	Different / weakly testable at the paper's length scale

Phenomenon	Paper	This dataset	Method	Evidence	Interpretation
Nutrient distributions	Log-normal macros; mean KS 0.073 vs worse alternatives	Best KS by nutrient: calories: lognorm; carbohydrates: gamma; protein: weibull; fat: lognorm; sugar: gamma; sodium: lognorm; incomplete coverage	KS on x>0	Nutrient n much smaller; serving unknown	Partially testable; not a reproduction of meal log-normality
Ingredient diversity	19,019 unique ingredients; 1.15M mentions	14,466 standard unique; hapax=73.5%	Normalized tokens	Smaller closed bar vocabulary	Consistent with constrained vocabulary, different scale
Recipe complexity	Unimodal 8–12 ingredients; mean ~10; ~9 steps	median L=4; mean=4.7; no steps	Unique ingredient count	No technique/step data	Different: cocktails are shorter; steps not testable

10. Major Findings

1. Ingredient usage is concentrated: the most common normalized ingredient is “vodka” (2,191 recipes; 13.0% of the corpus). 73.5% of the vocabulary occurs in only one recipe.

2. Rank–frequency curves are heavy-tailed. A log–log OLS Zipf exponent of $\alpha \approx 1.00$ was obtained on ranks 5–500 (descriptive $R^2 = 0.991$). A Clauset-style power-law fit to the frequency distribution is reported separately and is not treated as automatic confirmation of Zipf’s law.

3. Heaps’ law is clearly sublinear: mean $\beta = 0.738$ (100 permutations; 95% interval [0.725, 0.751]). This is close to the paper’s global $\beta \approx 0.56$ but is estimated on a smaller, cocktail-only corpus and is not a target to match.

4. Menzerath–Altmann: Spearman correlation between recipe size L and mean ingredient information y(L) is $\rho = 0.101$. The paper’s decreasing-then-rising global pattern is not assumed; the fitted MA parameters and R^2 are reported in §6.3.

5. Recipe size is small relative to the paper’s food corpus: mean unique ingredients = 4.67 (median 4.0; n = 16,872).

6. Ratings, where observed, are high and left-censored toward 5 stars (mean 4.32, median 4.45, n = 7,890). All 116 Rating=0 rows have Votes=0 and were treated as unrated.

7. Votes are heavy-tailed. Spearman association of votes with rating is $\rho = -0.058$ (n = 7878). This is an association, not evidence that votes cause ratings.

8. Cuisine cannot support cuisine-level laws: only 393 non-missing values (2.33% of the frequency corpus), with labels that mix drink types and nationalities.

9. Category is available for 6,925 recipes (41.0% of the frequency corpus); category comparisons are conditional on that subset.

10. Nutrient strings exist for 5,694 recipes (66.3% missing). Log-space KS comparisons are reported per nutrient; log-normality is not claimed globally. Best KS candidates include: calories: lognorm; carbohydrates: gamma; protein: weibull; fat: lognorm; sugar: gamma; sodium: lognorm.

11. Strongest complete-case Spearman associations among the pre-specified multivariate pairs: ingredient count vs calories (ρ=0.25, n=5683), ingredient count vs votes (ρ=-0.19, n=7878), ingredient count vs protein (ρ=0.14, n=4988).

12. Cooking techniques, utensils, instruction steps, and 26-cuisine stratification from the reference paper cannot be reproduced because those variables are absent.

11. Limitations

1. Missing values are extensive and structured; every analysis reports its own n.

2. Cuisine is too sparse and semantically inconsistent (Cocktail vs American vs Vegan) for cuisine-level laws.

3. Category incompleteness (about 59% missing) blocks globally representative category inference.

4. Nutrient fields mix compact and colon formats; units were harmonized only where the text supported it.

5. Ingredient normalization is partly subjective; sensitivity across four levels is provided but residual quantity/brand strings remain.

6. The file likely concatenates heterogeneous sources (rating sites, recipe blogs, nutrition-labelled pages).

7. Duplicate names may be variants or competing scrapes; only exact row and Name+Ingredients clones were dropped.

8. Sampling bias: web cocktails over-represent English-language, Instagram-era, and database-popular drinks.

9. Votes measure attention, not quality.

10. Ratings with few votes are unstable and piled at 5.

11. Observational correlations are not causal.

12. Cocktail-specific composition may not generalize to global cuisine—the paper's claim domain.

13. Corpus size (16,872 frequency recipes) is an order of magnitude below 118,083.

14. No reliable geographic or cultural metadata.

15. Prep times and nutrients may mix servings, infusions, and panel-generator artifacts; extremes were investigated, not auto-deleted.

12. Future Work

A cocktail-specific NER pass would align the corpus more closely with the paper's structured entities. Linking drinks to flavor-compound data would test whether co-occurrence clusters match chemistry. Serving-size standardization is required before nutrient log-normality can be compared to packaged-food or RecipeDB results. Larger multilingual bar corpora would show whether β and α are drink-domain constants. Historical menus could separate traditional formulas from modern riffs.

13. Conclusion

CocktailDB supports the paper's high-level claim that culinary compositions reuse a core vocabulary and accumulate novelty sublinearly: Zipf-like rank–frequency structure and Heaps $\beta \approx 0.74$ are visible after careful normalization. It does not support a blanket transfer of the global Menzerath–Altmann minimum at $L \approx 13$, cuisine-universal exponents, technique scaling, or fully log-normal meal macronutrients. The honest result is mixed—and that is the scientifically relevant one.

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Alstott, J., Bullmore, E. & Plenz, D. powerlaw: A Python package for analysis of heavy-tailed distributions. PLoS One 9, e85777 (2014).

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Menichetti, G. & Barabási, A.-L. Nutrient concentrations in food display universal behaviour. Nat. Food 3, 375–382 (2022).

Ahn, Y.-Y. et al. Flavor network and the principles of food pairing. Sci. Rep. 1, 196 (2011).

Newman, M. E. J. Power laws, Pareto distributions and Zipf's law. Contemp. Phys. 46, 323–351 (2005).

Egghe, L. Untangling Herdan's law and Heaps' law. J. Am. Soc. Inf. Sci. Technol. 58, 702–709 (2007).

Appendix

A. Reproducibility. Run python scripts/run_all.py from eda_project/. Numerical results live in tables/json/. Figures are written to figures/ as PNG and PDF. This report reads those files and does not hard-code exponents.

B. Duplicate policy. Exact rows: remove extras (no information). Name+Ingredients clones: remove extras for frequency (avoid double-counting the same formula). Same name, different ingredients: keep (variants).

C. Paper Methods notes used here. Ingredient frequency = recipe incidence; Zipf via rank–frequency plus Clauset MLE on frequencies with xmin by KS; Heaps $V(R)=K R^{\beta}$ by log–log OLS on permutations; MA $y(L)=a L^{b} \exp(cL)$ by nonlinear least squares; nutrients: log transform of positives, log-skewness, KS versus log-normal/gamma/Weibull/Gaussian/exponential.

D. Frequency corpus size = 16,872. Raw rows = 16,895. Unique names = 14,916. Unrated zeros = 116.

Modifications

Already in the report: The original research report occupies pages 1-39. Its pages and figures are retained in full.

Modification: This appended section responds to all seven observation points. New explanatory text is purple. Figures M01-M17 are new; original figure numbers remain unchanged. Page references below use PDF page numbers, which match the original printed numbers.

Scope and interpretation

The theme remains culinary composition: ingredient reuse, vocabulary growth, recipe structure, nutrient patterns and ingredient pairing. The new work tests what survives changes in rank range, scale, normalization and network filtering. It does not force the cocktail data to match the reference food corpus.

Observation	Original location	New pages
1. Zipf rank intervals 5-500 / 5-1000 / 5-2000	pp. 22-24; Figs. 15-16, 32	42-43
2. Boxplot for Figure 17	p. 25; Heaps curve (not the Zipf boxplot)	44
3. Interpret correlation colours and weak associations	p. 30, Fig. 25; pp. 32-35	45-47
4. Replot skewed distributions and scattered extremes	pp. 15, 17, 19-21, 29, 31, 35-37	48-53
5. Hierarchical co-occurrence view	pp. 32-34, Figs. 20-21	54-55
6. Normalized edge widths and node colours	p. 34, Fig. 21	56-57
7. Crude-network backbone; alpha and significance	pp. 32-34: no backbone analysis	58-64

Read this before comparing numbers

The CSV reproduces the original row/duplicate audit. However, the earlier scripts and cleaned ingredient tables were not attached. Ingredient-dependent results below use an explicitly reconstructed boundary-normalization pipeline. It yields 13,860 strings, versus the original 14,466. New counts are therefore a transparent reanalysis, not a claim to recover the earlier implementation exactly. Literal-string sensitivity is also reported. See page 41.

Evidence is described as supported, contradicted, or unresolved. A high R^2, a coloured cluster, or a small p-value alone does not establish a universal law or a causal mechanism.

M0. Reanalysis audit and definitions

Already in the report: Cleaning rules and baseline counts appear on pp. 3-5, 8-9 and 39.

Modification: Rebuild the analytical corpus directly from the attached CSV. Keep the original CSV and all original PDF pages unchanged.

Quantity	Original report	Reanalysis
Raw rows / columns	16,895 / 9	16,895 / 9
Name versus Name.1 mismatches	0	0
Exact duplicate extras	17	17
Name + Ingredients duplicate extras (includes exact)	23	23
Frequency-corpus recipes	16,872	16,872
Ingredient vocabulary: standard / reconstructed	14,466	13,860
Surface vocabulary: lowercase/whitespace	31,180	31,224 (also NFKC Unicode normalization)
Recipes containing vodka	2,191	2,381
Median / mean unique ingredients	4 / 4.67	4 / 4.674
Maximum recipe size	22	22

Reconstructed ingredient rule

Split on |; apply Unicode NFKC, lowercase and whitespace normalization; strip boundary quantities and measurement units (including trailing quantities); repeat boundary removal up to three times. Map Cointreau to orange liqueur and Kahlua to coffee liqueur. Retain internal brands, descriptors, fresh/not-fresh forms, lime versus lime juice, and unrecognized text. Count each resulting string at most once per recipe. Alphabetical tie-breaking makes ranks deterministic.

Meaning of “do not encode” and “crude”

Ingredient names remain readable strings. No arbitrary category/name numbers enter a correlation or Zipf fit. Binary recipe membership is only a counting device. Because the handwritten instruction is ambiguous, the unmerged quantity-bearing strings form a separate sensitivity analysis. “Crude network” means the full recipe projection before any top-40 selection, PMI cut-off or weight cut-off.

Other fields

Use the same 16,872-row corpus throughout new plots. Parse comma-separated votes. Treat the 116 joint zero rating/vote rows as unrated. Convert prep units to minutes and collapse whole repeated phrases. Parse colon and compact nutrient formats, including scientific notation; preserve kcal, g and mg. No imputation, serving conversion or outlier deletion. Original nutrient-only figures sometimes used all 16,895 rows, so their n can differ.

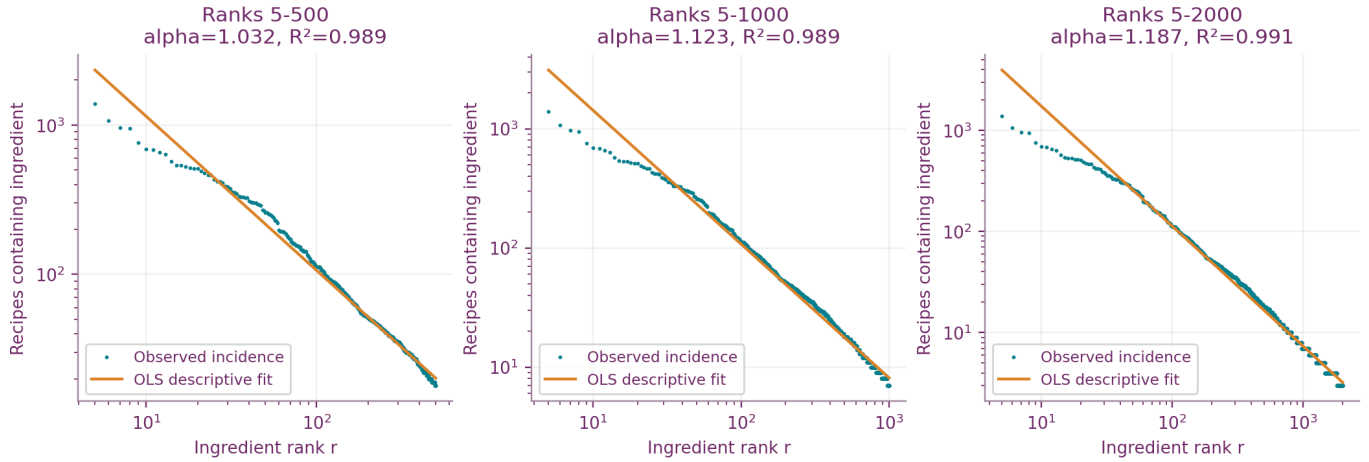
Limitation: differences in ingredient cleaning and the earlier undocumented Heaps sampling grid prevent exact recovery of every old exponent. They are not concealed by matching numbers by hand.

M1. Zipf fits in each requested interval

Already in the report: pp. 22-23: main fit used ranks 5-500 (alpha 0.998; R^2 0.991). Page 24, Fig. 32, used different rank strata for boxplots.

Modification: Fit the three specified inclusive rank windows separately: 5-500, 5-1000 and 5-2000. These are overlapping cumulative windows, not three disjoint bins.

Estimator: log10 f(r) = a - alpha × log10 r. Here f is the number of distinct recipes containing the ingredient; ranks follow decreasing f. All observed ingredients remain in the corpus; only the regression window changes. No name/category encoding is used.



Rank window	Rank points	Alpha	R^2	Frequency endpoints
5-500	496	1.032	0.989	1,392 to 18
5-1,000	996	1.123	0.989	1,392 to 7
5-2,000	1996	1.187	0.991	1,392 to 3

Evidence and inference

The exponent rises from 1.032 to 1.187 when lower-frequency ingredients are included. Ingredient reuse is strongly rank-ordered, and a restricted range is approximately Zipf-like. However, the slope is not constant across the requested windows. The corpus supports concentrated reuse and a rare tail, not one exact inverse-rank law over every scale.

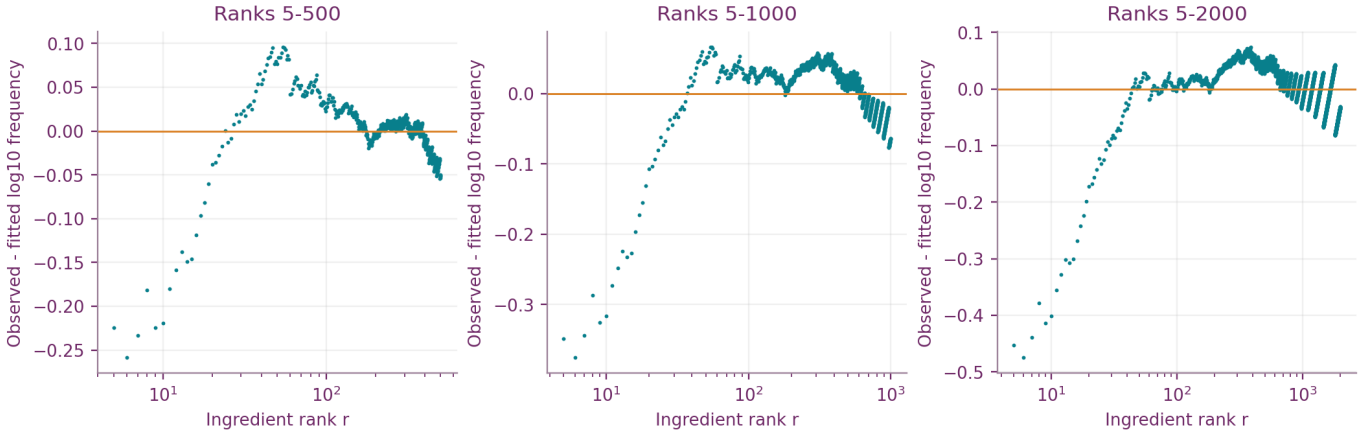
What this does not prove

All three R^2 values are near 0.99, but ranks are constructed from sorted, dependent counts. OLS is a descriptive rank-curve fit, not a valid power-law goodness-of-fit test [R2]. A frequency-distribution MLE exponent is a different quantity and must not be substituted for the rank exponent. The earlier Clauset-style fit is discussed on pp. 22-23; it has not been silently rerun on an unavailable old normalized table.

M1. Challenge the Zipf interpretation

Already in the report: pp. 16 and 22-23 already warn about curvature; only the 5-500 fit is shown in the old Figure 16.

Modification: Show residuals, test the effect of retaining quantity-bearing strings, and compare a fixed inverse-rank slope with the free-slope fit.



Window	Raw-string alpha / R^2	RMSE: free slope	RMSE: alpha=1	Curved log fit RMSE
5-500	0.618 / 0.974	0.0418	0.0435	0.0217
5-1000	0.683 / 0.982	0.0473	0.0685	0.0183
5-2000	0.750 / 0.984	0.0470	0.0908	0.0261

Residual bends persist rather than forming unstructured noise around zero. A quadratic in log-rank has lower in-sample RMSE, so even excellent linear R^2 hides curvature. This is a shape diagnostic; adding a parameter naturally improves fit and is not independent model validation.

Three genuinely disjoint segments

As an additional check, fitted slopes are 5-500: 1.032, 501-1000: 1.383, 1001-2000: 1.300. These segments confirm that the local decay rate changes; they do not replace the three professor-specified windows.

Recipe-resampling check (250 repeats)

Window	Observed alpha	2.5%-97.5% reranked bootstrap range
5-500	1.032	1.016 to 1.036
5-1000	1.123	1.096 to 1.115
5-2000	1.187	1.130 to 1.145

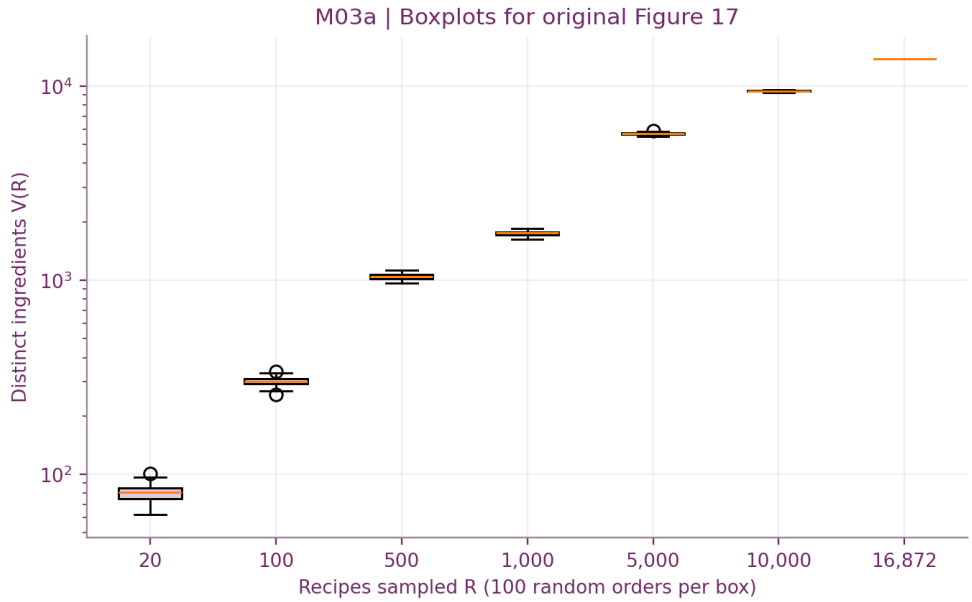
Each repeat samples 16,872 recipes with replacement and re-ranks the resulting vocabulary. These are sensitivity ranges, NOT confidence intervals or goodness-of-fit p-values. The observed long-window estimates lie outside their ranges: resampling changes rare-string visibility and re-ranking introduces bias. Therefore the ranges cannot certify an exponent. Literal strings also give much smaller slopes (0.618-0.750), showing that ingredient identity rules materially affect the scaling claim.

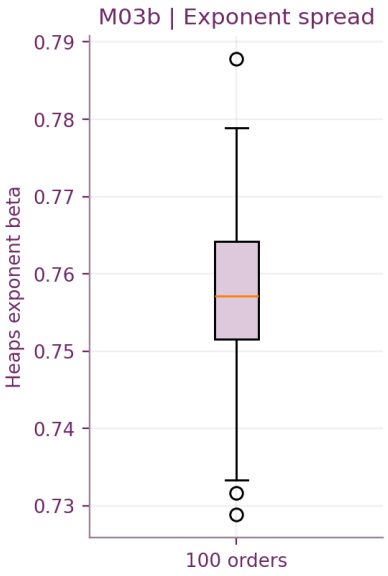
Verdict: approximate Zipf-like structure is supported in each window; one invariant exponent, exact alpha=1 over the full range, and preprocessing-independent universality remain unsupported.

M2. Boxplots for Figure 17 (Heaps)

Already in the report: p. 25, Fig. 17: mean vocabulary-growth curve with a 95% recipe-order band. Page 26, Fig. 18: histogram of fitted beta. Neither is a checkpoint boxplot.

Modification: At fixed recipe counts R, plot the distribution of vocabulary V(R) across 100 random recipe orders. Add a separate boxplot of the 100 fitted exponents. This directly extends the Heaps curve, not the unrelated Zipf boxplots in Fig. 32.





Recipes R	Q1 V(R)	Median V(R)	Q3 V(R)
20	75.0	81.0	85.2
100	293.8	300.0	310.2
500	1,018.0	1,046.5	1,067.2
1,000	1,716.8	1,750.5	1,778.2
5,000	5,629.5	5,680.5	5,723.2
10,000	9,394.8	9,439.5	9,480.2
16,872	13,860.0	13,860.0	13,860.0

Estimator and result. Use 155 unique log-spaced/checkpoint values from R=20 to 16,872. Fit log V versus log R in each order. Mean beta=0.757; SD=0.010; central 95% order range [0.736, 0.773]. Earlier beta was 0.738; normalization and fit-grid differences must be considered before comparing them.

Read the boxes. Boxes contain the middle 50%; centre lines are medians; whiskers extend to the last value within 1.5 IQR; farther observations are displayed. At R=N, every ordering contains the same recipes, so V(N)=13,860 exactly: the final box must collapse. This is a mathematical endpoint check, not unusually precise estimation.

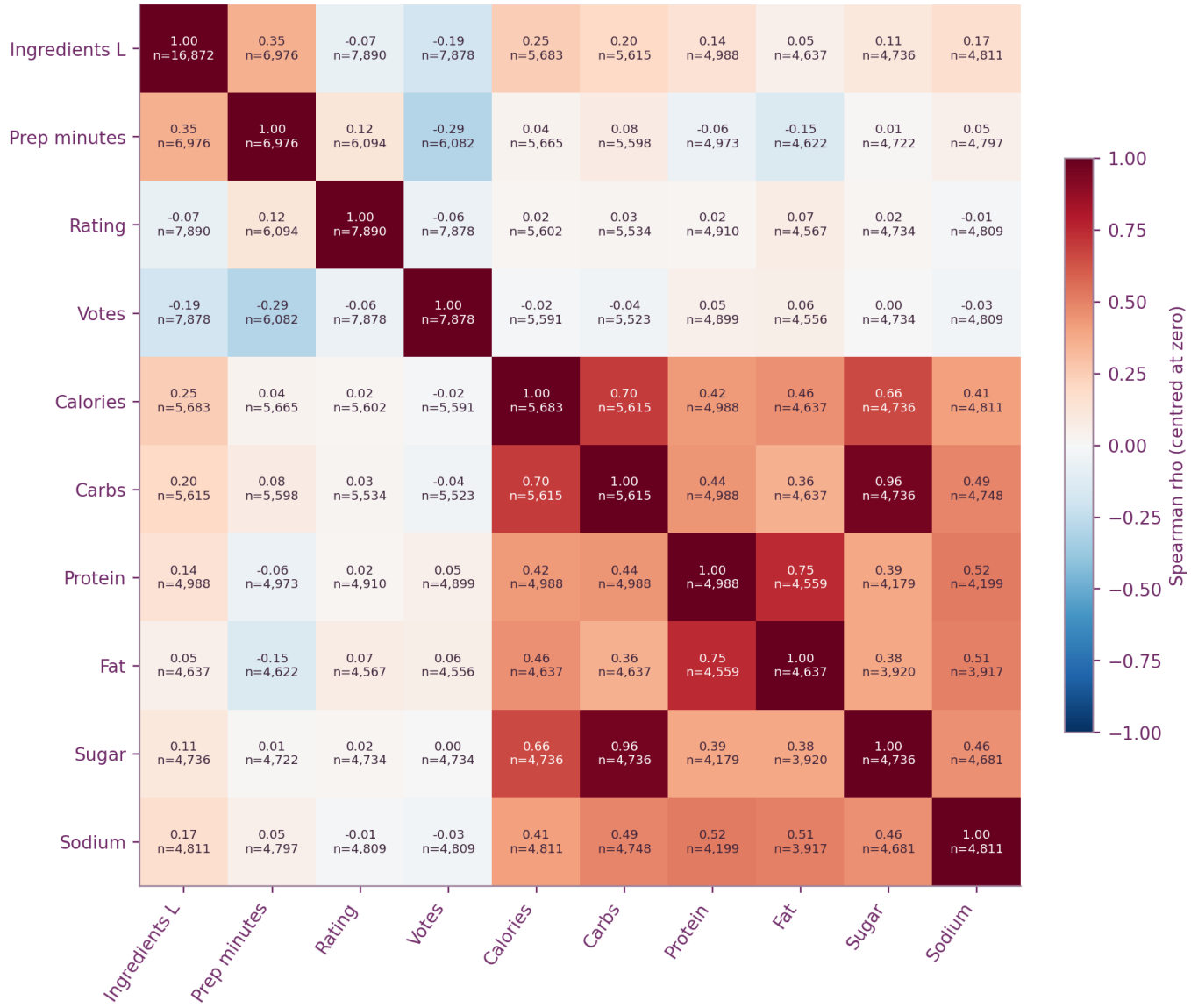
Inference. Vocabulary growth is sublinear on this fitted range. The permutations quantify order variability conditional on this fixed corpus. They do not measure uncertainty about all cocktail recipes or prove the same exponent as the food corpus. The log vertical axis improves visibility without transforming the raw V values used to construct the boxes.

M3. Read heatmap colour scales correctly

Already in the report: p. 30, Fig. 25 is a correlation matrix; p. 33, Fig. 20 uses a blue co-occurrence count scale; p. 32, Fig. 28 is a KS-distance matrix.

Modification: Add an annotated numeric Spearman matrix, with a fixed -1 to +1 colour range, zero at the centre, and the complete-case observation count in every cell.

M04 | Numeric associations: magnitude, sign and observation count



Blue is not inherently “no correlation”. In this new diverging scale, blue means negative association; white means near zero; red means positive association. In the original Fig. 20, darker blue instead means MORE shared recipes. Its pale cells cannot establish an absence of correlation. In Fig. 28, smaller KS distance means a closer candidate distribution fit, not weak association.

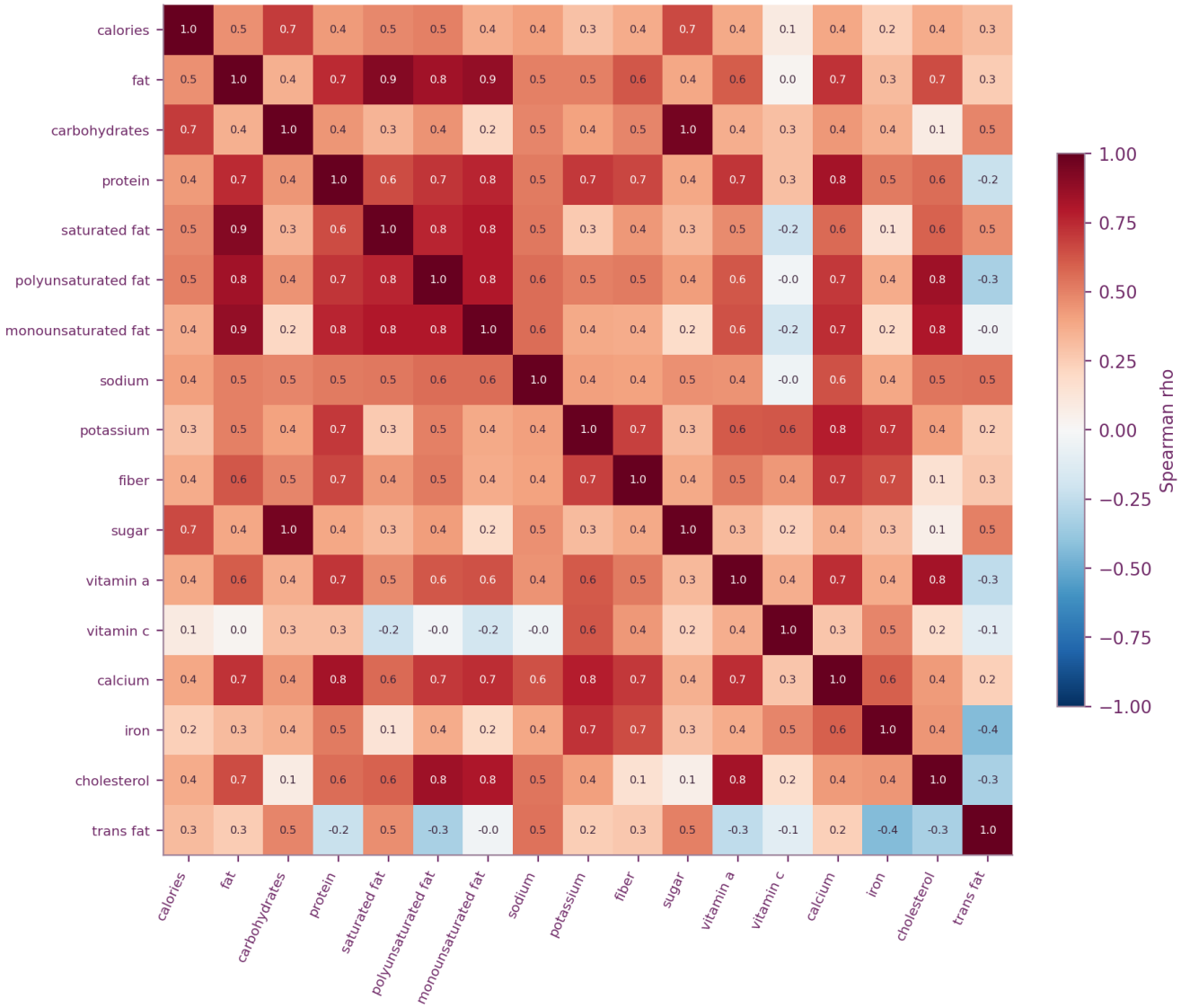
Spearman detects monotone association. A near-zero value does not rule out a nonlinear, subgroup-specific or otherwise hidden relationship. Pairwise deletion changes the sample from cell to cell; the n values show which comparisons have little supporting data. Names and categories are excluded because arbitrary numeric codes would create meaningless distances.

M3. Nutrient correlations are often strong

Already in the report: p. 30, Fig. 25 already contains strong red nutrient cells, although its numerical values and pairwise n are absent.

Modification: Recompute and annotate the nutrient matrix. Nutrient labels qualify with at least 80 reported values, matching the earlier selection rule; a particular pair can still have fewer joint observations.

M05 | Reconstructed Figure 25: nutrient rank correlations



Measured examples. Carbohydrates versus sugar: rho=0.957 (n=4,736). Fat versus saturated fat: rho=0.916 (n=3,334). Calories versus carbohydrates: rho=0.696 (n=5,615). These contradict a blanket claim that all matrices show almost no association.

Why this pattern is plausible. Sugar is part of carbohydrate mass, and saturated fat is part of total fat. Their strong association is partly a part-whole relationship, not evidence that one causes another. Absolute nutrient amounts can also rise together for larger recipes or servings. Pairwise n ranges from 57 to 5,683; weak cells with small joint samples deserve less weight.

Limits. Neither serving count nor standardized alcohol mass is available. Panel composition, rounding and selective nutrient reporting can affect coefficients. The part-whole explanation is consistent with the field definitions; it is not a tested biological mechanism. The appendix makes no claim that every nutrient relationship is strong.

M3. Why some associations remain weak

Already in the report: pp. 16-21 and 34-35 report popularity, preparation and complexity associations, but give limited measured explanations for weak structure.

Modification: Separate measured constraints from possible explanations. The table uses 400 recipe bootstrap resamples per pair; intervals are conditional on the observed subset and do not remove missingness bias.

Pair	n	rho	Bootstrap 95% interval
L vs prep	6,976	0.352	0.331 to 0.373
prep vs votes	6,082	-0.291	-0.320 to -0.263
rating vs votes	7,878	-0.058	-0.081 to -0.029
L vs calories	5,683	0.247	0.223 to 0.275
L vs votes	7,878	-0.191	-0.212 to -0.171
L vs protein	4,988	0.144	0.114 to 0.171

1. Bounded ratings and tied prep times (measured)

84.7% of analyzed ratings are at least 4 stars. The scale ends at 5. Prep time equals its modal value of 3 minutes for 69.7% of observed recipes. Ceiling concentration and many ties leave limited rank resolution. This can restrict observable association, but does not prove what the latent, uncensored association would be.

2. Ingredient count omits ingredient amount (definition)

L gives the same weight to a small garnish and a large cream or spirit measure. It cannot represent volume, serving count, ingredient composition or energy directly. A weak L-nutrient association is therefore compatible with large nutrient differences between recipes of equal size; the hexbin plots on page 52 show this spread.

3. Source-format mixing (tested sensitivity)

Subset by nutrient text format	Rating-vote n	Rating-vote rho
Colon-style panels	4,871	0.004
Compact panels	720	-0.406
Nutrient field absent	2,287	-0.306

The pooled rho=-0.058 hides heterogeneity across observed formats. These labels are format proxies, not verified source identities. This supports checking source mix; it does not establish a particular scraping or user-behaviour cause. Ingredient-count/calorie rho stays positive within both colon (0.235) and compact (0.335) subsets.

4. Missingness and effect size

Observed subsets are selective (original pp. 8-9). Popularity measures attention rather than ingredient design. A tiny p-value is possible for a weak effect when n is large: rating-vote p=2.96×10^-7 but |rho|=0.058. Transforming values cannot manufacture a stronger Spearman association; log1p preserves ranks exactly.

M4. Exact transformation register

Already in the report: Log views already occur in Fig. 8 (p. 17), Fig. 10 (p. 19), Figs. 15-17 (pp. 16, 23, 25) and Fig. 27 (p. 31). Other figures remain compressed by extremes.

Modification: Name each transformed original figure. Use paired raw and ln(1+x/u) views on identical finite, nonnegative observations, where u is one original unit (one vote, minute, kcal, g or mg). This log1p transform retains zero.

Original figure / page	New figure / page	Exact change
Fig. 14 / p. 15: ingredient frequency	M06 / p. 49	ln(1+recipe frequency), zero absent
Fig. 8 / p. 17: votes	M06 / p. 49	Same log1p idea; paired raw view and explicit n
Fig. 10 / p. 19: prep time	M06 / p. 49	Use log1p, retaining 44 zero-minute rows
Figs. 23, 26 / pp. 29, 31: calories	M07 / p. 50	Raw versus log1p, same recipe subset
Figs. 24, 26 / pp. 29, 31: carbs, protein, fat; sugar, sodium	M07-M08 / pp. 50-51	Raw versus log1p; zeros counted, not discarded
Fig. 11 / p. 20	M09 / p. 52	Prep axis becomes ln(1+minutes)
Fig. 12 / p. 21	M09 / p. 52	Log1p prep and votes; retain complete cases
Unnumbered multivariate figure / p. 35	M09 / p. 52	Log1p nutrient/vote axes; hexbin density
Figs. 29-30 / pp. 36-37	M17 / p. 53	PCA on log1p nutrients then standardize; compare raw PCA

What stays on its meaningful scale

Keep bounded star ratings and short integer recipe sizes on their original scales. Missingness percentages, category bars, correlations and KS distances are not heavy-tailed physical measurements needing an automatic log. The scaling plots already use log axes. Mean ingredient information in Fig. 19 is already built from -log probabilities. PCA scores can be negative, so they are not logged directly.

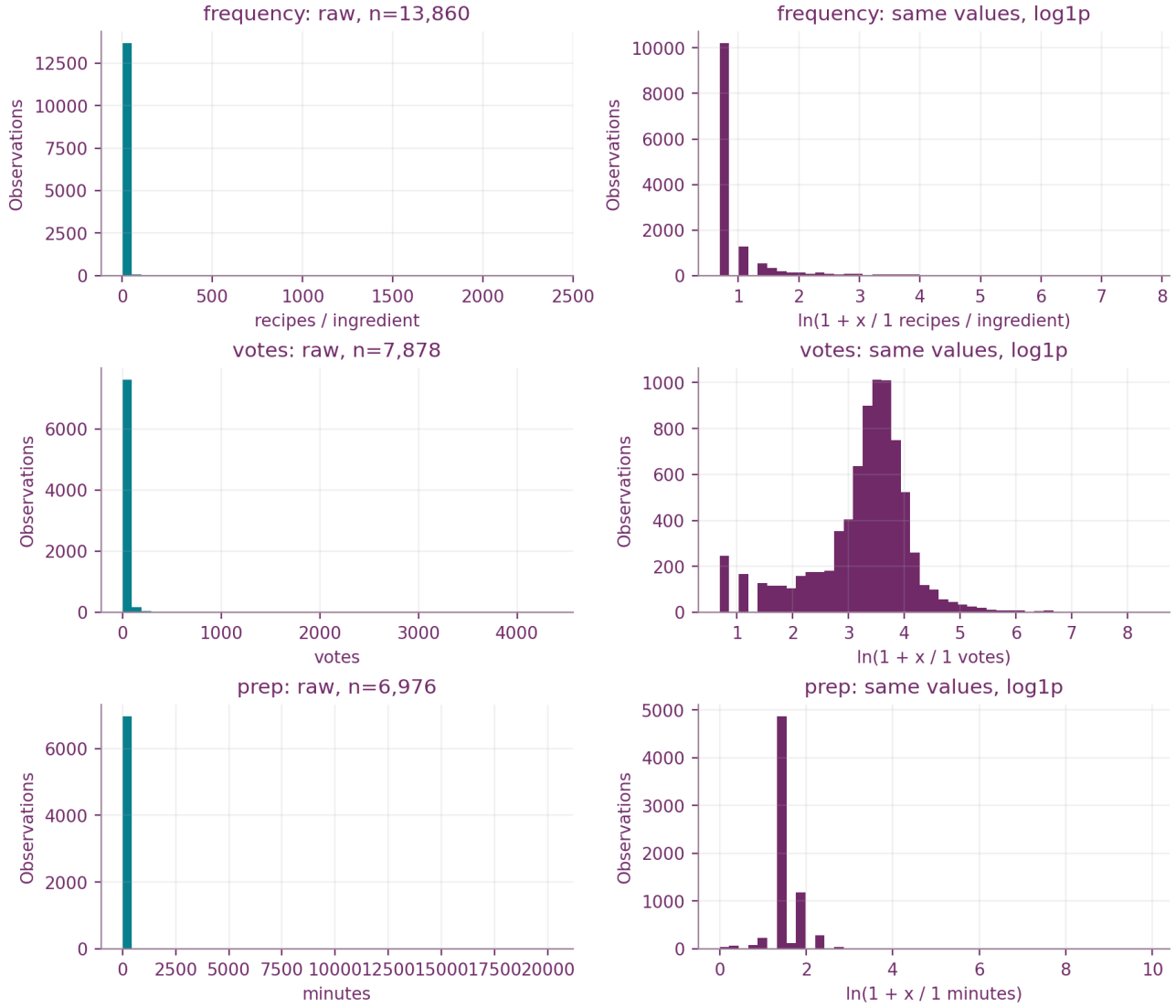
What is and is not concluded

Log1p makes the central mass visible and compresses extreme values; it does not validate them or standardize servings. It may leave skew or create negative skew. Symmetry of log1p(x) does not prove log-normality of x: the earlier positive-only ln(x) analysis is a different operation. No outlier is deleted or silently winsorized.

M4. Frequency, votes and preparation time

Already in the report: Fig. 14 (p. 15) compresses rare ingredients and the far tail. Figs. 8 and 10 (pp. 17, 19) already add log views for votes/time.

Modification: M06 supplies matched raw/log1p views and retains all finite observations. The left panels deliberately expose how the original linear scale hides the central mass.



Ingredient reuse. 10,210 / 13,860 strings (73.7%) occur once. The median frequency is 1; the maximum is 2,381. The prominent left bar and thin far-right tail are the actual pattern, rather than a sudden process beginning at large x. Log1p still shows a dominant singleton mass; it does not make the distribution Gaussian.

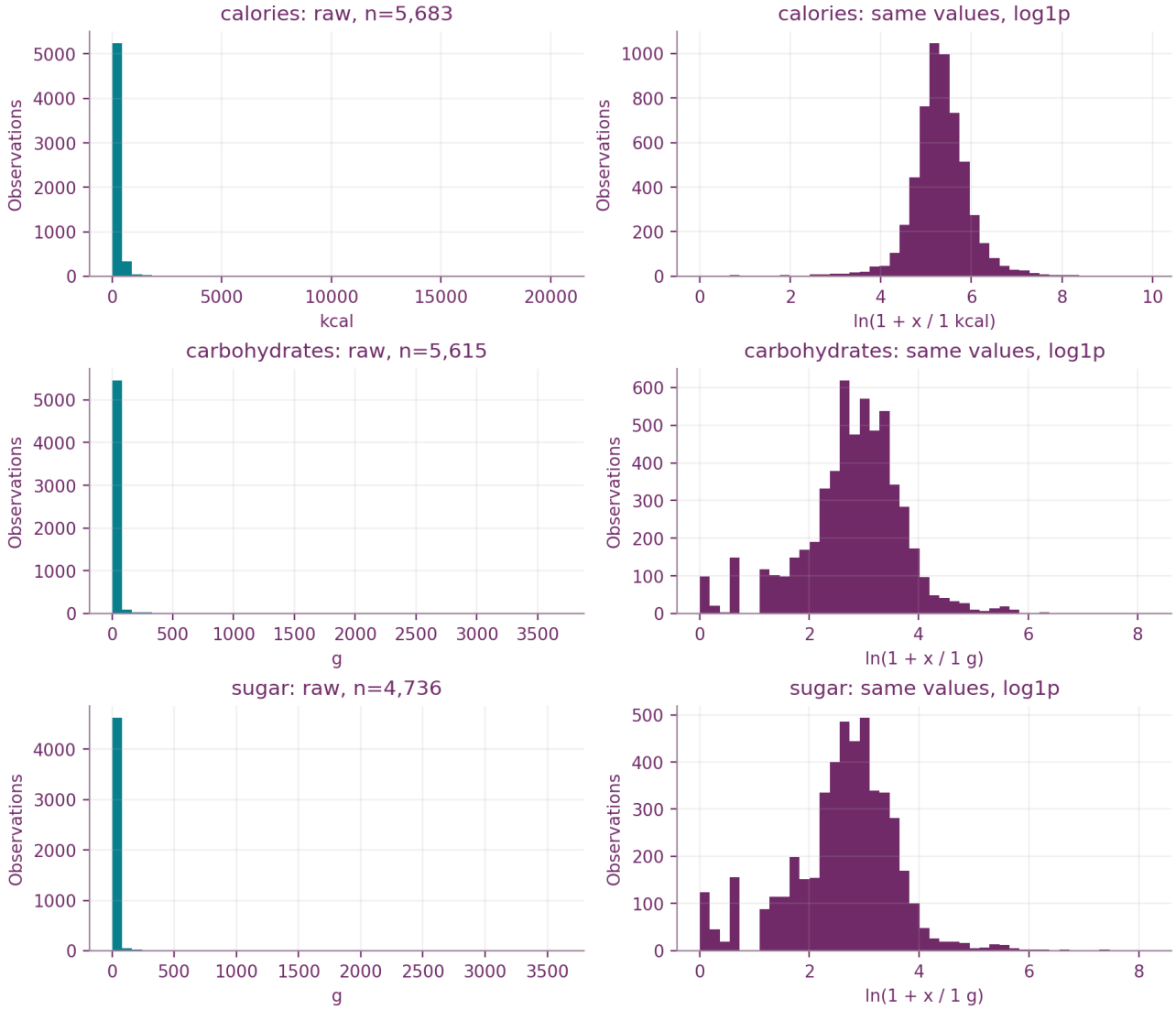
Votes. Median 30, maximum 4,348, raw skewness 25.10. After log1p, skewness is -0.68: the extreme right tail contracts, revealing concentration around tens of votes. This describes attention in the rating-observed subset, not quality.

Prep time. Median 3 minutes, 95th percentile 10, maximum 20,170, with 44 zeros. Raw skewness 78.78 falls to 4.58 but remains large. The maximum belongs to Lemongrass-Ginger Infused Tequila (336 hrs 10 mins). It may combine infusion and active time; the file cannot separate them. Keep it flagged rather than declaring it a typical cocktail or a confirmed error.

M4. Calories, carbohydrate and sugar tails

Already in the report: Figs. 23-24 (p. 29) and Fig. 26 (p. 31) use original nutrient scales; Fig. 27 (p. 31) logs positive values only.

Modification: M07 adds matched raw/log1p histograms using the deduplicated corpus. Retain zeros and give the same sample in each raw/log pair.



Field	n / zeros	Median	95th percentile	Maximum
Calories (kcal)	5,683 / 2	201	545.6	20,506
Carbohydrates (g)	5,615 / 12	17	58	3,699
Sugar (g)	4,736 / 0	14	48	3,616

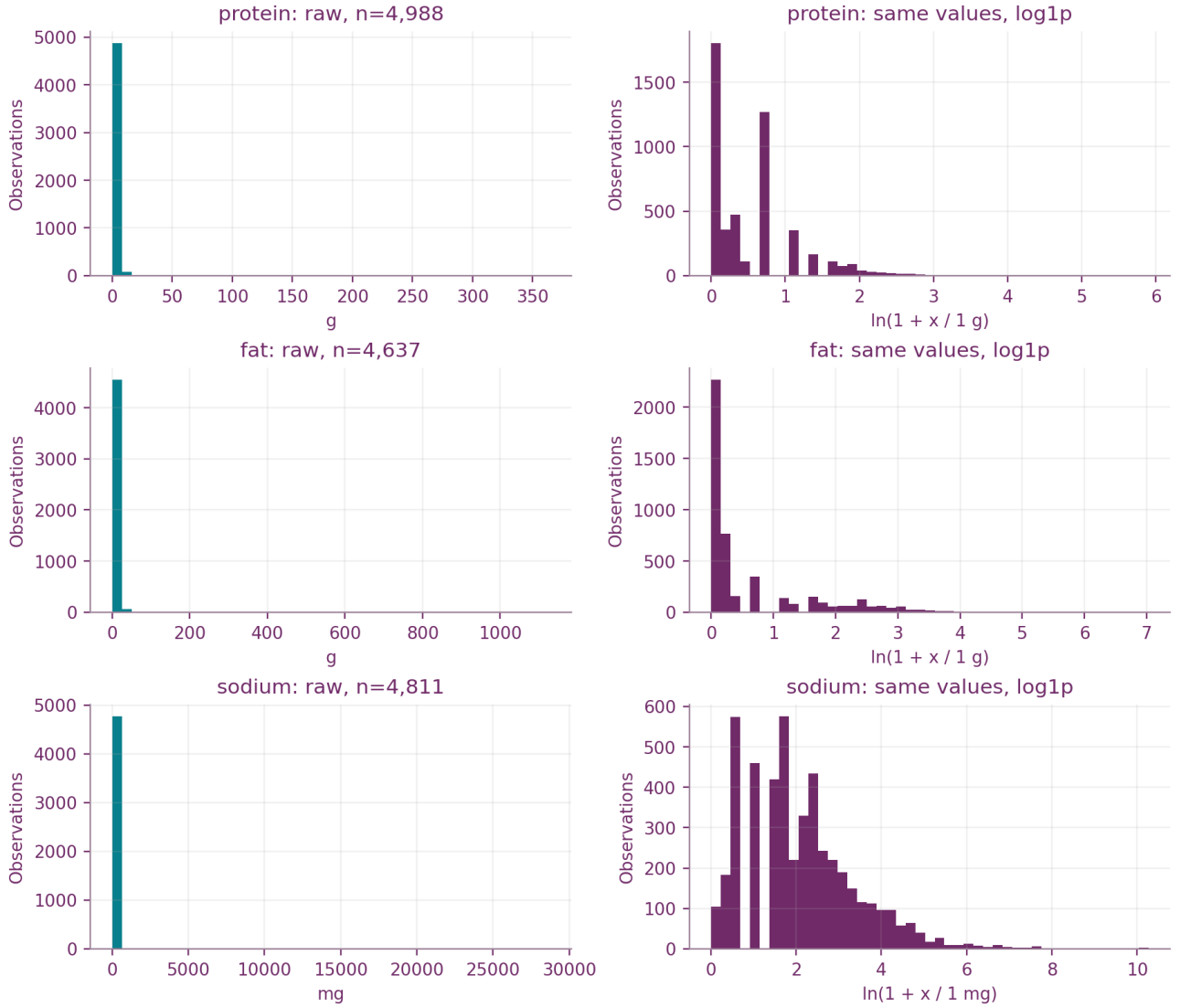
The new scale exposes the typical recipe values hidden by a handful of large panels. Log1p skewness is -0.77 for calories, -0.30 for carbohydrates and -0.29 for sugar. Therefore the claim that “all graphs remain strongly right-skewed after log” is contradicted.

Extremes checked against source rows. Homemade Coquito Bueno reports 20,506 kcal, 1,747 g carbohydrate, 364 g protein and 1,127 g fat. Medd reports 13,895 kcal and 3,699 g carbohydrate. These values are present in the original strings; they are not created by this parser. Batch totals and source errors are possible, but serving metadata are absent, so neither explanation is proven. Sugar can even exceed reported total carbohydrate in a panel, consistent with rounding or inconsistent sources; do not use these as precise per-glass nutrition.

M4. Protein, fat and sodium tails

Already in the report: Figs. 24 and 26 (pp. 29, 31) compress these variables; the older positive-only logs omit zero protein/fat.

Modification: M08 keeps zero values using log1p. State the units and zero counts rather than replacing zero with an arbitrary positive constant.



Field	n / zeros	Median	95th percentile	Maximum
Protein (g)	4,988 / 408	0.4	6	364
Fat (g)	4,637 / 506	0.2	15	1,127
Sodium (mg)	4,811 / 0	6	91	28,703

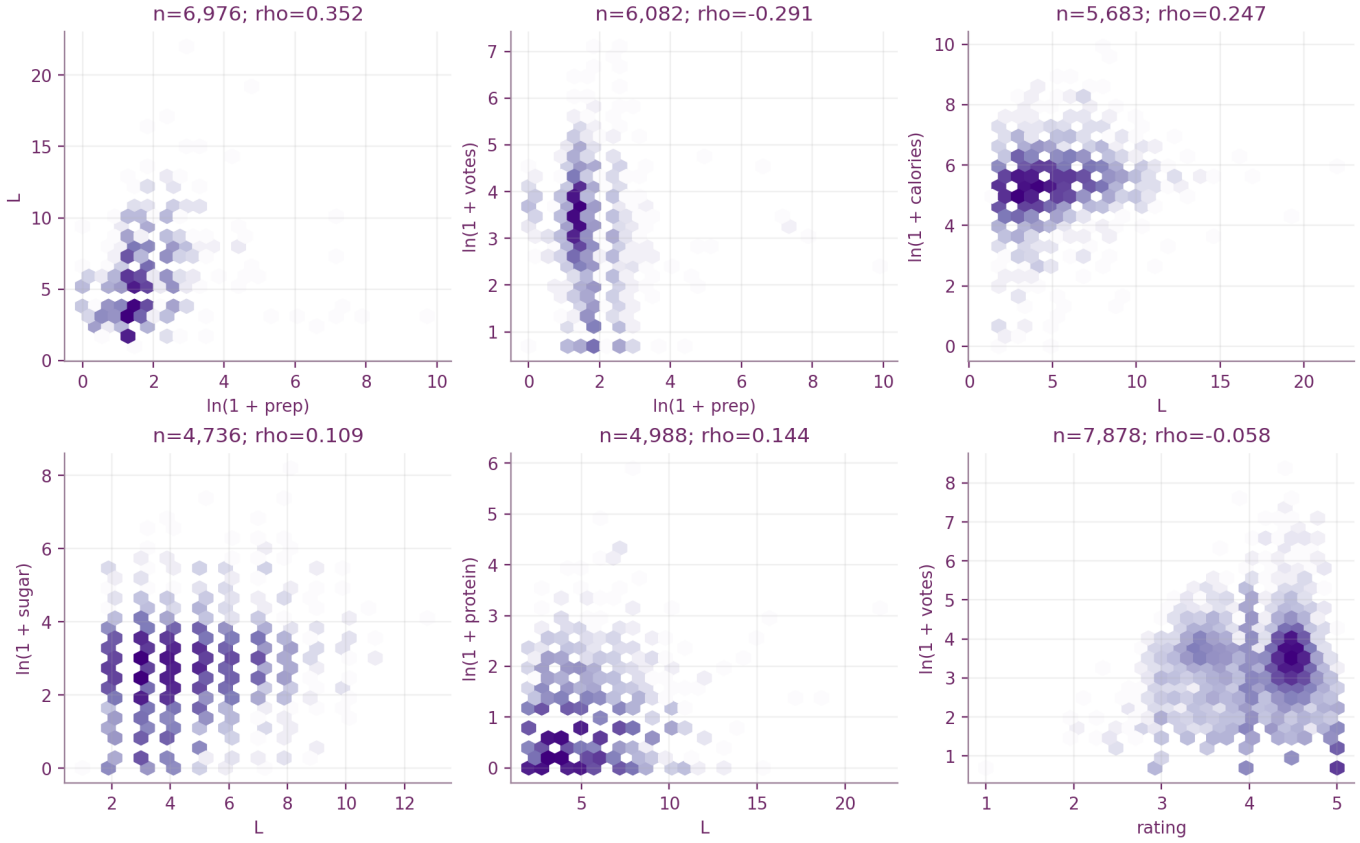
Protein and fat retain a near-zero concentration and a substantial right tail after log1p (skewness 1.84 for each). Sodium also remains positively skewed (1.18). The transformation helps reveal distributions; it does not remove their distinct structure. Low protein/fat and occasional cream, dairy or egg ingredients are a plausible compositional explanation, not a measured ingredient-to-nutrient causal result.

Sodium audit. Eric`s Bloody Bull and Bloody Bull report 28,703 and 28,512 mg sodium. The source text explicitly says mg. These unusual panels are retained and flagged. Their names suggest a savoury recipe family, but neither high salt per serving nor a unit error can be established from the CSV alone. Non-reporting also differs from zero: absent nutrient labels remain missing.

M4. Read relationships after compression

Already in the report: Fig. 11 (p. 20), Fig. 12 (p. 21), and the unnumbered multi-scatter figure (p. 35) are dominated by large time/nutrient values.

Modification: M09 uses log1p on long-tailed axes and hexbin counts to reduce overplotting. Darker hexagons contain more recipes, with logarithmic bin-count colour. L and ratings remain on their original scales.



Prep and complexity. Once the long time tail is compressed, the concentration around 3-5 minutes and small ingredient counts becomes visible. Rho=0.352 remains moderate, not zero. Multiple prep times occur at equal L, so ingredient count is an incomplete summary of the process.

Popularity and prep. Rho=-0.291 remains negative after transforming both axes. The large vertical stack at 3 minutes and variable vote counts argue against treating time as a direct popularity mechanism. The source-format heterogeneity on page 47 remains a competing explanation.

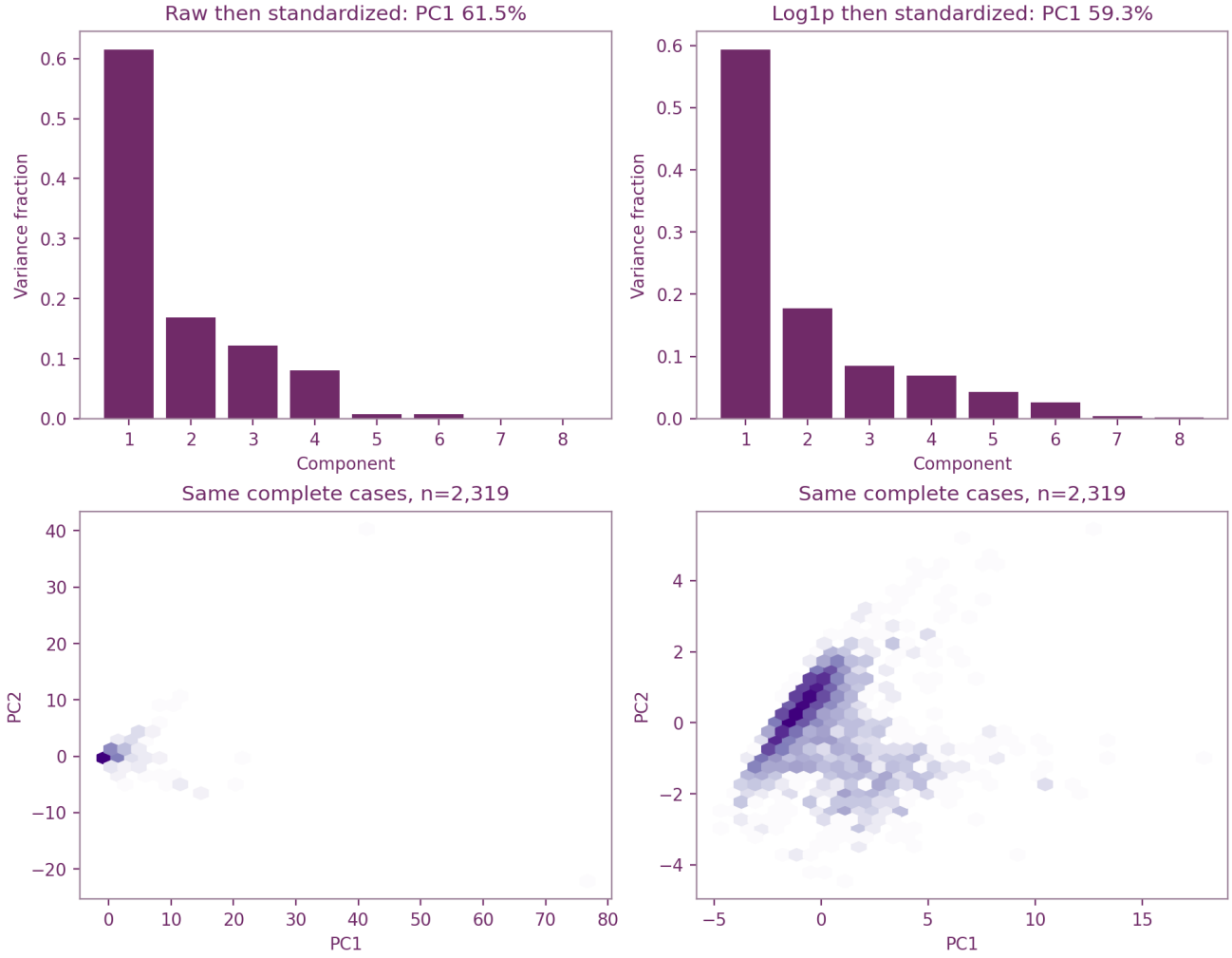
Nutrition and complexity. Calories at fixed L span a wide range; protein often stays near zero. Rho is 0.247 for calories and 0.144 for protein. Showing this spread is more useful than claiming the original empty-looking plot means no structure.

Rating and attention. Rho=-0.058 remains very weak. Compressed votes reveal low-vote rating extremes without proving that more votes improve or lower quality. Strictly increasing log1p leaves all ranks and ties unchanged, so Spearman coefficients are numerically identical before and after every corresponding transform.

M4. PCA sensitivity to skewed inputs

Already in the report: Figs. 29-30 (pp. 36-37) standardize eight original-scale nutrients. The old projection contains widely separated extreme points.

Modification: M17 transforms the nutrient inputs with log1p before standardization and PCA. Use exactly the same 2,319 complete recipes and eight variables. Never log negative component scores.



Cross-check. The original-scale rerun recovers PC1=61.46% and PC2=16.83%, matching the old 61.5% and 16.8%. Log1p-input PCA yields PC1=59.30% and PC2=17.74%. This is an exact matched-subset sensitivity comparison, not an ingredient-normalization discrepancy.

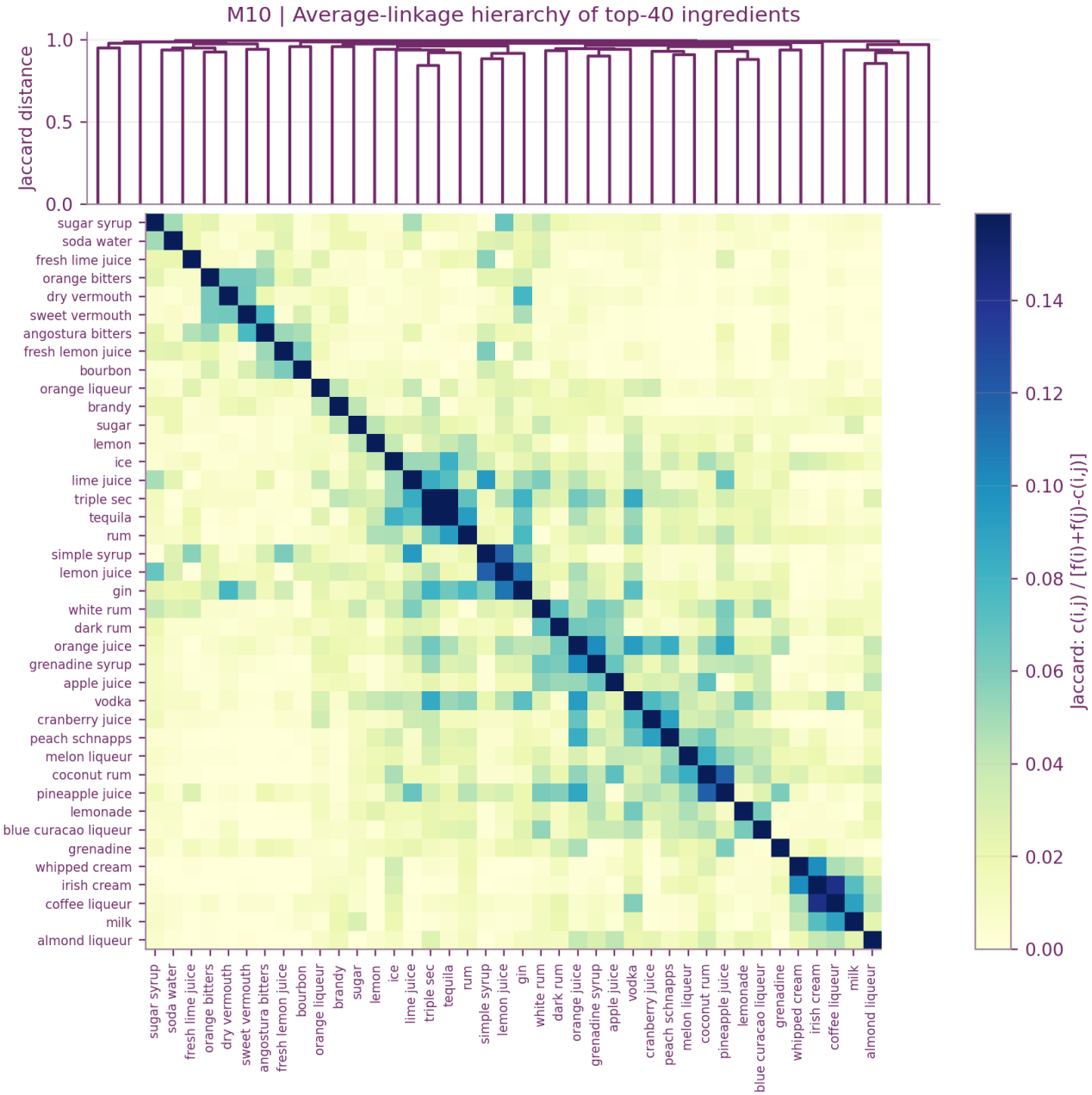
Interpretation. Large nutrient panels have less leverage after log compression. PC1 still captures a broad positive amount-related direction; its loadings have the same sign. Sodium's PC1 loading rises from 0.084 to 0.329, showing that raw-scale variation previously gave it a smaller role. Loadings and axes depend on the transformation; a component is not a universal health or recipe-quality index.

Limits. Standardization equalizes variable variance but does not itself neutralize outliers. Log compression changes the geometry and therefore may change component directions. Only the complete-panel subset is represented. Neither projection validates a natural cluster, geographical cuisine grouping or health score.

M5. Hierarchical co-occurrence view

Already in the report: Fig. 20 (p. 33) orders the top ingredients mainly by popularity; Fig. 21 (p. 34) shows a thresholded network. No hierarchy is estimated.

Modification: M10 clusters the top 40 reconstructed ingredients using Jaccard similarity of recipe membership. Distance=1-Jaccard. Average-linkage merges groups by mean pairwise distance. The same ingredient order is used on both heatmap axes.



Normalization. J(i,j)=c(i,j)/[f(i)+f(j)-c(i,j)] compares the recipes containing both with recipes containing either. It reduces the automatic advantage of globally common ingredients. The diagonal is self-similarity 1 and is saturated in the display; the colour scale expands the off-diagonal range for reading pairs.

Read the tree as an exploratory hierarchy of co-use, not a hierarchy of chemical similarity or descent. Most off-diagonal similarities are small. Even the strongest pair, triple sec-tequila, has J=0.159: common co-use still allows many alternative recipes. The next page reports the cut rule, group members, and measured limitations.

M5. What the coloured groups support

Already in the report: The old network discussion on pp. 32-34 names plausible pairing families but does not test a hierarchical grouping.

Modification: Evaluate tree cuts k=2 through 8 with silhouette on the same precomputed Jaccard distance. The maximum is only 0.020 at k=3; use that cut for consistent colours in M11-M14. Near-zero silhouette means weak global separation. These colours are exploratory co-use groups, not validated cocktail categories.

Group / colour	Ingredients (human-readable)
1 / teal	fresh lime juice, angostura bitters, fresh lemon juice, sweet vermouth, sugar syrup, soda water, dry vermouth, orange bitters, bourbon
2 / amber	vodka, lime juice, lemon juice, orange juice, gin, simple syrup, pineapple juice, triple sec, ice, tequila, cranberry juice, white rum, coconut rum, lemonade, sugar, peach schnapps, grenadine syrup, rum, orange liqueur, brandy, apple juice, melon liqueur, dark rum, grenadine, lemon, blue curacao liqueur
3 / purple	coffee liqueur, irish cream, milk, whipped cream, almond liqueur

Check against observed recipe labels

Group	Recipes containing ≥1 group item	With Category	Most common observed label / share
1	3,056	995	Spirit-Forward: 93/995 (9.3%)
2	10,954	4,684	Shots & Shooters: 336/4684 (7.2%)
3	1,705	592	Shots & Shooters: 163/592 (27.5%)

Count each recipe once per group; a recipe may contain items from several groups. The broad amber group is mixed. The cream/coffee group has 163 of its 592 labelled recipes marked Shots & Shooters (27.5%): this is descriptive support for shared use, not a pure “dessert” class. Labels remain 59% missing overall and mix ingredient names, styles and serving forms.

Pair-level support

Pair	Joint recipes	Jaccard
triple sec + tequila	224	0.159
coffee liqueur + irish cream	140	0.143
pineapple juice + coconut rum	163	0.119
lemon juice + simple syrup	300	0.118

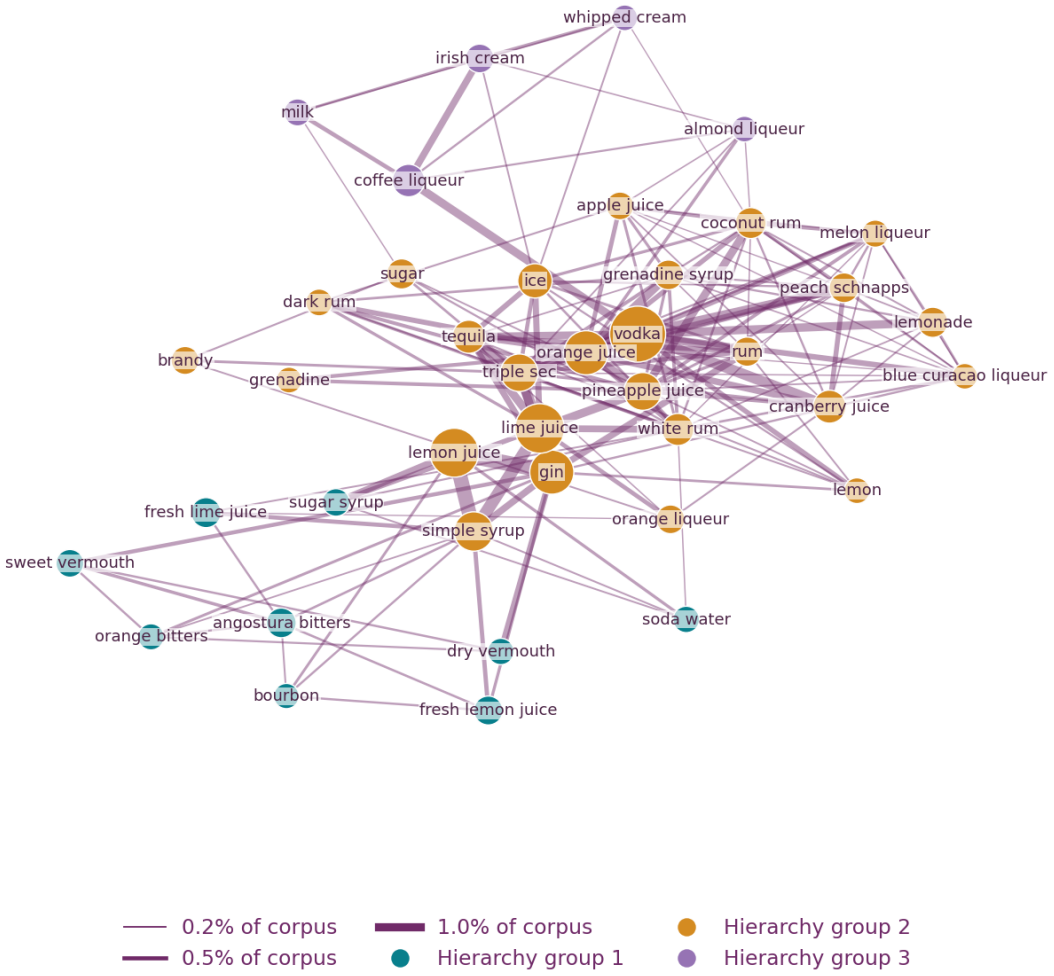
Interpretation. Familiar tequila/citrus/liqueur and dairy/coffee pairings are consistent with co-use patterns. The fresh-versus-unspecified juice split may also reflect wording or source conventions. No ingredient role, taste compatibility, or origin is established by these clusters. Category evidence with tiny n is unreliable: almond liqueur has only one labelled recipe, so its 100% modal-label share must not be treated as category validation.

M6. Network with measurable edge widths

Already in the report: p. 34, Fig. 21: top-40 positive-PMI network with minimum joint count 25; caption specifies PMI>0.4. Edge-width mapping and category colours are not quantified.

Modification: M11 keeps that visualization rule (c≥25 and natural-log PMI>0.4) on the reconstructed data, but makes every visible edge width proportional to corpus-normalized joint frequency and uses the hierarchy colours from page 55.

M11 | Revised Figure 21: 159 edges; top-40 view



The new view has 159 edges versus the old 166; ingredient normalization differs. Page 32 described “positive PMI”, while the old Figure 21 caption used >0.4. Under this reanalysis, merely PMI>0 with c≥25 would retain 208 edges. Threshold wording therefore matters.

Visual mapping. Edge prevalence=c(i,j)/16,872. Width in points=6×prevalence/p_max, where p_max is the largest pair prevalence among all top-40 pairs (320/16,872). A 0.5% edge is exactly half the width of a 1.0% edge. Node area in points²=40+0.18 f(i); colours are the three measured co-use groups. Positions are a fixed seeded spring layout; distances are not statistical estimates.

Read the structure. Wide ties represent common recipe pairs, while the separate PMI selection identifies more-than-independent co-use among frequent pairs. A high-PMI pair need not be the widest edge. Node colour represents a descriptive group, not nutrient content or a reliable cuisine label.

M6. Frequency is not association strength

Already in the report: pp. 33-34 compare a count heatmap with a PMI network. The quantities answer different questions.

Modification: Keep the denominator and edge-width scale explicit, and verify the mapping using real pairs. “Normalized frequency” here means the share of all 16,872 corpus recipes containing both items, not row-normalized counts or PMI.

Pair	c	Prevalence	Width (pt)	PMI (ln)
vodka + orange juice	320	1.897%	6.00	0.482
lemon juice + gin	314	1.861%	5.89	0.764
triple sec + tequila	224	1.328%	4.20	1.758
coffee liqueur + irish cream	140	0.830%	2.62	2.050
irish cream + whipped cream	72	0.427%	1.35	2.076

Three distinct normalizations

Quantity	Formula	Question answered
Corpus prevalence	c(i,j)/N	How common is this pair in the entire corpus?
Jaccard similarity	c(i,j) / [f(i)+f(j)-c(i,j)]	How much do the two recipe sets overlap?
PMI	ln[c(i,j) N / (f(i) f(j))]	Is shared use greater than expected from separate ingredient prevalences?
Disparity local share	c(i,j) / sum_k c(i,k)	How much of one node’s pair weight is assigned to this neighbour?

Why this matters

Vodka-orange juice occurs 320 times and receives the maximum 6-point width. Irish cream-whipped cream occurs 72 times, so its width is 1.35 points, even though its PMI is much higher. This distinguishes everyday pair coverage from disproportionately specific co-use. Neither metric measures taste quality.

Verification and limits

Pair counts were independently checked by scanning recipe sets for selected named pairs. Every edge satisfies $c \leq \min(f(i),f(j))$ and prevalence≤ 1. Within-recipe repeated ingredients never create extra co-occurrences. Keeping a common denominator means prevalence is invariant to which nodes happen to be displayed. Do not recompute N from the top-40 subgraph.

No reliable ingredient-role or cuisine categories are supplied. Rather than colouring nodes by guessed taxonomies, the appendix uses data-derived groups and reports their weak silhouette and observed label distributions. The network is suitable for hypotheses about recipe design; chemistry, substitution and causal compatibility need separate data.

M7. Backbone extraction: precise method

Already in the report: pp. 32-34 use a top-40/minimum-count/PMI display. That is a selected association network, not a multiscale backbone of the crude graph.

Modification: Build the full undirected graph from every deduplicated recipe, before any top-N, PMI or weight filtering. Each recipe contributes one unit to every unordered pair of its distinct ingredient strings. Compute the disparity filter on this full graph [R3].

Full graph definition	Boundary-normalized	Literal quantity-bearing strings
Nodes	13,860	31,224
Observed pair edges	99,039	152,293
Pair-incidence weight sum	172,300	173,077
Pre-filtering	None	None

Disparity-filter null and decision rule [R3]

Let w(i,j)=c(i,j), s(i)=Σ_j w(i,j), k(i)=number of neighbours and p(i,j)=w(i,j)/s(i). For k(i)>1, the endpoint tail probability is a(i,j)=[1-p(i,j)]^[k(i)-1]. Under the null, node strength is partitioned uniformly among its k neighbours. Retain an undirected edge if min[a(i,j),a(j,i)] < alpha: significance at either end (OR rule). Lower alpha is stricter.

Degree-one convention. If only one endpoint has degree one, test the other endpoint; do not make the leaf automatically significant. Isolated two-node components are kept for alpha>0 as an explicit convention, not as a statistical test. The normalized graph has 34 such dyads. Alpha=0 retains nothing. An AND-rule sensitivity is also tabulated.

What alpha does not mean

Alpha is a threshold under a local strength-allocation null. A retained edge is not proven “real”, and a non-retained edge is not proven absent or unimportant. The filter does not directly test independence of ingredient occurrences across recipes. Edge tests are numerous and dependent; the unadjusted alpha sweep is exploratory. Page 64 adds multiplicity and a different null comparison.

Why use raw counts as weights?

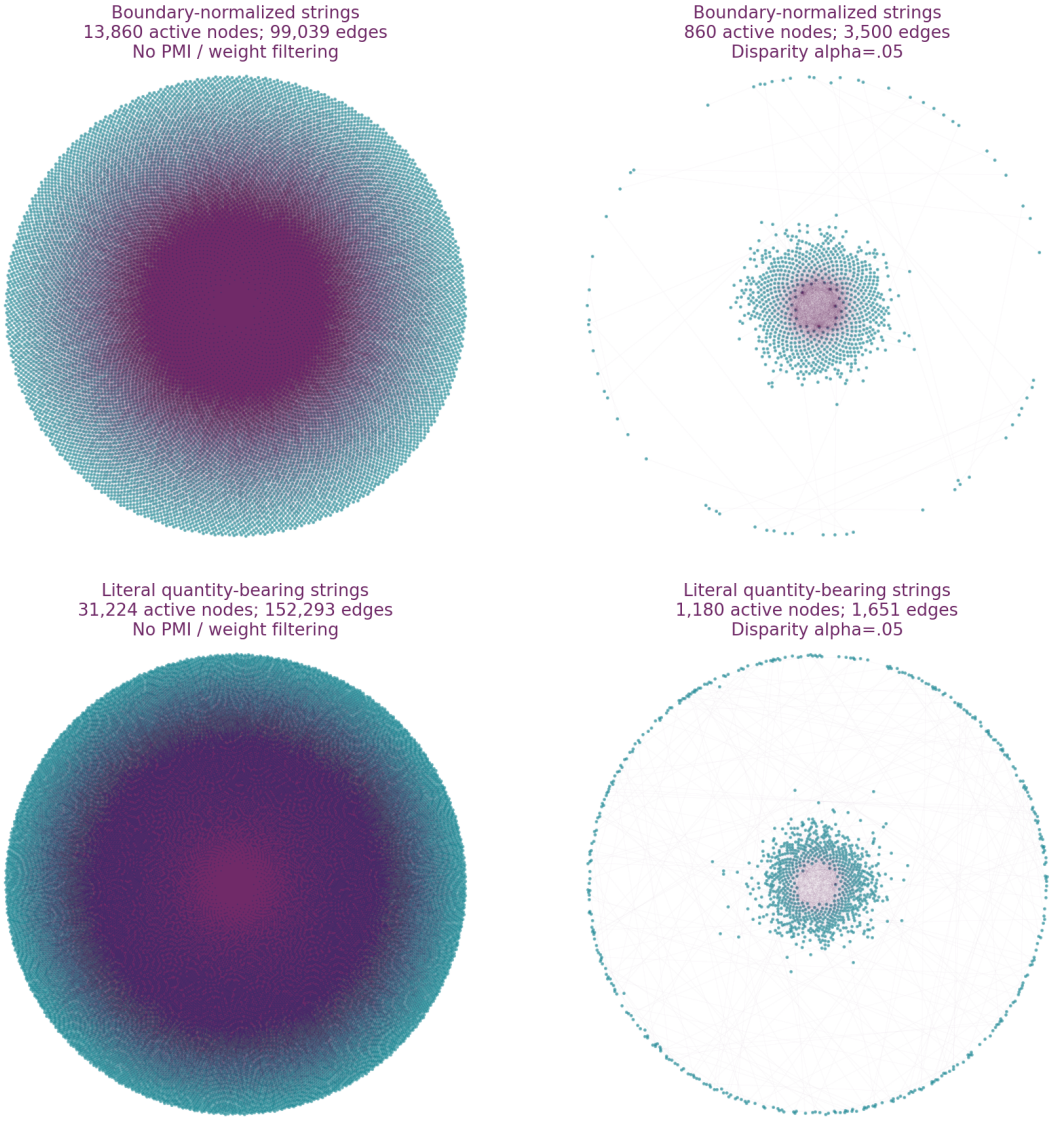
Counts preserve pair-incidence meaning and include rare strings rather than beginning with the old PMI selection. Dividing every weight by the same N leaves local shares and filter decisions unchanged. The literal-string graph tests sensitivity to no quantity stripping. These are different nodes, so their alpha results cannot be treated as the same network.

M7. Full crude graphs and backbones

Already in the report: The network on p. 34 contains only 40 selected ingredients; the complete recipe projection was not displayed.

Modification: M16 shows every active node and observed edge in both crude representations, beside the alpha=0.05 backbone. No top-N or PMI preselection is used in constructing or testing these graphs.

M16 | Full crude graphs and extracted backbones



All active nodes and edges shown. Rank-spiral coordinates are illustrative; spatial proximity has no meaning.

Layout is deliberately illustrative. A rank-based spiral assigns identical coordinates within each raw/backbone pair; centre-to-edge position reflects weighted-degree order. Spatial proximity is not a similarity estimate, and the spiral is not an inferred community structure. This full-graph view shows scale and pruning; named-pair interpretation uses the labelled top-40 view on page 56.

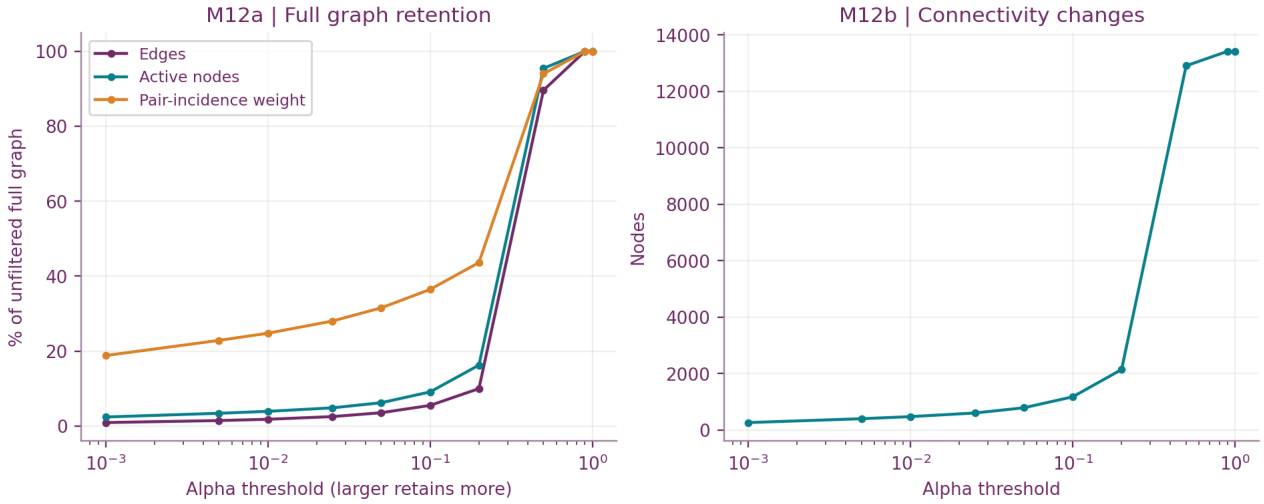
The literal-string version fragments identities such as different amounts of vodka. It retains much less original pair weight at alpha=0.05 than boundary normalization does (6.03% versus 31.53%). Thus crude tokenization is not a neutral choice. Neither graph is a complete semantic ingredient ontology; both are sensitivity representations of the same recipe file.

M7. Alpha sweep on the full graph

Already in the report: pp. 32-34 use one display threshold; no alpha sensitivity is available there.

Modification: Test alpha=0, .001, .005, .01, .025, .05, .1, .2, .5, .9 and 1. Report full-graph results rather than extrapolating from 40 visible nodes. Alpha is continuous; these values sample strict, intermediate and permissive regimes.

Alpha	Edges (% full)	Active nodes	Weight %	Components / largest
0	0 (0.00%)	0	0.00	0 / 0
0.001	895 (0.90%)	332	18.79	38 / 258
0.005	1,432 (1.45%)	469	22.84	38 / 395
0.01	1,766 (1.78%)	541	24.75	37 / 469
0.025	2,478 (2.50%)	670	28.00	37 / 598
0.05	3,500 (3.53%)	860	31.53	39 / 783
0.1	5,433 (5.49%)	1,259	36.44	43 / 1,171
0.2	9,873 (9.97%)	2,250	43.61	56 / 2,138
0.5	88,751 (89.61%)	13,229	93.99	89 / 12,908
0.9	99,039 (100.00%)	13,860	100.00	128 / 13,421
1	99,039 (100.00%)	13,860	100.00	128 / 13,421



Main result. At alpha=.05, only 3.53% of edges, 6.20% of nodes and 31.53% of pair-incidence weight survive. This is a compact recurring core, but it loses most rare ingredients. It cannot represent the whole creative vocabulary. No claim from an airport or food-web example can substitute for these measured retention values.

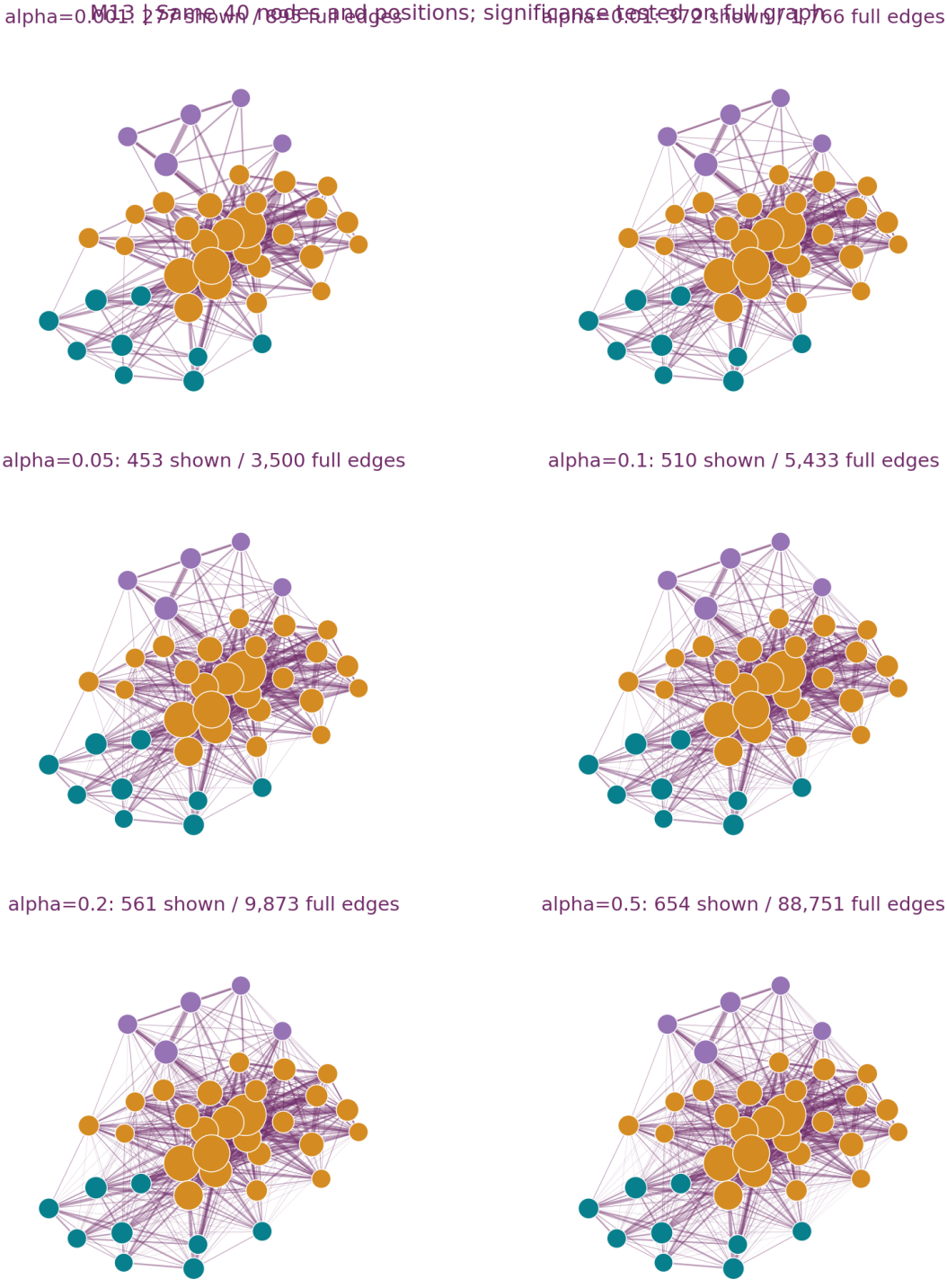
A sharp transition occurs between .2 and .5: edge retention jumps from 9.97% to 89.61%. Many low-count neighbours have similar local weights and cross the threshold together. In this graph 83.15% of edges have weight one. The full graph has 128 components and a largest component of 13,421 nodes; high retention does not imply universal connectivity. At .9 and 1 the observed edges are all retained under the stated dyad convention.

M7. How the visible backbone changes

Already in the report: p. 34 presents one fixed top-40 association network; it cannot show the dependence on alpha.

Modification: M13 keeps the same 40 most frequent nodes, colours, positions and edge-width mapping in every panel. Edge significance was computed with full-graph degrees and strengths before taking this induced view. All observed edges are eligible here; the old PMI/c≥25 cut is not applied.

M13 | Same 40 nodes and positions; significance tested on full graph



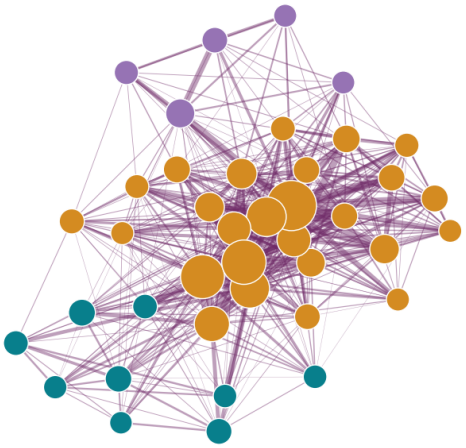
Panel titles distinguish displayed top-40 edges from the full retained edge count. Isolated displayed nodes remain visible for comparison. A disappearing line is a filtering decision under the chosen null, not evidence that a recipe pair never occurs. The selected view over-represents common ingredients, so use page 60 for corpus-wide coverage.

M7. Retained versus non-retained ties

Already in the report: p. 34 hides weak ties with fixed thresholds, but does not compare their measured properties with a significant backbone.

Modification: M14 splits the same 713 observed top-40 pairs at alpha=.05. The retained panel contains 453 edges and the other 260. Zero-count pairs are absent from BOTH panels; they are not “non-significant observed edges”.

M14 | Retained at alpha=.05
453 displayed edges



M14 | Not retained at alpha=.05
260 displayed edges



Pair	Count	Min endpoint p	Decision at .05
fresh lemon juice + dark rum	7	0.0387	Retained
rum + soda water	7	0.0430	Retained
coconut rum + irish cream	14	0.0524	Not retained
melon liqueur + almond liqueur	12	0.0647	Not retained

Local significance is not simply large count. Some seven-recipe ties survive because they occupy a sufficiently large share at one endpoint, whereas coconut rum-irish cream has 14 shared recipes but p=0.0524 and is not retained at .05. The decision is also threshold-sensitive: this example would survive at alpha=.1.

Scope of comparison. Across the full normalized graph, 3,500 edges survive and 95,539 observed edges do not at .05. Retained edges carry 31.53% of the total pair-incidence weight. They capture repetitive core combinations but lose much of the rare-string tail. “Insignificant” is used only as shorthand for not passing this particular test, never as proof of irrelevance.

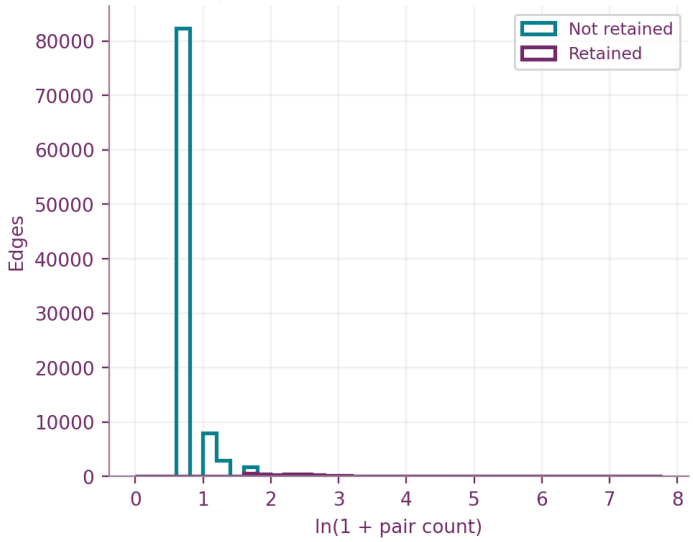
Colours retain the hierarchy grouping from page 55, and widths still show normalized prevalence. Neither attribute changes according to the significance result. This keeps frequency, grouping and test status as separate, interpretable data features.

M7. Compare weights and filter choices

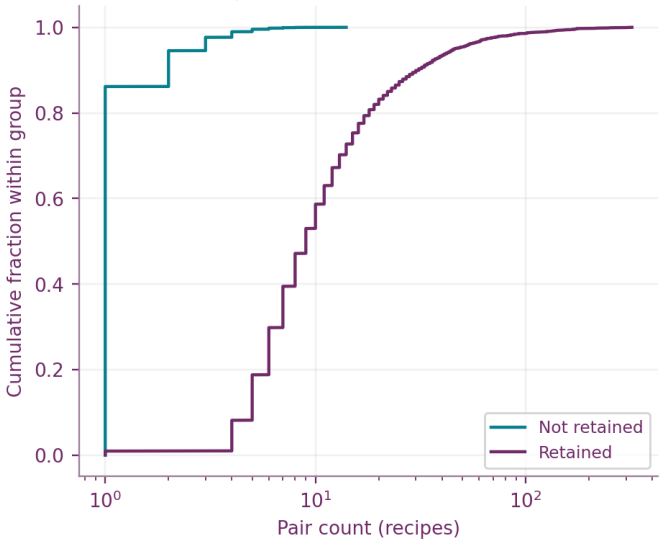
Already in the report: The original report makes no full-graph comparison of surviving versus removed edge weights or of local versus global filtering.

Modification: M15 shows the full observed-edge weight distributions for retained and non-retained edges. Log count axes make both low- and high-weight ties visible.

M15a | Both groups include low-count edges



M15b | Count and significance differ



Filter at matched edge budget	Retained edges	Active nodes	Retained weight
Local disparity, alpha=.05	3,500	860	31.53%
Largest-count edges, same 3,500-edge budget	3,500	577	32.20%

The global baseline orders all edges by descending count, using alphabetical pair order to resolve ties (boundary count=5). It keeps slightly more total weight but fewer nodes. The local filter reaches more low-strength regions for the same edge budget. It is therefore useful for diversity of coverage, not guaranteed superior on every criterion.

What fails under a universal-backbone interpretation

At .05 the local filter still retains only 6.20% of all ingredient nodes. Therefore a readable core is not evidence that the full corpus has been faithfully preserved. Both weight distribution and vocabulary coverage must be reported. The CDFs show relative weight distributions within groups; the histogram shows absolute edge counts, so the two views answer different questions.

Recommendation. Use alpha=.05 as a conventional exploratory core, .01 as a stricter sensitivity, and .1-.2 for broader coverage; show the complete sweep rather than declaring any threshold intrinsically optimal. No alpha was tuned to recover a desired culinary story. Literal-string and multiple-testing sensitivity on page 64 restrict stronger claims.

M7. Challenge significance and robustness

Already in the report: The original network discussion (pp. 32-34) is exploratory and does not control multiple testing or compare null models.

Modification: Compare endpoint rules, literal-string input, multiplicity adjustment, and a different recipe-level null. These checks can disprove interpretations that a single network picture seems to support.

Alpha	Normalized OR / AND edges	Literal-string OR edges	Literal weight %
0.001	895 / 459	385	1.90%
0.01	1,766 / 892	686	3.30%
0.05	3,500 / 1,676	1,651	6.03%
0.1	5,433 / 2,508	3,087	8.77%
0.2	9,873 / 4,361	8,481	15.49%
0.5	88,751 / 41,087	149,496	98.38%

OR versus AND

At .05 the OR rule retains 3,500 normalized edges; requiring significance at BOTH endpoints retains 1,676. OR allows a tie important to just one ingredient; AND asks for mutual local dominance. These are different questions and should not be interchanged.

Multiplicity sensitivity

Apply Benjamini-Hochberg adjustment [R4] at q=.05 to 197,838 testable endpoint probabilities, then retain an edge if either endpoint passes: 628 edges survive, versus 3,500 unadjusted. Degree-one dyads are not included as tested discoveries. This is a stricter sensitivity display, not a guarantee of FDR control under arbitrary endpoint/recipe-projection dependence. Local-null probabilities are model-based and many edges share recipes.

Different null: ingredient-occurrence independence

For each of all 780 top-40 pairs, use a one-sided Fisher exact test on the recipe-level 2×2 presence table. BH-adjust across all 780 tests, including zero-count pairs. This asks whether the two ingredient indicators are enriched relative to their marginal prevalences; it is not the disparity null.

Top-40 comparison	Fisher q<.05	Fisher q≥.05
Disparity p<.05	178	275
Disparity p≥.05	0	327

Example: vodka-lemon juice appears 173 times and passes disparity, but PMI=-0.368 and Fisher enrichment q=1. Its frequency is locally prominent while still lower than an independence expectation. Thus “disparity significant” cannot be renamed “positive pairing preference”. Fisher also assumes exchangeable recipes; variants and varying recipe lengths limit that null. A size/degree-preserving recipe-incidence randomization would be a stronger future check.

Revised conclusions and explicit corrections

Already in the report: The original executive summary (p. 5), main findings (p. 38), conclusion and limitations (p. 39) remain visible as historical results.

Modification: Use the following statements when interpreting the original claims alongside the new work. Ingredient-dependent numerical differences reflect the documented reconstruction, not a silent edit of the old figures.

Claim tested	New evidence	Status
One invariant Zipf exponent	Alpha 1.032 / 1.123 / 1.187 across required windows; curved residuals	Not supported globally
Heaps sublinear vocabulary growth	Mean beta 0.757; all 100 recipe-order fits; endpoint boxes collapse	Supported on fitted corpus/range
All heatmaps show little correlation	Sugar-carbs 0.957; fat-saturated fat 0.916; count and KS matrices are not correlations	Contradicted
Log transforms strengthen rank relationships	Same Spearman rho before and after strictly increasing log1p	Contradicted
Hierarchy defines clear cocktail classes	Maximum silhouette 0.020; incomplete mixed Category labels	Not established
Backbone .05 preserves most recipe diversity	3.53% edges, 6.20% nodes, 31.53% weight retained	Contradicted for vocabulary coverage
Retained edge means positive pairing enrichment	275 top-40 pairs pass disparity but not Fisher enrichment at stated levels	Contradicted as an equivalence

Textual and mathematical corrections found during review

Page 18, Figure 9: the caption says “positive” but reports rho=-0.058. Read this as a weak NEGATIVE association. Pages 5, 16-17, 38: ratings concentrated near 5 exhibit a ceiling; “left-censored toward 5” is not a demonstrated censoring mechanism.

Pages 5 and 38: beta≈0.74 and the cited food beta≈0.56 are both sublinear, but calling them numerically close lacks an agreed criterion. Pages 26-27: the stated MA form has derivative of log y equal to b/L+c. The quoted reference pair b=-0.26,c=-0.019 cannot have a positive-L minimum. The cocktail b=-0.157,c=0.042 implies a model minimum near L=3.74. This is an internal equation check; verify the reference-paper parameter transcription before repeating its claimed L≈13 comparison.

Page 32 versus p. 34: “positive PMI” and “PMI>0.4” specify different edge sets. Pages 27, 31-32: candidate KS rankings, visual symmetry and high fit scores are descriptive; fitted-parameter distribution tests require appropriate calibration. The new log1p plots do not replace the old ln(x), positive-only distribution tests.

Methods, references and verification record

Already in the report: pp. 2-4 and 39 describe an earlier eda_project pipeline, but that source code is not among the attached files.

Modification: The new section was computed directly from the supplied CSV with a deterministic reconstruction, then appended to the original PDF. All numerical tables were generated from computed results. Original pages are preserved rather than regenerated.

Reproduction specification

Recipe corpus: drop exact duplicates, then Name+Ingredients clones, keep same-name variants. Ingredient definitions: page 41. Zipf: inclusive ranks, deterministic alphabetical ties, log10 OLS; 250 recipe bootstrap repeats re-rank each sample. Heaps: 100 random orders, unique rounded geomspace(20,N,150) values plus fixed checkpoints, one log-log fit per order. Fixed NumPy seed=2024050. Correlation CIs: 400 recipe resamples per listed pair; Spearman uses average tied ranks.

Nutrient parsing recognizes colon labels and compact value/unit/name strings; Carbs becomes carbohydrates. Scientific notation such as 5.0E-5 g is preserved. Preparation parsing converts seconds/minutes/hours/days, averages numeric ranges and collapses whole repeated phrases. Ambiguous composite time strings are summed, consistent with the stated earlier rule, but cannot be separated into active versus elapsed time.

Hierarchy: top 40 by incidence, average linkage on 1-Jaccard, k=2..8 silhouette sweep; figures use k=3. Networks: all unordered ingredient pairs, no self-loops; natural-log PMI for the M11 display only. Full-graph disparity, endpoint conventions, alpha grid, BH sensitivity and Fisher comparison are defined on pp. 58-64. M16 uses non-inferential rank-spiral coordinates. PCA: same eight complete-case nutrients; compare raw versus log1p inputs after z-standardization.

Checks completed

Verified row/duplicate counts and name-column identity; direct named-pair recounts; symmetric pair matrix; c≤min(f(i),f(j)); total strength=2×edge weight; pair-weight sum=Σ_recipe L(L-1)/2; monotonically growing Heaps curves and fixed final vocabulary; nested alpha edge sets; unchanged Spearman after log1p; original PCA variance/loadings reproduced. Every figure uses its stated subset. Original pages were compared after merging; all new pages were rendered for layout review.

References for new methods

[R1] Bagler et al. (2026). Universal statistical laws governing culinary design. arXiv:2604.28021v1. Framework: culinary reuse, vocabulary growth and composition. arxiv.org/abs/2604.28021

[R2] Clauset, Shalizi & Newman (2009). Power-law distributions in empirical data. SIAM Review 51, 661-703. OLS is not a power-law validation test. arxiv.org/abs/0706.1062

[R3] Serrano, Boguñá & Vespignani (2009). Extracting the multiscale backbone of complex weighted networks. PNAS 106, 6483-6488. doi:10.1073/pnas.0808904106. arxiv.org/abs/0904.2389

[R4] Benjamini & Hochberg (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. JRSS B 57, 289-300. doi:10.1111/j.2517-6161.1995.tb02031.x. BH implemented as a sensitivity analysis; dependence caveat stated on p. 64.

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